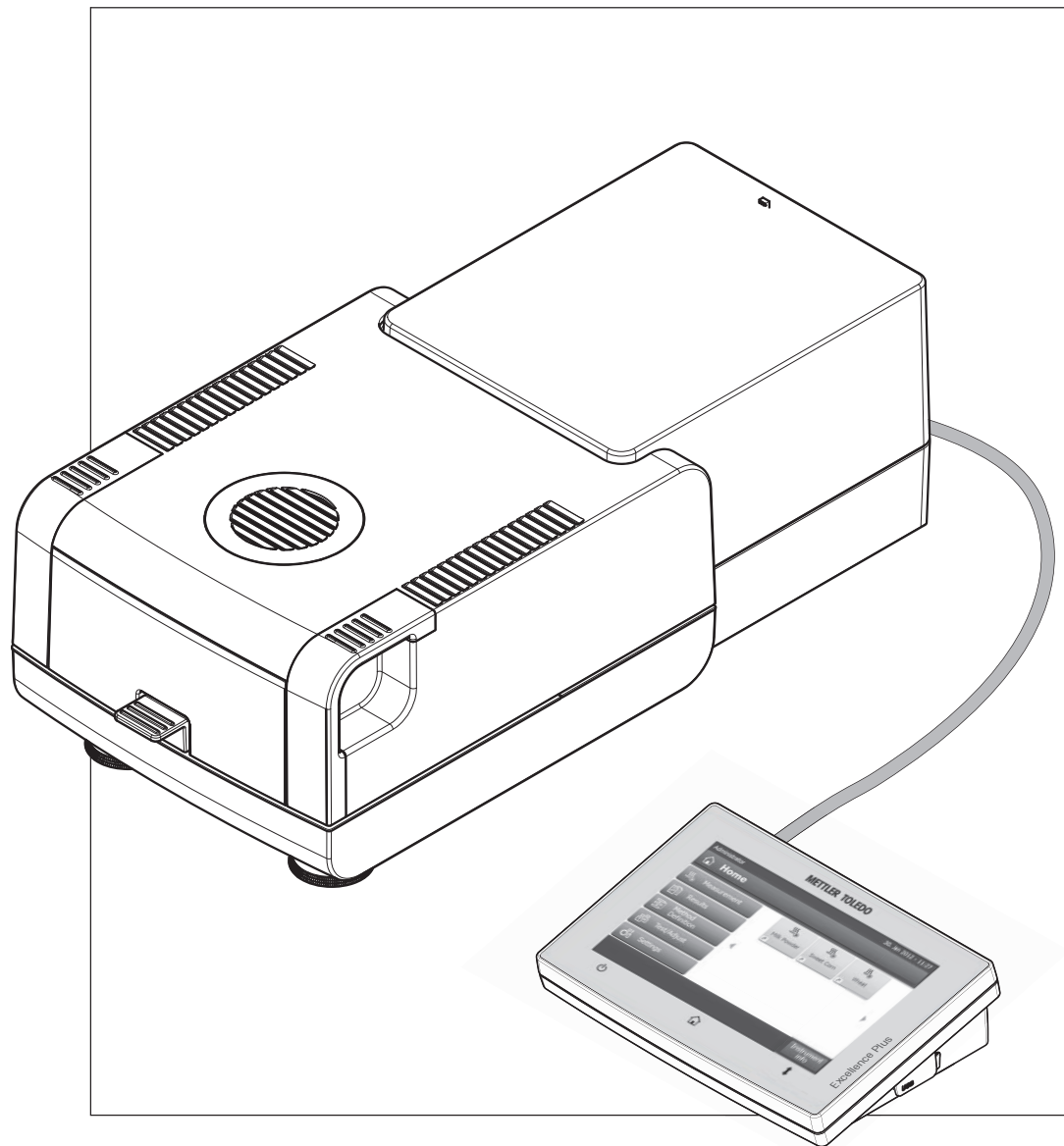


Moisture Analyzer Excellence Plus HX204



METTLER TOLEDO

Table of Contents

1	Introduction	7
	1.1 Conventions and Symbols Used in these Operating Instructions	7
2	Safety Information	8
	2.1 Definition of Signal Warnings and Symbols	8
	2.2 Product Specific Safety Notes	8
3	Design and Function	11
	3.1 Overview	11
	3.1.1 Drying Unit	11
	3.1.2 Terminal	12
	3.1.3 Keys on the Terminal	14
	3.2 User Interface	15
	3.2.1 User "Home" Screen	16
	3.2.2 Basic Elements on Touch Screen	17
	3.2.3 Icon and Buttons	17
	3.2.3.1 Status Message Icons	17
	3.2.3.2 Title Bar Icons	17
	3.2.3.3 Action Bar Buttons	17
	3.2.4 Input Dialogs	18
	3.2.4.1 Entering Text and Numbers	18
	3.2.4.2 Entering Numerical Values	19
	3.2.4.3 Changing Date and Time	20
	3.2.5 Lists and Tables	21
	3.2.6 Work Screen	22
	3.2.7 Graphics View	23
4	Installation and Putting into Operation	24
	4.1 Scope of Delivery	24
	4.2 Location	24
	4.3 Opening and Closing Sample Chamber	25
	4.4 Connecting Instrument	25
	4.5 Setting up Drying Unit	26
	4.6 Leveling Drying Unit	27
	4.7 Anti-theft Device	28
	4.8 Setting Reading Angle of Terminal	28
	4.9 Setting Date, Time and Country	28
	4.10 Adjustment	29
	4.11 Install METTLER TOLEDO RHT sensor	29
5	My First Measurement	31
6	Settings	34
	6.1 User Preferences	35
	6.1.1 Languages	35
	6.1.2 Screen	35
	6.1.3 Sound	36
	6.1.4 Password	36

6.2	System Settings	37
6.2.1	Regional Settings	37
6.2.2	Peripherals	37
6.2.3	Default Settings	39
6.2.3.1	Default Languages	39
6.2.3.2	Default Screen Settings	40
6.2.4	Company Information	40
6.2.5	Touch Screen Adjustment	40
6.2.6	Network Interfaces	41
6.3	Applications Settings	44
6.3.1	Identifications Management	44
6.3.2	Print and Export Management	45
6.4	Quality Management	48
6.4.1	Test / Adjustment settings	48
6.4.1.1	Equipment	48
6.4.1.2	Test Settings	49
6.4.1.3	Adjustment Settings	51
6.4.1.4	Print and Output Settings	51
6.4.1.5	FACT (Fully automatic adjustment)	52
6.5	User Management	53
6.5.1	Groups	53
6.5.2	Users	55
6.5.3	Account Policies	56
6.6	System and Data Management	57
6.6.1	Result Management	57
6.6.2	Export / Import	57
6.6.3	Back up / Restore	59
6.6.4	Export event logs	59
6.6.5	Reset	60
6.6.6	Update	60
7	Test/Adjust	62
7.1	Adjustments	62
7.1.1	Weight Adjustment - Internal	63
7.1.2	Weight Adjustment - External	63
7.1.3	Temperature Adjustment	64
7.1.4	Adjustment Printout Examples	66
7.2	Tests	67
7.2.1	Weight test - internal	67
7.2.2	Weight test - external	67
7.2.3	Temperature test	68
7.2.4	SmartCal test	69
7.2.5	Test Printout Examples	71
7.3	History	72
8	Method Definition	73
8.1	Main Measuring Parameters	75
8.1.1	Settings of Drying Program	75
8.1.1.1	Settings of Standard drying	75
8.1.1.2	Settings of Rapid drying	76

	8.1.1.3	Settings of Gentle drying	77
	8.1.1.4	Settings of Step drying	79
	8.1.1.5	Settings of Switch-off Criterion (SOC)	80
	8.1.1.6	Settings of SOC Delay	82
	8.1.2	Settings of Display Mode	82
	8.1.3	Settings of Start Weight	85
	8.2	Result and Value Handling	86
	8.2.1	Control limits	86
	8.2.2	Resolution	87
	8.2.3	Free Factor	88
	8.3	Workflow Handling	89
	8.3.1	Starting Mode	89
	8.3.2	Preheating	89
	8.4	General Method Properties	91
	8.4.1	Method name	91
	8.5	Testing Methods	92
	8.5.1	Testing a Method	92
	8.5.2	Testing with AutoMet	93
9	Measurement		94
	9.1	Performing a Measurement	94
	9.2	Work with Weighing-in Aid	96
	9.3	Using Shortcuts	97
10	Results		99
	10.1	Graphical Evaluations of Measurement Results	99
	10.2	Information on Printouts	101
	10.3	Exporting Results	103
	10.3.1	Single Result Export	103
	10.3.2	Multiple Result Export	104
	10.4	Information on Exports	105
11	Maintenance		107
	11.1	Cleaning	107
	11.1.1	Sample Chamber	108
	11.1.2	Heating Module	108
	11.1.3	Fan Grill	110
	11.2	Dust Filter	110
	11.3	Replacing Power Line Fuse	110
	11.4	Disposal	111
12	Troubleshooting		112
	12.1	Error Messages	112
	12.2	Status Messages	113
	12.3	What to Do if...	114
13	Technical Data		116

13.1	General Technical Data	116
13.1.1	Explanatory note to periodic inspections in accordance with EU Directive 2001/95/EC	118
13.2	Dimensions	119
13.2.1	Drying Unit	119
13.2.2	Terminal	120
13.3	Interfaces Specification	120
13.3.1	RS232C	120
13.3.2	USB Host	121
13.3.3	USB Device	121
14	Accessories and Spare Parts	122
15	Appendix	127
15.1	How to Obtain Best Results	127
15.1.1	Measurement Principle of Halogen Moisture Analyzer	127
15.1.2	Notes on Adjustment of Balance and Heating Module	127
15.1.3	Optimum Sample Preparation	128
15.1.4	Further Information on Moisture Determination	128
15.2	Recommended Printer Settings (Strip printer)	128
15.3	Installing USB Device Driver	129
15.4	Third Party License/Notice	130
	Index	131

1 Introduction

Thank you for purchasing a Halogen Moisture Analyzer from METTLER TOLEDO.

Your Moisture Analyzer is fast and reliable. It offers a high level of operating convenience and useful functions to facilitate determination of the moisture content of your samples.

Behind your instrument stands METTLER TOLEDO, a leading manufacturer of not only balances and scales for the lab and production, but also analytical measuring instruments. A customer service network covering the entire globe with well trained personnel is your service at all times, whether you are choosing accessories or require guidance for a specific application to ensure optimum utilization of your instrument.

Your Halogen Moisture Analyzer is used for determining the moisture content of almost any substance. The instrument works on the thermogravimetric principle. At the start of the measurement the Moisture Analyzer determines the weight of the sample, the sample is then quickly heated by the integral halogen heating module and the moisture vaporizes. During the drying process the instrument continually measures the weight of the sample and displays the reduction in moisture. Once drying has been completed, the moisture or solids content of your sample is displayed as the final result.

Of decisive importance in practice is the rate of heating and even heating of the surface of the sample. In comparison with conventional infrared heating or the drying oven method, for example, the halogen heating module of your instrument needs a shorter time to reach its maximum heating power. It also allows use of high temperatures, an additional factor in shortening the drying time. Uniform heating of the sample material ensures good repeatability of the drying results and makes it possible to use a smaller amount of sample.

The Moisture Analyzer conforms with all common standards and directives. It supports stipulations, work techniques and result records as demanded by all international quality assurance systems, e.g. GLP (Good Laboratory Practice), GMP (Good Manufacturing Practice). The instrument has a CE declaration of conformity and METTLER TOLEDO as the manufacturer has been awarded ISO 9001 and ISO 14001 certification. This provides you with the assurance that your capital investment is protected in the long term by a high product quality and a comprehensive service package (repairs, maintenance, servicing, adjustment service).

Finding More Information

► www.mt.com/hxhs

Software version

These operating instructions are based on the initially installed firmware (software) version V2.10.

1.1 Conventions and Symbols Used in these Operating Instructions

Key and button designations are indicated by a picture or text in square brackets (e.g. [**Save**]).

These symbols indicate an instruction:

- prerequisites
- 1 steps
- 2 ...
- ⇒ results

2 Safety Information

2.1 Definition of Signal Warnings and Symbols

Safety notes are marked with signal words and warning symbols. These show safety issues and warnings. Ignoring the safety notes may lead to personal injury, damage to the instrument, malfunctions and false results.

Signal Words

WARNING	for a hazardous situation with medium risk, possibly resulting in severe injuries or death if not avoided.
CAUTION	for a hazardous situation with low risk, resulting in damage to the device or the property or in loss of data, or minor or medium injuries if not avoided.
Attention	(no symbol) for important information about the product.
Note	(no symbol) for useful information about the product.

Warning Symbols

	General hazard		Electrical shock
	Hot surface		Inflammable or explosive substance
	Toxic substance		Acid / Corrosion

2.2 Product Specific Safety Notes

General Safety Information

Your instrument meets the state of the art technology and complies with all recognized safety rules, however, certain hazards may arise in extraneous circumstances. Do not open the housing of the instrument: It does not contain any parts which can be maintained, repaired or replaced by the user. If you ever have problems with your instrument, contact your authorized METTLER TOLEDO dealer or service representative.

Always operate and use your instrument only in accordance with the instructions contained in this manual. The instructions for setting up your new instrument must be strictly observed.

If the instrument is not used according to these Operating Instructions, protection of the instrument may be impaired and METTLER TOLEDO assumes no liability.

Intended Use

Your Moisture Analyzer is used for determining the moisture in samples. Use the instrument exclusively for this purposes. Any other type of use and operation beyond the limits of technical specifications without written consent from Mettler-Toledo AG, is considered as not intended.

Moisture determination applications must be optimized and validated by the user according to local regulations. Application-specific data provided by METTLER TOLEDO is intended for guidance only.



It is not permitted to use the instrument in explosive atmosphere of gases, steam, fog, dust and flammable dust (hazardous environments).

Staff Safety

The Moisture Analyzer may be operated only by trained personnel who are familiar with the properties of the samples used and with the handling of the instrument.

In order to use the instrument, you must have read and understood the operating instructions. Keep the operating instructions for further reference.

Never make any modifications to the instrument and use only original spare parts and optional equipment from METTLER TOLEDO.

Protective Clothing

It is advisable to wear protective clothing in the laboratory when working with the instrument.



A lab coat should be worn.



A suitable eye protection such as goggles should be worn.



Use appropriate gloves when handling chemicals or hazardous substances, checking their integrity before use.

Safety Notes



WARNING

Risk of electric shock

Your instrument is supplied with a 3-pin power cable with an equipment grounding conductor. Only extension cables which meet this relevant standards and also have an equipment grounding conductor may be used. Intentional disconnection of the equipment grounding conductor is prohibited.



CAUTION

The Halogen Moisture Analyzer works with heat!

- a) Ensure sufficient free space around the instrument to avoid heat accumulation and overheating (approx. 1 m free space above the heating module).
- b) The vent over the sample must never be covered, plugged, taped over or tampered with in any other way.
- c) Do not place any combustible materials on, under or next to the instrument since the area around the heating module may be hot.
- d) Exercise caution when removing the sample. The sample itself, the sample chamber, the draft shield and any sample vessels used may still be very hot.
- e) During operation, you should never open the heating module itself as the ring-shaped heating reflector or its protective glass can reach 400 °C! If you have to open the heating module e.g. for maintenance, disconnect the instrument from the power supply and wait until the heating module has cooled down completely.
- f) No modifications must be made within the heating module. It is particularly dangerous to bend any components or remove them or to make any other changes.

Certain samples require special care!

With certain types of samples, there is a possibility of danger to personnel or damage of property. Please note that the user always has the responsibility and liability for damage caused by use of any types of samples!



CAUTION

Fire or Explosion

- Flammable or explosive substances.
 - Substances containing solvents.
 - Substances which evolve flammable or explosive gases or vapors when heated.
- a) In cases of doubt, perform a careful risk analysis.
 - b) Work at a drying temperature that is low enough to prevent the formation of flames or an explosion.
 - c) Wear protective goggles.
 - d) Work with small amounts of sample.
 - e) **Never leave the instrument unattended!**
-



WARNING

Substances which contain toxic or caustic components

Toxic gases produced during drying could cause irritations (eyes, Skin, breathing), illness or death.

- Such substances may be dried only in a fume cupboard.
-



CAUTION

Corrosion

Substances which evolve corrosive vapors when heated (e.g. acids).

- Work with small amounts of samples as the vapor can condense on cooler housing parts and cause corrosion.
-

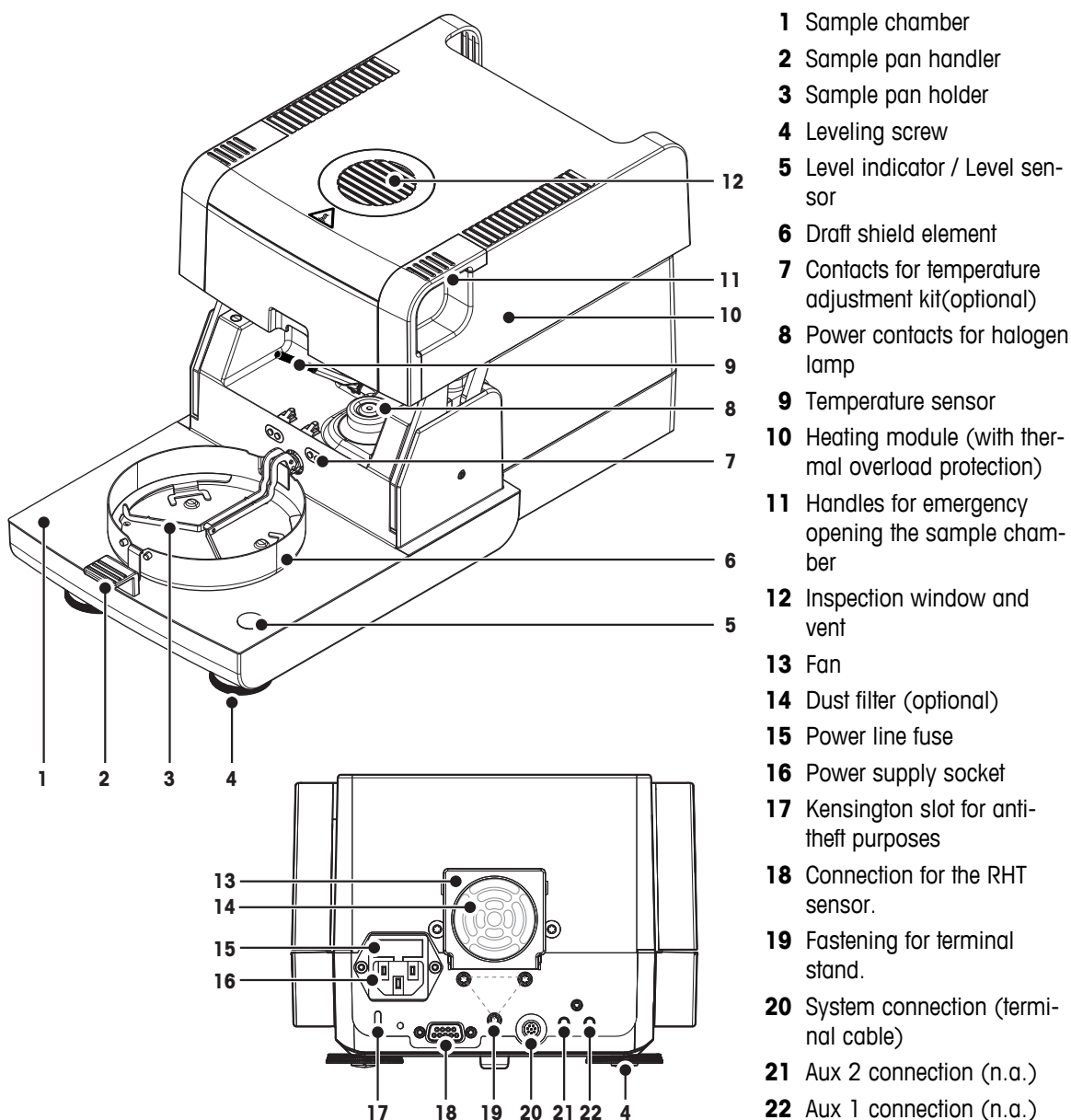
3 Design and Function

In order to work with your Moisture Analyzer, you need to familiarize yourself with the instrument layout, drying unit, graphical user interface of the touch screen as well as the terminal.

3.1 Overview

3.1.1 Drying Unit

Description of the components and the external connectors on the back of the drying unit (Weighing unit with heating module).



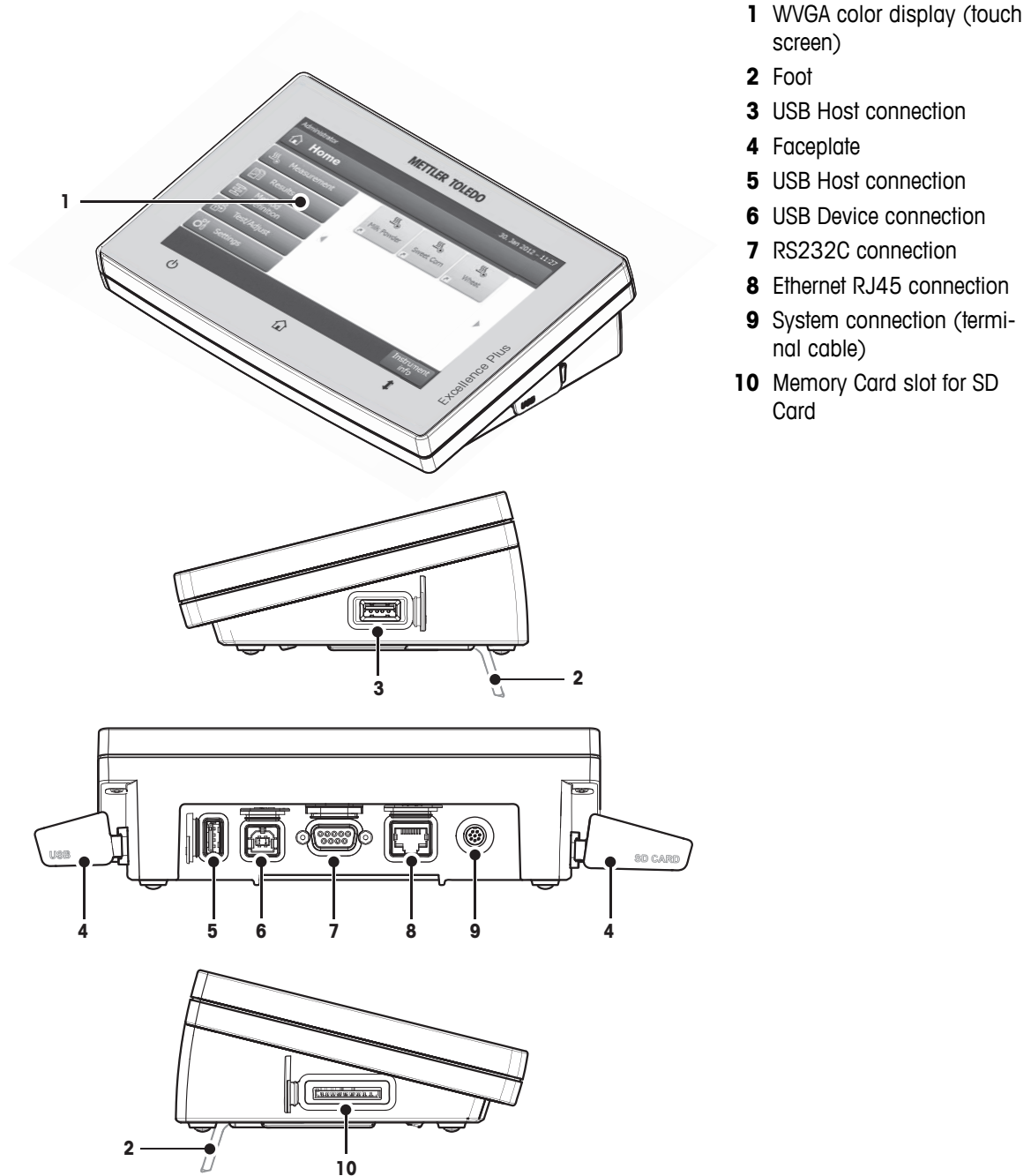
Connectivity

20	System connection	Interface for the communication between terminal and drying unit.
16	Power supply socket	Power supply inlet socket for the country-specific power cable. The socket has a built in fuse compartment containing a replacement fuse.

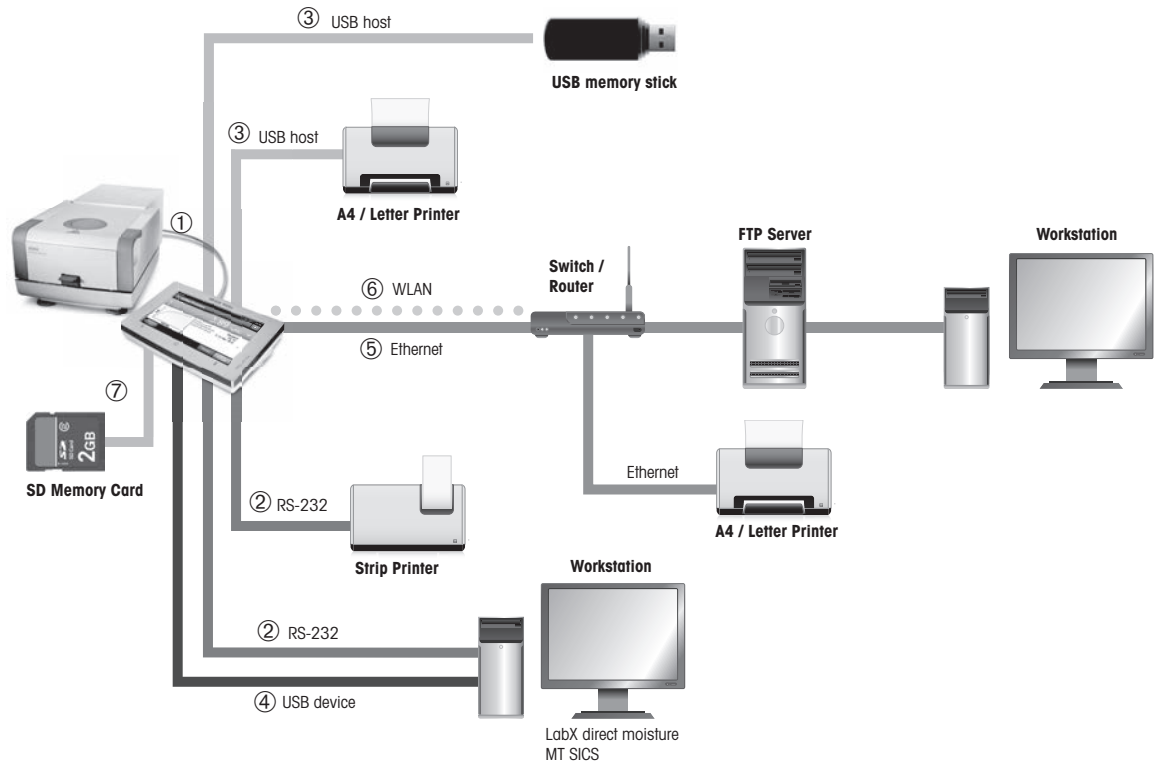
18	RHT sensor connection	Connection for the RHT sensor from METTLER TOLEDO for detecting the current room temperature and relative humidity for SmartCal™ application. Attention Do not use this port for other purposes!
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3.1.2 Terminal

Description of the connections at the terminal (Input and output unit with interfaces).

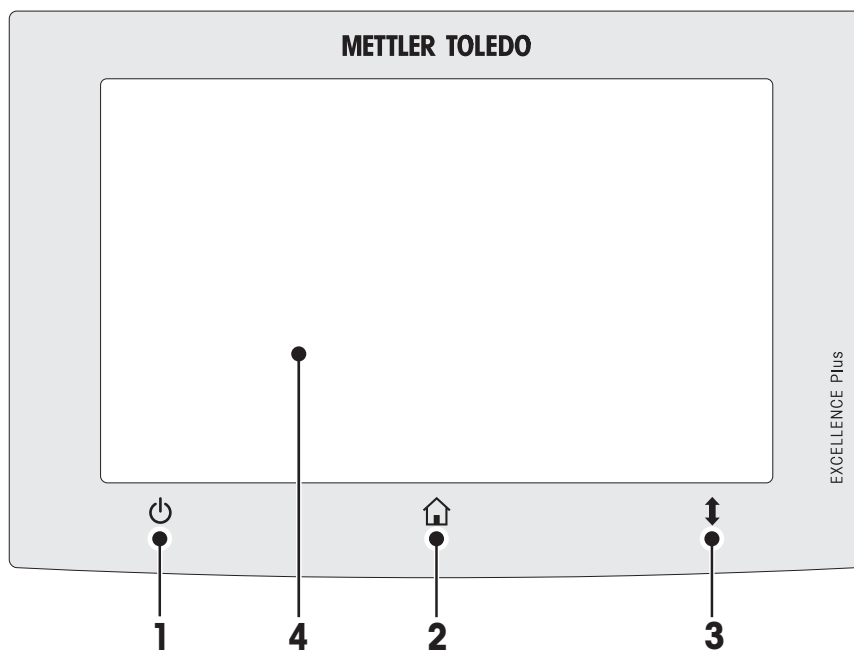


Connectivity



1	System connection	Interface for the communication between terminal and drying unit.
2	RS232C connection	RS232C interface for connecting the instrument to a peripheral device (e.g. printer or a PC with a 9-pin male connector).
3	USB Host connection	USB interface for connecting a peripheral device to the instrument (e.g. memory stick up to 32 GB). Type A Socket.
4	USB Device connection	USB interface for connecting the instrument to a peripheral device with an USB host connection (e.g. PC). Type-B socket.
5	Ethernet TCP/IP connection	Ethernet interface for connecting the instrument to a TCP/IP network. The most simple network can be implemented by connecting the instrument directly to the PC with a cross-over cable (RJ45 Socket). DHCP and fix IP addresses are supported.
6	WLAN connection	Wireless LAN interface for connecting the instrument to a wireless TCP/IP network. DHCP and fix IP addresses are supported. Note This function is not available in every country.
7	SD Memory Card slot	The memory card slot fits Secure Digital Memory Cards. SD and SDHC types as well as sizes up to 32 GB are supported. You can use it as an external storage device e.g. to store measured results. Note SDXC types are not supported.

3.1.3 Keys on the Terminal



		Name	Explanation
1		ON/OFF	To switch on or off (standby operation) the instrument. Note Do not disconnect the instrument from the power supply except if you will not be using the instrument for an extended period.
2		Home	To return from any menu level, or other window to the user home screen.
3		Open/Close	To open or close the sample chamber motorized. Attention Do not open the sample chamber manually. Only open manually in emergency situations. In all other cases use always the [↑] key. See Opening and Closing Sample Chamber (page 25)
4		Touch screen	

3.2 User Interface

The colored touch screen is a touch-sensitive WVGA monitor. The touch screen not only displays information, it also allows you to enter commands by touching certain areas on its surface: You can choose the information displayed on the screen, change terminal settings or perform certain operations on the instrument.

Only those elements which are available for the current dialog appear on the display.

Buttons on the Touch Screen

Buttons are software elements on the touch screen (Soft keys).



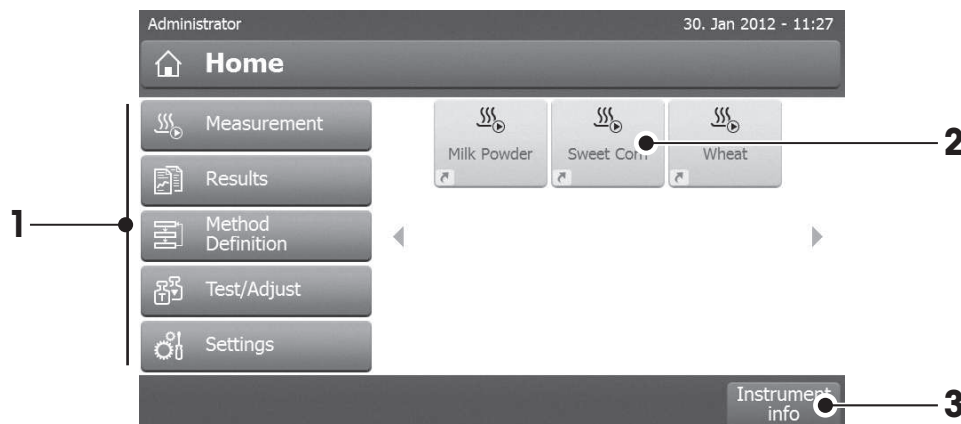
CAUTION

Do not touch the touch screen with pointed or sharp objects!

This may damage the touch screen.

3.2.1 User "Home" Screen

The user **Home** screen is the main screen and appears after startup or login of the instrument. All screens of the user interface can be accessed from the home screen. Returning to the **Home** screen is possible from all other screens of the user interface by pressing the [⏠] key or tapping the [Home] button.

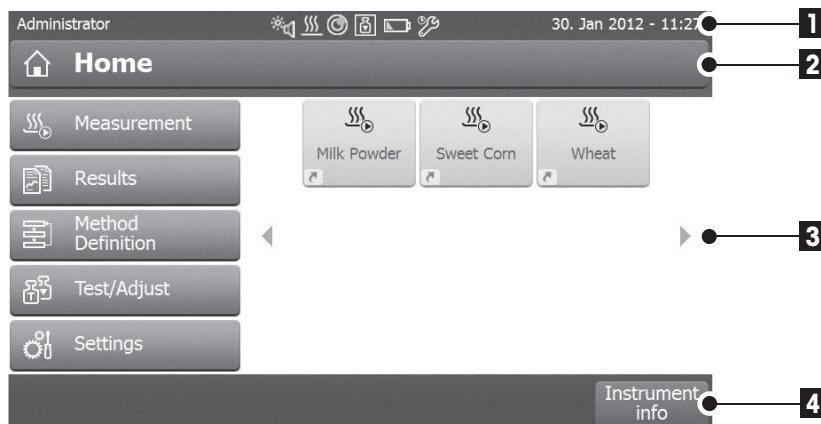


	Name	Explanation
1	Measurement	Start measurement Prerequisite: Method is defined and all necessary settings are made. For further information see My First Measurement (page 31) and Measurement (page 94).
	Results	Show results Under this menu item, all measurement results can be displayed, printed or exported. For further information see Results (page 99).
	Method Definition	Define, edit or delete a method All actions using in a drying method, can be defined under this menu item. Description of the menu item see Method Definition (page 73).
	Test/Adjust	Adjust or test the instrument Under this menu item, the integrated balance and the heating module can be adjusted or tested. Description of the menu item see Test/Adjust (page 62).
	Settings	Define general settings All settings of the system can be defined, e.g, system settings, user management or user preferences. These settings are usually made during installation of the instrument. Description of the menu item see Settings (page 34)
2		User shortcuts User-specific shortcuts for frequently used methods. Shortcuts are saved in the user profile. How to create shortcuts see Using Shortcuts (page 97).
3	Instrument info	Shows general info about the instrument like type, serial number, software version.

3.2.2 Basic Elements on Touch Screen

The graphical user interface consists of the following basic elements.

Screen Areas and their Elements



	Name	Explanation
1	Status Bar	The status bar contains the user name, status icons as well as date and time. See Status Message Icons (page 17).
2	Title Bar	The title bar contains elements for the user's orientation and information, as well as for special functions. See Title Bar Icons (page 17).
3	Content Area	The content area is the main working area of the menus and the applications, content depends on application or performed action. The content area also shows graphic illustrations such as the curve diagram of a data set (e.g. drying curve).
4	Action Bar	The action bar contains action buttons for triggering the actions which are required and available in the current dialog (e.g. Back , ->0/T<- , Print , Save , Delete , OK). See Action Bar Buttons (page 17).

3.2.3 Icon and Buttons

3.2.3.1 Status Message Icons

Status messages are displayed by means of small icons in the status bar. For the most important messages **see** Status Messages (page 113).

By tapping the symbols, the explanation of the symbols will appear in a separate window. By tapping the quick access icon , brightness or sound can be set.

3.2.3.2 Title Bar Icons

The title bar icons only appear when they are active. The icons which appear in the title bar, are listed with their function below:

	Explanation
	Create a shortcut and add it to your user home screen. Edit or remove an existing shortcut see Using Shortcuts (page 97).

3.2.3.3 Action Bar Buttons

The action bar contains action buttons for triggering any of the actions which are required and available in the current dialog (e.g. **Back**, **->0/T<-**, **Print**, **Save**, **Delete**, **OK**). There are up to six action buttons available for a work step. The functions of these buttons are self explanatory and therefore not discussed further.

3.2.4 Input Dialogs

3.2.4.1 Entering Text and Numbers

The keyboard dialog allows you to enter characters including letters, numbers and a variety of special characters. The country-specific layout of the keyboard is in accordance with the selected language, **see** Regional Settings (page 37).



	Name	Explanation
1	Input field	
2	Explanation field	The maximum number of characters allowed is indicated.
3	Shift	You can enter lower and upper case letters by toggling  . Switch the keyboard by tapping the corresponding button in the action bar, for entering numbers, special characters or letters

– Confirm with [OK] (To exit keyboard tap [Cancel]).

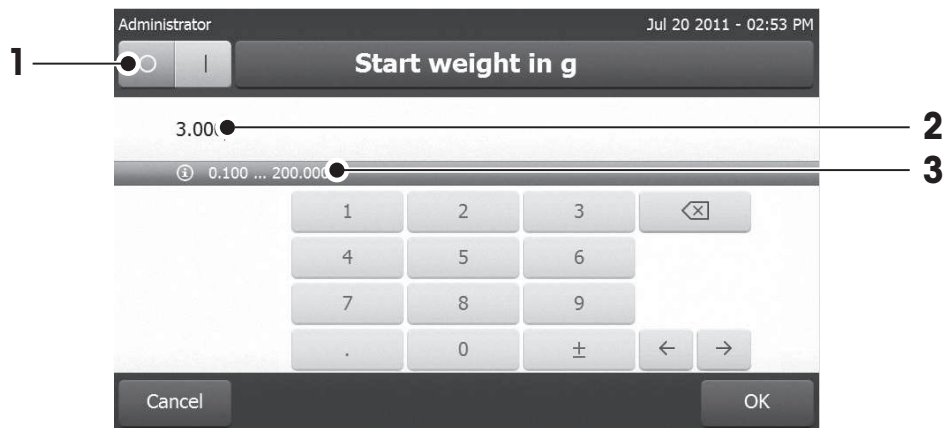
	Function
	Shift-Lock
	Delete last character
	Cursor left
	Cursor right

Note

It is also possible to place cursor directly in input field by tapping the respective position.

3.2.4.2 Entering Numerical Values

The numerical keypad dialog allows you to enter numerical values.



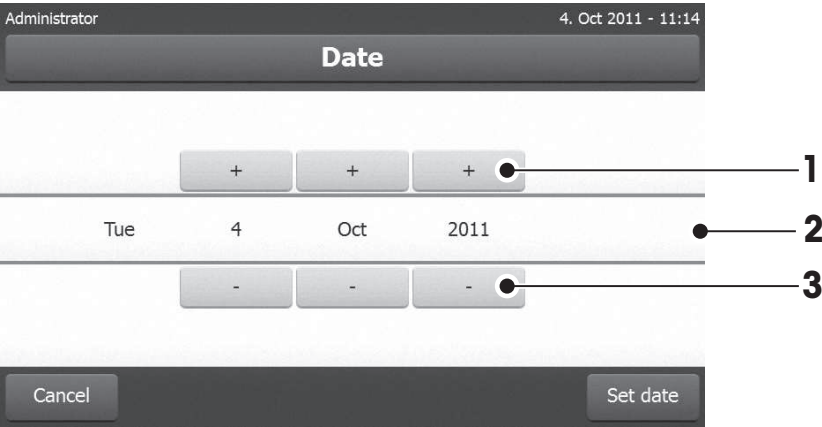
	Name	Explanation
1	Switch	If a switch appears, located on the left side in the title bar; it must be switched on for the entry of a value: 0 = OFF , 1 = ON
2	Input field	
3	Explanation field	The valid input range is indicated.

- Confirm with [**OK**].
To exit keypad, tap [**Cancel**].

	Function
	Delete last character
	Cursor left
	Cursor right

3.2.4.3 Changing Date and Time

This dialog (Picker view) allows you to set the date and time by picking from the built-in calendar/clock.



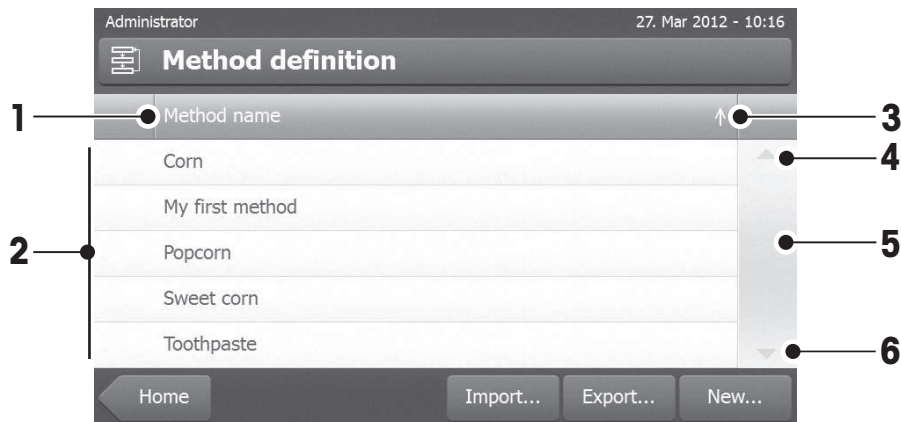
	Name	Explanation
1	Pick button	Pick button upwards (input control)
2	Explanation field	Current value
3	Pick button	Pick button downwards (input control)

- Confirm with [**Set date**]. (To exit without changing, tap [**Cancel**].)

3.2.5 Lists and Tables

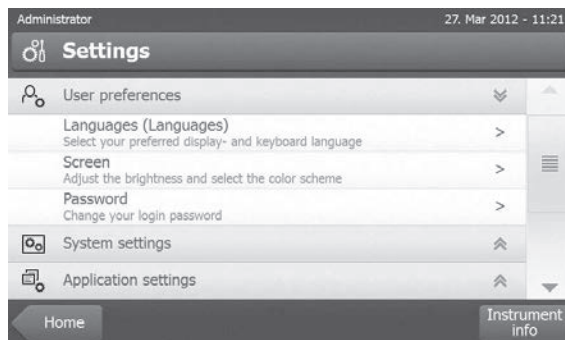
Simple list view

Basic elements of a simple list including a content title. If necessary, the content area can be scrolled vertically and can be arranged in a different order.

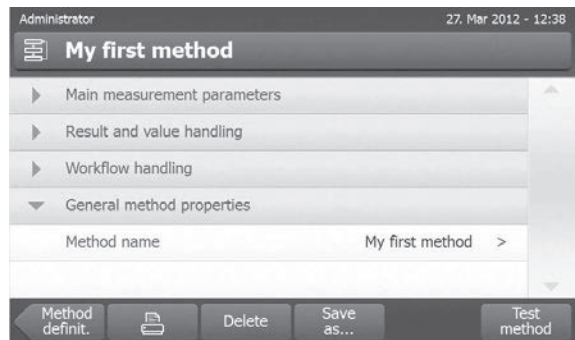


	Name	Explanation
1	Content title	
2	Content area	
3	Arrow icon	[↓] Showing the list in ascending order. [↑] Showing the list in descending order.
4	Scrollbar	Scrolling up
5		Scroller
6		Scrolling down

Further types of lists:



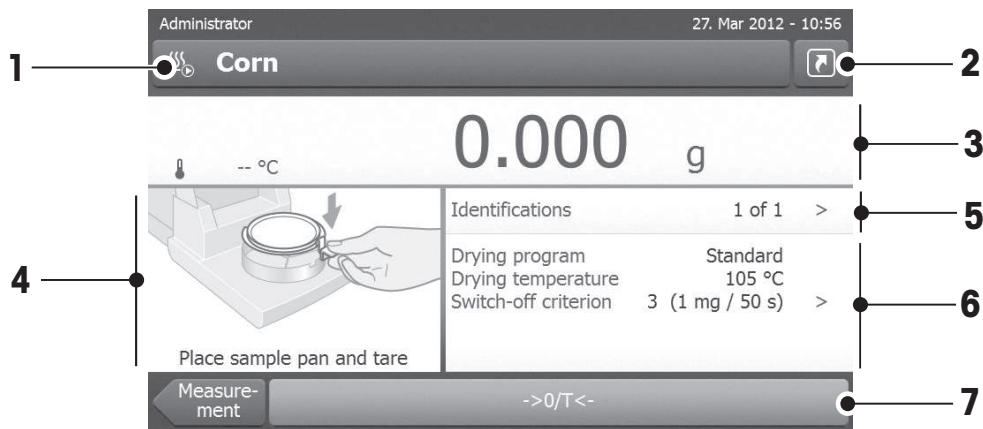
Accordion panels can be opened or closed by tapping [^].



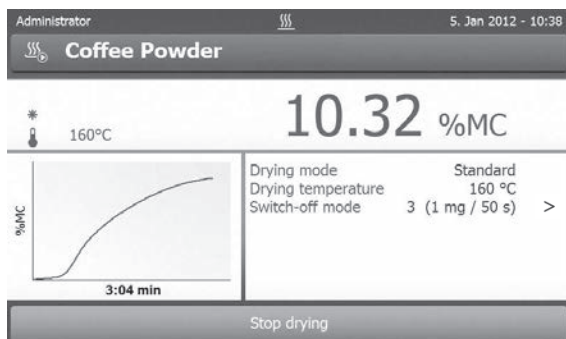
Closable panels can be opened or closed by tapping [>].

3.2.6 Work Screen

The work screen is used for performing main tasks e.g. measurements.

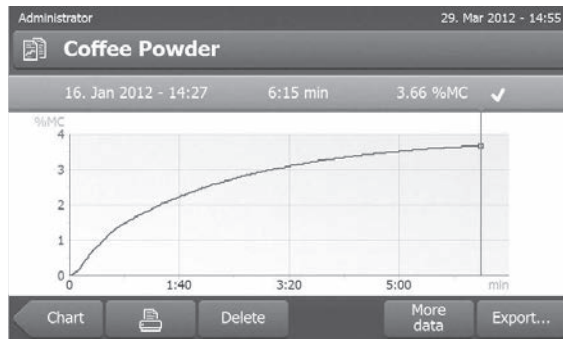


	Name	Explanation
1	Name field	Name of current Method
2	Shortcut button	Adding/editing a shortcut to the home screen for this method
3	Value panel	Displays the current measured values of the work process
4	Graphics panel	E.g. graphical illustrations of the drying curves, instructions for the user for performing tasks, and weighing-in aid
5	ID panel	Identifications (IDs) appear after tapping the ID panel for entering or editing values (comments). ID panel appears only if identification input is activated in the menu.
6	Parameter panel	Displays the parameters of the current work process. A detailed overview of the method parameters appears after tapping the parameter panel.
7	Action buttons	Depending on the current context



Measurement in progress

3.2.7 Graphics View



Graphical view of single measurement

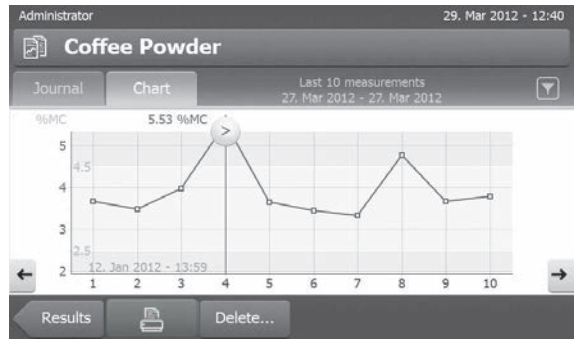


Chart view of series of measurements

For further information **see** Graphical Evaluations of Measurement Results (page 99)

4 Installation and Putting into Operation

In this section you will find information how to put your new instrument into operation.

4.1 Scope of Delivery

Open the package and remove drying unit, terminal and the accessories. Check the completeness of the delivery. The following accessories are part of the standard equipment of your new Moisture Analyzer:

- 80 aluminum sample pans
- 1 Sample handler
- 1 Sample pan holder
- 1 Draft shield
- 1 Specimen sample (circular, absorbent glass fibre filter)
- 1 RHT sensor
- 1 Power cable (country specific)
- 1 Terminal cable (Interface Terminal – drying unit)
- 1 SmartCal sampler
- 1 Operating instructions
- 1 Application brochure «Guide to Moisture Analysis»
- 1 Production certificate
- 1 Declaration of conformity

Remove the packaging from the instrument. Check the instrument for transport damage. Immediately inform your METTLER TOLEDO representative if you have any complaints or parts are missing.

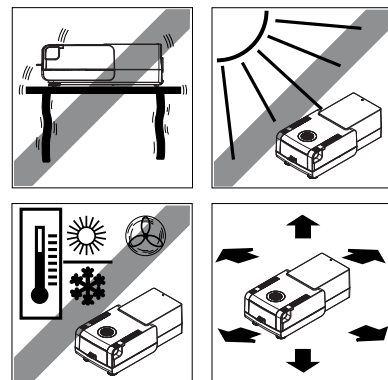
Note

Store all parts of the packaging. This packaging guarantees the best possible protection for the transport of your instrument.

4.2 Location

Your Moisture Analyzer is a precision instrument. An optimum location guarantees accuracy and dependability. Make sure that the following environmental conditions are met:

- Operate the instrument only indoors and at an altitude of less than 4000 m above sea level.
- Before switching on the instrument, allow all its parts to reach room temperature (+5 to 30°C).
Make sure that the relative humidity is between 20% and 80% and non-condensing conditions are met.
- The power plug must be easily accessible.
- Firm, horizontal location as free from vibrations as possible.
- Avoid direct sunlight.
- No excessive temperature fluctuations.
- No powerful drafts.
- Surroundings as free from dust as possible.
- Sufficient clearance around the instrument to allow warm air to dissipate.
- Sufficient distance from heat-sensitive materials in the vicinity of the instrument.





WARNING

Substances which contain toxic or caustic components

Toxic gases produced during drying could cause irritations (eyes, Skin, breathing), illness or death.

- Such substances may be dried only in a fume cupboard.
-

4.3 Opening and Closing Sample Chamber

Open and close the sample chamber exclusively by pressing the [↑] key.

Attention

Do not open the sample chamber manually.

Only manually opening the sample chamber in emergency situation:

- 1 Hold the handles on the heating module **with both hands** and pull the module firmly upwards to open the sample chamber.
- 2 Pull out the power plug.

Note

It is necessary to reboot the system, after an emergency opening.

4.4 Connecting Instrument



WARNING

Risk of electric shock

- a) Use only the 3-pin power cable with equipment grounding conductor which was supplied with your instrument.
 - b) Only a 3-pin grounded outlet for connecting your instrument must be used.
 - c) Only extension cables which meet this relevant standards and also have an equipment grounding conductor may be used.
 - d) Intentional disconnection of the equipment grounding conductor is prohibited.
-

Attention

- First, check whether the voltage printed on the type plate of the drying unit matches your local line voltage. If this is not the case, on no account connect the drying unit to the power, but contact your METTLER TOLEDO sales representative or dealer.

Two different versions of drying units with country-specific power cable are available (110 V AC or 230 V AC).

- The power plug must be always accessible.
- Before operating, check all cables for damage.
- Guide the cables so that they cannot become damaged or interfere with the measuring process.

Connecting the Terminal to the Drying Unit

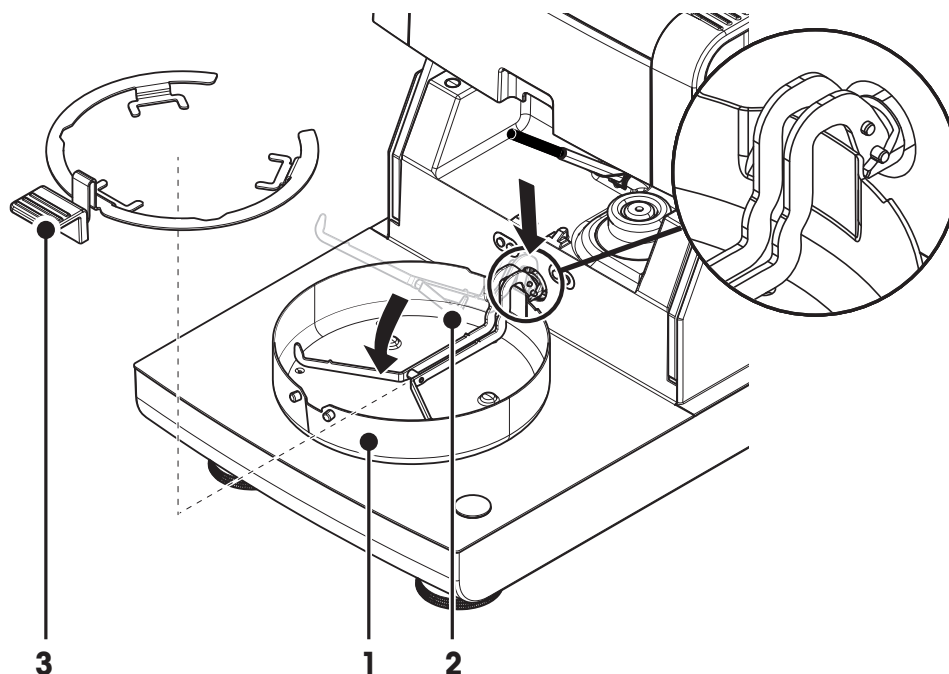
- Drying unit and terminal are at the final location.

- 1 Connect one end of the supplied terminal cable to the system connection socket on the terminal, **see** Terminal (page 12).
- 2 Screw the connector firmly.
- 3 Connect the other end to the system connection socket on the drying unit, **see** Drying Unit (page 11).
- 4 Screw the connector firmly.

Connecting the Drying Unit to the Power Supply

- ▶ Terminal – drying unit interface is connected.
- 1 Connect one end of the power cable to the power supply socket on the drying unit, **see** Drying Unit (page 11).
- 2 Connect the other end to the power line outlet.

4.5 Setting up Drying Unit



- ▶ Terminal and drying unit are connected.
- 1 Switch on the Instrument by pressing [ϕ].
- 2 Press [⌂].
 - ⇒ the motorized automatic sample chamber opens.
- 3 Place the draft shield element (1). Only one position is possible.
- 4 Insert the sample pan holder (2) carefully. Check that the sample pan holder is hooked correctly.
- 5 Insert the sample pan handler (3).

Attention

The drying unit is specially designed for motorized opening or closing of the sample chamber. Do not open the sample chamber manually. Only open manually in emergency situations. In all other cases use always the [⌂] key, **see** Keys on the Terminal (page 14).

Note

When putting into operation for the first time or if the instrument is disconnected from the power over a longer period, leave the instrument connected to the power supply for at least 5 hours to allow the built-in rechargeable battery to charge up! This battery ensures that the date and time are not lost when the instrument is disconnected from the power. This battery can not be replaced by the user. Please contact a METTLER TOLEDO representative.

4.6 Leveling Drying Unit

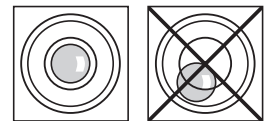
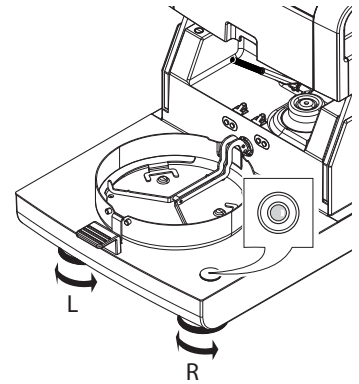
Exact horizontal positioning and stable installation are prerequisites for repeatable and accurate results. To compensate for small irregularities or inclinations ($\pm 2\%$) at the location, the instrument must be leveled.

For exact horizontal positioning, the drying unit has a level indicator (level) and 2 leveling screws. When the air bubble in the level indicator is exactly in the center, the instrument is standing perfectly horizontally. To level it, proceed as follows:

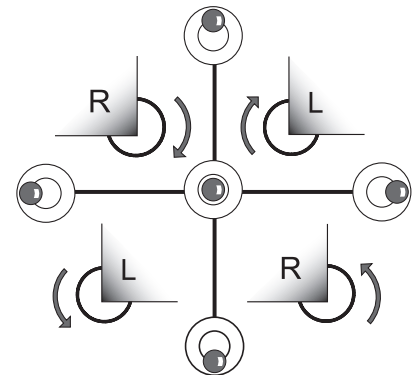
- 1 Position your Moisture Analyzer at the selected location.
- 2 Turn the two leveling feet until the air bubble is in the center of the level indicator.

L = left foot

R = right foot



Air bubble at	"12 o'clock"	turn both feet clockwise
Air bubble at	"3 o'clock"	turn left foot clockwise, right foot counterclockwise
Air bubble at	"6 o'clock"	turn both feet counterclockwise
Air bubble at	"9 o'clock"	turn left foot counterclockwise, right foot clockwise



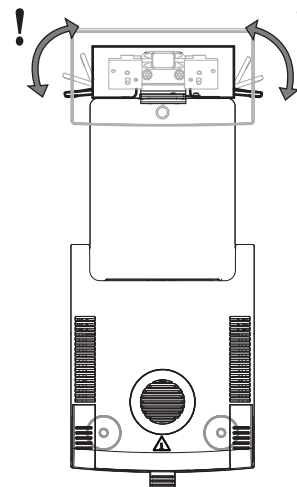
Note

The drying unit should be re-leveled each time its location is changed.

Leveling the drying unit with optional stand for terminal mounted.

Stand for terminal **see** Accessories and Spare Parts (page 122).

- ▶ The drying unit is positioned at the selected location.
- 1 To loose the clamps for the safety feet of the stand, turn them outwards.
 - 2 Level the drying unit as previously described.
 - 3 To lock the clamps for the safety feet, turn them inwards as far as they will go.



Level sensor

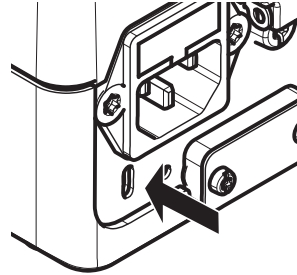
The built-in level sensor constantly monitors the correct horizontal alignment of the drying unit. When the level is incorrect, the status icon  appears in the display, **see** Status Message Icons (page 17).

Note

The level sensor cannot be switched off.

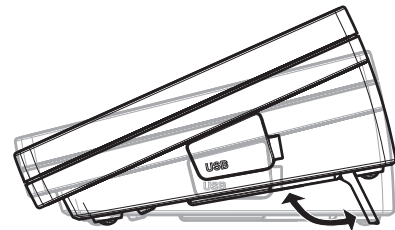
4.7 Anti-theft Device

For protection against theft, your drying unit is fitted with a Kensington security slot for an anti-theft device. To find a suitable anti-theft cable **see** Accessories and Spare Parts (page 122).



4.8 Setting Reading Angle of Terminal

- To change the reading angle, fold out the two tilt feet.



4.9 Setting Date, Time and Country

Navigation: Home > Settings > System settings > Regional Settings

When you put your new instrument into operation for the first time, you should enter the current date and time as well as the country in which the instrument is operated. These settings are retained even if you disconnect your instrument from the power supply.

Note

Various languages are available for the dialog with your instrument and can be selected in the menu:

Navigation: Settings > User preferences, see User Preferences (page 35).

Set the current date

- ▶ **Regional Settings** is selected.

- 1 Tap **Date**.
- 2 Set day, month and year.
- 3 Confirm with [**Set date**].

Set the current time

- ▶ **Regional Settings** is selected.

- 1 Tap **Time**.
- 2 Set Hours and minutes.
- 3 Confirm with [**Set time**].

Set the country in which the instrument is operated.

► **Regional Settings** is selected.

- 1 Tap **Country**.
- 2 Set the country (region).
- 3 Confirm with [**Save**].

Note

For changing the format for date and time, **see** Regional Settings (page 37).

4.10 Adjustment

To obtain accurate measuring results, it is necessary to adjust the integrated balance as well as the heating module.

Adjusting is necessary:

- before the instrument is used for the first time.
- at regular intervals.
- after a change of location.

The following adjustment options are possible:

- Fully automatic balance adjustment FACT.
- Balance adjustment with internal weight.
- Balance adjustment with external weight (accessory).
- Temperature adjustment with temperature adjustment kit (accessory).

For more information to perform these operations **see** Test/Adjust (page 62).

4.11 Install METTLER TOLEDO RHT sensor

The RHT sensor from METTLER TOLEDO allows the automatic detection of relative humidity and temperature values for SmartCal tests, see SmartCal test (page 69).

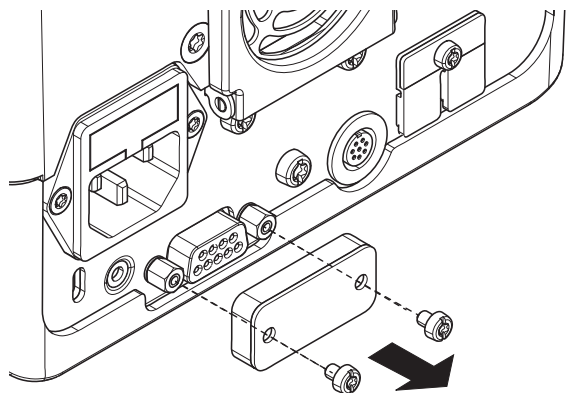
Note

Not delivered with every model.

Installation on the Drying Unit

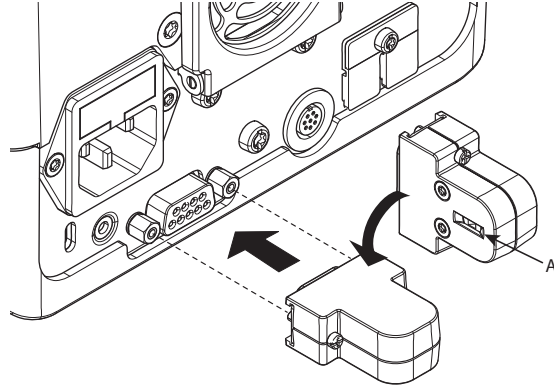
1

- 1 Loosen the two screws.
- 2 Remove screws and cover.
- 3 Retain these for later use.



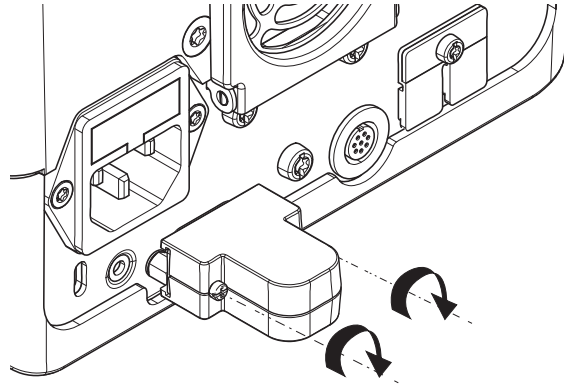
2

- 1 Take the sensor out of its packing.
- 2 Retain the packing for later use.
- 3 Connect the RHT sensor to the drying unit with sensor opening (A) facing down.



3

- Tighten the two screws to secure the RHT sensor.



Attention

Please observe the following instructions to avoid wrong SmartCal measurements.

- Do not install the RHT sensor on the terminal.
- Be sure that the sensor opening is clean.
- Do not touch the sensor opening with sharply pointed objects.
- After installing, RHT sensor must be acclimatized in working area for about one hour, before using.
- Exposure to acids, bases or volatile compounds (e.g. solvents) at high concentration or long exposure time must be avoided.
- The RHT sensor must be stored in the original, sealed anti-static bag.
- For storage temperature shall be in the range of 10°C...50°C and humidity at 20...60%rH.
- **We recommend to remove the RHT sensor from the drying unit after a SmartCal measurement and to store it in the original, sealed anti-static and bag.**
- Settings **see** Equipment (page 48).

5 My First Measurement

After you have successfully put your new Moisture Analyzer into operation for the first time, you can immediately perform your first measurement. In doing so, you will become familiar with the instrument.

Use the supplied specimen sample (absorbent glass fiber filter) for your first measurement to determine the moisture content. During your first measurement the instrument operates with the factory settings.

Switching on

- ▶ The instrument is connected to the mains.
- To switch on, press [ϕ].



Create a method

- 1 Tap **Method Definition**.
 - ⇒ **Method Definition** menu appears.
- 2 To define a new method, tap [**New...**].
 - ⇒ The keypad appears.
- 3 Enter a name for your first method e.g. **My first method**.
- 4 Confirm with [**OK**].
- 5 To store your new method with factory settings, tap [**Save**].
- 6 Tap [**Home**].
 - ⇒ User home screen appears.

Select the method for measuring

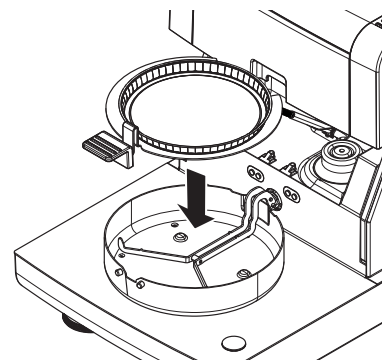
- 1 Tap [**Measurement**].
 - ⇒ The methods list is displayed.
- 2 Tap **My first method**.
 - ⇒ The work screen of the method **My first method** appears.
 - ⇒ The sample chamber opens automatically.

Placing sample pan

- ▶ The display now prompts you to load the empty sample pan and tare the balance.
- 1 Place the empty sample pan in the sample pan handler.
 - 2 Place the sample pan handler in the sample chamber. Ensure that the tongue of the pan handler fits exactly in the slot of the draft shield element. The sample pan must lie flat in the pan holder.

Note

We advise you to work with the sample pan handler at all times. The pan handler is ergonomic, automatic positioning, safe and provides protection against possible burns due to the hot sample pan.



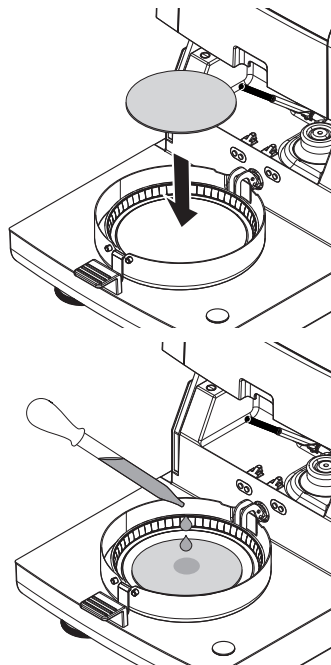
Taring balance

- Tap [**->0/T<-**].
 - ⇒ The sample chamber closes automatically for taring.
 - ⇒ When taring is completed, the sample chamber opens automatically.



Start measuring

- ▶ Following taring, the display prompts you to add the sample to the sample pan.
 - 1 Place the specimen sample (glass fibre filter) in the sample pan.
 - 2 Wet the specimen sample with a few drops of water.
 - 3 Tap [**Start drying**]
 - ⇒ The sample chamber closes automatically.
 - ⇒ The drying process starts automatically.



Drying process

You can follow the measurement process on the display, **see** Work Screen (page 22).

- The drying process is continuously displayed graphically.
- The current temperature in the heating module is displayed as well as the elapsed drying time and the current drying value.
- The display shows the selected settings.
- The drying process can be aborted by tapping [**Stop drying**].

At the end of the drying process you can read the moisture content of your sample on the display. If you have defined the control limits in the method, passed, warned or failed status is shown.

Remove Sample



CAUTION

Danger of burns

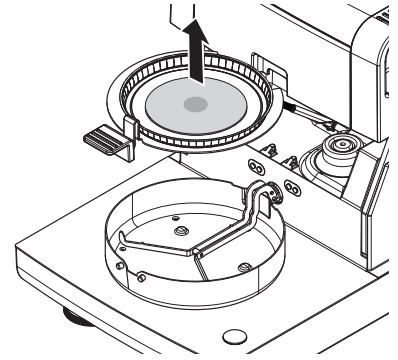
Sample, sample pan and sample pan holder may still be hot.

- ▶ The drying process is completed.
 - ▶ The sample chamber is open (opens automatically after the measurement is completed).
- 1 Carefully remove the sample pan handler from the sample chamber.

Note

To remove the sample pan from the handler, lift the pan slightly from below and remove it from the handler.

- 2
 - To perform a further measurement with the current method, tap [**Next Sample**].
 - To perform a measurement with a new method, tap [**Measurement**].
 - To return to the home screen, press [].



Congratulations!

You have just performed your first measurement with your new Moisture Analyzer. In the following sections you will find more information about the extensive functions of this instrument.

6 Settings

Navigation: Home > Settings

In this section you will find information how to define the general settings of your instrument.

Menu structure

Main menu	Submenu	Further information
User preferences	Languages	see Languages (page 35)
	Screen	see Screen (page 35)
	Sound	see Sound (page 36)
	Password	see Password (page 36)
System settings	Regional Settings	see Regional Settings (page 37)
	Peripherals	see Peripherals (page 37)
	Network interfaces	see Network Interfaces (page 41)
	Default settings	see Default Settings (page 39)
	Company information	see Company Information (page 40)
	Touch screen adjustment	see Touch Screen Adjustment (page 40)
Application settings	Identifications management	see Identifications Management (page 44)
	Print and export management	see Print and Export Management (page 45)
Quality management	Test / Adjustment settings	see Test / Adjustment settings (page 48)
User management	Groups	see Groups (page 53)
	Users	see Users (page 55)
	Account policies	see Account Policies (page 56)
System and data management	Result management	see Result Management (page 57)
	Export / Import	see Export / Import (page 57)
	Back up / Restore	see Back up / Restore (page 59)
	Export event logs	see Export event logs (page 59)
	Reset	see Reset (page 60)
	Update	see Update (page 60)

6.1 User Preferences

Navigation: Home > Settings > User preferences

This menu item defines the basic settings for each user. This settings are stored under the active user profile and apply when working within this profile. When a user profile is called up (login user account), the relevant settings are loaded automatically. **See** User Management (page 53).

You can define the following parameters:

Menu item	Explanation	Further Information
Languages	Defines your preferred display and keyboard language.	see Languages (page 35)
Screen	Defines the brightness and the color scheme of the display	see Screen (page 35)
Sound	Defines audible feedback	see Sound (page 36)
Password	Defines user password	see Password (page 36)

6.1.1 Languages

Navigation: Home > Settings > User preferences > Languages

Select your preferred language for operating the terminal.

Factory setting: Generally, the language of the destination country (if available) or English is set.

You can define the following parameters:

Parameter	Explanation	Values
Display language	Defines user-specific screen language.	English Deutsch Italiano Français Español Português Polski Magyar Český Русский 日本語 中文 한국어/조선말
Keyboard language	Defines user-specific layout of the Keypad.	English Deutsch Italiano Français Español Português Polski Magyar Český

6.1.2 Screen

Navigation: Home > Settings > User preferences > Screen

You can define the following parameters:

Parameter	Explanation	Values
Brightness	Defines user-specific brightness of the display.	10...100 % (80 %)*
Color scheme	Defines user-specific display color scheme.	Blue* Green Orange Grey

* Factory setting

6.1.3 Sound

Navigation: Home > Settings > User preferences > Sound

You can define the following parameters:

Parameter	Explanation	Values
Sound	Activate / deactivate of audible feedback and adjust the Volume . Audible feedback: • when tapping a button. • upon measurement completed. • before automatic closing of the sample chamber. • when an error message appears.	OFF* ON 0...100 % (50 %)*

* Factory setting

6.1.4 Password

Navigation: Home > Settings > User preferences > Password

In this menu item you can change the user password. It is also possible to create a password for protection or to deactivate the password protection.

You can define the following parameters:

Parameter	Explanation	Values
Password	Change an existing (Old password) and create a New password . ON = for entry of a new login password. OFF = no password protection.	ON OFF any

6.2 System Settings

Navigation: Home > Settings > System settings

This menu item defines the basic settings for the system. The system settings apply to the entire system and therefore to all user profiles and applications.

You can define the following parameters:

Menu item	Explanation	Further Information
Regional settings	Defines the settings of date & time, keypad and country options.	see Regional Settings (page 37)
Peripherals	Defines the configuration of printers and other devices.	see Peripherals (page 37)
Network interfaces	Defines network connections.	see Network Interfaces (page 41)
Default settings	Defines the default settings of your instrument.	see Default Settings (page 39)
Company information	Defines company specific information.	see Company Information (page 40)
Touch screen adjustment	Adjusts the touch screen.	see Touch Screen Adjustment (page 40)

6.2.1 Regional Settings

Navigation: Home > Settings > System settings > Regional Settings

This menu item defines the settings of date & time.

You can define the following parameters:

Parameter	Explanation	Values
Date	Sets the date of the clock of the instrument.	Day Month Year
Time	Sets the clock time of the instrument.	Hours Minutes
Date format	Defines the date format. D = Day M = Month Y = Year	D. MMM YYYY* MMM D YYYY DD.MM.YYYY MM/DD/YYYY YYYY/MM/DD YYYY-MM-DD
Time format	Defines the time format. M = Minutes	24:MM* 12:MM 24.MM 12.MM
Country	Defines the country in which the instrument is operated.	Europe China Japan US Other *

* Factory setting

Note

Date and time of a connected printer will be ignored. Date and time of instrument is relevant.

6.2.2 Peripherals

Navigation: Home > Settings > System settings > Peripherals

In this menu item you can configure one printer and one host connection.

PCs

The **MT-SICS** host connection (remote client) is enabled by default and only usable with the METTLER TOLEDO specific protocol MT-SICS.

MT-SICS commands and functions can be transmitted via the RS232 interface (COM1 port) or the USB device interface (COM4 port). The use of the USB device interface requires the installation of an USB device driver on the PC, **see** Installing USB Device Driver (page 129).

Commands and further information please refer to the Reference Manual **MT-SICS Interface Commands for Halogen Moisture Analyzers** downloadable from the internet.

► www.mt.com/hxhs

Printers / PCs

No printer is installed by default. To install a printer:

- The desired printer is properly connected to the terminal.

Note

Do not attempt to connect a printer to the drying unit!

- 1 Tap [**New...**]

⇒ **Peripheral type** list appears.

- 2 Select device.

Note

Currently supported devices from METTLER TOLEDO with preset parameters:

- **RS-P42 Strip Printer**, see RS232 parameters
- **RS-P25 Strip Printer**, see RS232 parameters
- **Local PCL printer**, see USB parameters
- **Network PCL printer**, see network parameters
- **LabX Direct Moisture**, see RS232 parameters

- 3 Set the parameters.

- 4 To store the settings, tap [**Save**].

- 5 To activate the interface, tap [**ON**].

⇒ Status changes from **Disabled** to **Enabled**.

Note

- A printer or host connection can also be deleted.
- For strip printer settings **see** Recommended Printer Settings (Strip printer) (page 128).
- For defining the printouts **see** Print and Export Management (page 45).
- Only one printer can be installed. To install another printer, you have to delete the old printer first.

RS232 parameters

You can define the following parameters:

Parameter	Explanation	Values
Device Name	Defines a name for the device.	any
Device Port	Defines the device port. COM1 : RS232 / COM4 : USB Device (Driver required) Note It is possible to use an USB-to-Serial adapter. If such an adapter is connected, these devices are shown as e.g. SOU1 , SOU2 (SOU stands for serial over USB)	COM1* COM 4
Baudrate	Defines the speed of the data transmission (data transfer rate / Baud rate).	1200 2400 4800 9600* 19200 38400 57600 115200
DataBits	Defines the number of data bits.	7 8*
Parity	Defines the variant of parity bit.	None* Odd Even
StopBits	Defines the stop bits of the transmitted data.	1 StopBit* 2 StopBits

Flow Control	Defines the type of flow control (Handshake).	None Xon/Xoff*
---------------------	---	-------------------------

* Factory setting

USB parameters

- Supported A4 Printer: OKI B411 USB
- Supported printer command language: PCL 3 or higher

You can define the following parameters:

Parameter	Explanation	Values
Device Name	Defines a name for the device.	any

Network parameters

You can define the following parameters:

Parameter	Explanation	Values
Device Name	Defines a name for the device.	any
IP address	Defines the IP address of the printer. Default address*: 192.168.000.004	Default address* any
Port	Defines the printer's port. Note Usually the default value is used.	9100 * any

* Factory setting

6.2.3 Default Settings

Navigation: Home > Settings > System settings > Default settings

Define the default settings of your instrument.

You can define the following parameters:

Menu item	Explanation	Further Information
Default languages	Defines a default display and keyboard language.	see Default Languages (page 39)
Default screen settings	Defines the default brightness and color scheme.	see Default Screen Settings (page 40)

6.2.3.1 Default Languages

Navigation: Home > Settings > System settings > Default settings > Default languages

Select the default language for operating the terminal. Factory setting: Generally, the language of the destination country (if available) or English is set.

You can define the following parameters:

Parameter	Explanation	Values
Default display language	Defines default screen language.	English Deutsch Italiano Français Español Português Polski Magyar Český Русский 日本語 中文 한국어/조선말
Default keyboard language	Defines default layout of the keyboard.	English Deutsch Italiano Français Español Português Polski Magyar Český

6.2.3.2 Default Screen Settings

Navigation: Home > Settings > System settings > Default settings > Default screen settings

You can define the following parameters:

Parameter	Explanation	Values
Brightness	Defines default brightness of the display.	10...100 % (80 %)*
Color scheme	Defines default display color scheme.	Blue* Green Orange Grey

* Factory setting

6.2.4 Company Information

Navigation: Home > Settings > System settings > Company information

In this menu item you can define optional specific company informations for using on printouts and reports.

You can define the following parameters:

Parameter	Explanation	Values
Company name	Entry of the company name.	any
Department	Entry of the company department.	any
Instrument ID	Entry of the company specific instrument identification	any
Instrument location	Entry of the location of the instrument.	any
Instrument responsible	Entry of the name of the responsible person for the instrument.	any

6.2.5 Touch Screen Adjustment

Navigation: Home > Settings > System settings > Touch screen adjustment

If the alignment of the touch-sensitive areas of the screen correspond not exactly with the position of the buttons e.g. after a software update was performed, then you can solve this error by executing this function.



CAUTION

Do not touch the touch screen with pointed or sharp objects!

This may damage the touch screen.

► Menu **System settings** appears.

1 Choose **Touch screen adjustment** and tap [**Start**].

⇒ Adjustment screen appears.

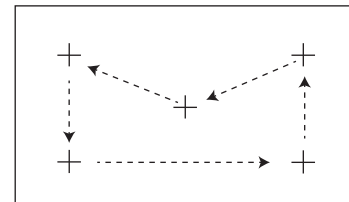
2 Touch and hold a touch screen stylus (resistive) on the center of the target until it moves to the next position on the screen.

3 Repeat this procedure until a message appears (**New calibration settings have been measured**).

4 Tap the screen to confirm or wait for 30 seconds for canceling (timeout).

⇒ A message box appears.

5 To store the setting, tap [**OK**].



Note

- Tap the center of the target as accurately as possible otherwise the process has to be repeated several times.
- Make sure not to touch any other locations on the screen and the fix key areas while doing this adjustment.
- Make sure not to touch the screen with your hand.
- Keep accurate angle for your usage.
- The adjustment process cannot be aborted.

6.2.6 Network Interfaces

Navigation: Home > Settings > System settings > Network interfaces

In this section you will find information how to define network connections and other terminal interfaces.

Menu structure

Main menu	Submenu	Further information
Host name	Defines the host name of the instrument	see Ethernet (page 41)
Ethernet	Activates and sets Ethernet connection.	see Ethernet (page 41)
WLAN	Activates and sets wireless local area network (WLAN) connection.	see WLAN (page 42)

Host Name

Navigation: Home > Settings > System settings > Network interfaces > Host name

The Host name is the name of your instrument and is a unique identifier on a network (Ethernet, WLAN). If using multiple devices, you should ensure that the host name of each instrument is unique. If necessary, contact the network administrator of your IT department or your IT support.

You can define the following parameters:

Parameter	Explanation	Values
Host name	Defines the host name. The host name must be unique on a network. Default host name*: Model name and serial number of the instrument.	any

* Factory setting

Ethernet

Navigation: Home > Settings > System settings > Network interfaces > Ethernet

Ethernet interface for connecting the instrument to a TCP/IP network. The most simple network can be implemented by connecting the instrument directly to the PC with a cross-over cable (RJ45 Socket). DHCP and fix IP addresses are supported.

Note

- We recommend to use the default setting DHCP (automatic network settings).
- Ensure that the host name of the instrument is unique, see Host Name (page 41).
- For installing and configuring an Ethernet interface you need to have basic knowledge of TCP/IP-based networks and of network technology in general. If necessary, contact the network administrator of your IT department or your IT support.

You can define the following parameters:

Parameter	Explanation	Values
Configuration	Defines whether the settings are made automatically (dynamic, DHCP) or manually (static).	DHCP assignment Manual assignment

DHCP assignment	Dynamic Host Configuration Protocol to assign all necessary network parameters on this interface automatically (Dynamic assignment).	none
Manual assignment	Manually assign all the necessary network parameters on this interface. (Static assignment)	IP address Subnet
IP address	Defines the IP address of this instrument. Only setable when DHCP is disabled. Default address*: 192.168.002.100 The IP address unique numeric identifier for a device which participates on a network. If DHCP is enabled, the automatically assigned IP address is displayed.	Default address* any
Subnet	Defines the subnet mask within the ethernet interface is addressable. Only setable when DHCP is disabled. Default subnet mask*: 255.255.255.000	Default value* any

* Factory setting

Note

The addresses are written as four dot-separated decimal integers in the range from 0 to 255. Leading zeros need not be written.

WLAN

Navigation: Home > Settings > System settings > Network interfaces > WLAN

Wireless LAN interface for connecting the instrument to a wireless TCP/IP network. DHCP and fix IP addresses are supported.

Note

WLAN is not available in every country.

The following **WLAN** authentications are supported:

- Open
- WEP
- WPA-PSK
- WPA2-PSK (tkip only)

You can define the following parameters:

Parameter	Explanation	Values
WLAN	Activates or deactivates WLAN radio transmitter.	OFF* ON

* Factory setting

Note

After activating WLAN, it takes some time until WLAN is active.

WLAN Network Settings

Navigation: Home > Settings > System settings > Network interfaces > WLAN > Settings > WLAN configurations

- We recommend to use the default setting DHCP (automatic network settings).
- Ensure that the host name of the instrument is unique, see Host Name (page 41).
- For installing and configuring an Ethernet interface you need to have basic knowledge of TCP/IP-based networks and of network technology in general. If necessary, contact the network administrator of your IT department or your IT support.

You can define the following parameters:

Parameter	Explanation	Values
Configuration	Defines whether the settings are made automatically (dynamic, DHCP) or manually (static).	DHCP assignment Manual assignment
DHCP assignment	Dynamic Host Configuration Protocol to assign all necessary network parameters on this interface automatically (Dynamic assignment).	none
Manual assignment	Manually assign all the necessary network parameters on this interface. (Static assignment)	IP address Subnet
IP address	Defines the IP address of this instrument. Only setable when DHCP is disabled. Default address*: 192.168.002.100 The IP address unique numeric identifier for a device which participates on a network. If DHCP is enabled, the automatically assigned IP address is displayed.	Default address* any
Subnet	Defines the subnet mask within the ethernet interface is addressable. Only setable when DHCP is disabled. Default subnet mask*: 255.255.255.000	Default value* any

* Factory setting

Note

The addresses are written as four dot-separated decimal integers in the range from 0 to 255. Leading zeros need not be written.

Networks

Receivable networks are displayed.

Entries for WLAN:

Parameter	Explanation	Values
WLAN name (SSID)	Displays the name of the selected WLAN (SSID, service set identifier)	not editable
Password	Entry for the required password.	any
Autoconnect	Activate or deactivate automatic connecting to the preferred network upon startup of the device. Note Only for one network possible.	☐ (deactivated)* ☒ (activated)

Establish a connection:

- 1 Touch the desired network.

Note

If the device is connected to a network, the WiFi icon indicates the signal strength. The more bars displayed, the stronger the connection.

- 2 Enter your password.
- 3 Enable automatic connection if needed.
- 4 Touch [connect] to establish the connection.

6.3 Applications Settings

Navigation: Home > Settings > Application settings

You can define the following parameters:

Menu item	Explanation	Further Information
Identifications management	Defines identifications and their use in order to comment measurements.	see Identifications Management (page 44).
Print and export management	Defines the default settings for the printouts.	see Print and Export Management (page 45).

6.3.1 Identifications Management

Navigation: Home > Settings > Application settings > Identifications management

Identifications (ID) contain descriptive text for individual measurements, which enables samples to be easily allocated to specific tasks or customers. This feature allows you to define identifications in order to comment measurements, such as company id, id badge, id sample. In the ID panel on the work screen, the values (comments) can be entered or changed before measurement, **see** Work Screen (page 22). IDs have an impact on all measurements, regardless of methods. The IDs are included on printouts or can be exported to a PC.

Menu structure

Main menu	Submenu	Further information
Identifications usage	Identifications input	see below
	Clear input after measurement	
Identifications	no submenu	

You can define the following parameters:

Parameter	Explanation	Values
Identifications input	Defines the method for entering comments. ON = activates ID panel on work screen. Free input = the IDs appear after tapping on the ID panel (work screen) for entries or changes of values (comments). Guided input = each ID is displayed individually in order of definition (ID1 to ID10) before measurement, to make entries or changes of values (comments).	OFF* ON Free input Guided input
Clear input after measurement	Defines the ID whose value (comment) is deleted after measurement.	OFF* ON ID selection: ☐ ☒
Identifications	Defines general identifications in order to comment measurements. + = New identifications (max. 10 IDs, 1...20 characters) - = Remove identifications (selectable)	- +* any

* Factory setting

6.3.2 Print and Export Management

Navigation: Home > Settings > Application settings > Print and export management

You can define the following parameters:

Menu item	Explanation	Further Information
Print management (Strip)	Defines the settings for printouts on strip printers.	see Print management (page 45).
Print management (A4)	Defines the settings for ISO A4 printouts on PCL printers.	see Print management (page 45).

Print management

Print management (Strip)

Navigation: Home > Settings > Application settings > Print and export management > Print management (Strip)

If a strip printer is connected and activated, results can be automatically printed out. The interim values are printed out in the pre-selected intervals and the final results are logged at the completion of the measurement. The printout contains information to the instrument, time, date as well as the method and results, depending of the protocol. The level of detail depends on the selected type of printout (short, standard) and the activated protocol items (settings or method), **see** Information on Printouts (page 101).

Note

- The definitions in this menu item are the default settings for the printouts. These parameters are displayed at every initialization of a Printout and can be modified again.
- The setting for printout language in this menu item has no impact on test or adjustment printouts, **see** Print and Output Settings (page 51).
- Not all languages are compatible with some printers, and also depending on the settings of the printer, **see** Recommended Printer Settings (Strip printer) (page 128).
- For installing and defining a printer **see** Peripherals (page 37).

You can define the following parameters:

Parameter	Explanation	Values
Printout type	Selects a detailed printout or in a short form.	Standard* Short
Printout language	Defines the language for general printouts. Note No impact on test or adjustment printouts.	English Deutsch Italiano Français Español Português Polski Magyar Český
Print results automatically after measurement	Print results automatically after measurement if activated.	☐ (deactivated)* ☒ (activated)
Include intermediate results to printout	Defines the time intervals at which interim results are transferred to the printer.	OFF* 10 s 30 s 1 min 2 min 10 min

* Factory setting

Print management A4

Navigation: Home > Settings > Application settings > Print and export management > Print management (A4)

If a PCL printer is connected and activated, results can be manually or automatically printed out on ISO A4 paper size. The level of detail depends on the selected export file, **see** Information on Exports (page 105).

Note

- The definitions in this menu item are the default settings for the printouts. These parameters are displayed at every initialization of a Printout and can be modified again.
- The setting for printout language in this menu item has no impact on test or adjustment printouts, **see** Print and Output Settings (page 51).
- For installing and defining an A4 printer, **see** Peripherals (page 37).

You can define the following parameters:

Parameter	Explanation	Values
Printout language	Defines the language for general printouts. Note No impact on test or adjustment printouts.	English Deutsch Italiano Français Español Português Polski Magyar Český Русский 日本語 中文 한국어/조선말
Print results automatically after measurement	Print results automatically after measurement if activated.	☐ (deactivated)* ☒ (activated)

Export management

Navigation: Home > Settings > Application settings > Print and export management > Export management

If a PC or WLAN is connected and activated, final results can be automatically exported. The export contains information to the instrument, time, date, method, result as well as the graphical curve.

You can define the following parameters:

Parameter	Explanation	Values
File name prefix	Defines the name of the new data file. Date and time are automatically added by the system.	any
File type	Defines the type of the data file.	CSV* PDF
Printout language	Defines the language for general printouts. Note No impact on test or adjustment printouts.	English Deutsch Italiano Français Español Português Polski Magyar Český Русский 日本語 中文 한국어/조선말
Export management	Export results automatically after measurement if activated.	Selection: ☐ * ☒
Export target	Defines the destination of data transfer.	Peripheral Network (FTP)

* Factory setting

Export target: Peripheral

Parameter	Explanation	Values
Location	Finds the location on the storage device.	Browse and select

* Factory setting

Export target: Network (FTP)

For data transfer over a network (Ethernet and WLAN) the File Transfer Protocol FTP is used. The receiving server is defined by entry of the appropriate IP address. If necessary, contact the network administrator of your IT department or your IT support.

Parameter	Explanation	Values
IP address	Defines the IP address of the receiving server.	any
Port	Defines the Port for communication with the server. Note Usually the default value is used.	21* any
Password	Entry for the password needed for server login.	any
User name	Entry for the user name needed for server login.	any

* Factory setting

6.4 Quality Management

Navigation: Home > Settings > Quality management

In this menu you can define all settings related to the adjustment and Test of the instrument. Periodic adjustments or tests ensure accurate measurement results.

You can define the following parameters:

Menu item	Explanation	Further Information
Test / Adjustment settings	Defines the equipment for adjusting and the workflow for testing or adjusting.	see Test / Adjustment settings (page 48)

6.4.1 Test / Adjustment settings

Navigation: Home > Settings > Quality management > Test / Adjustment settings

This section describes all the menu options available for defining the parameters relating to the adjustment and testing of your instrument.

For performing test and adjustments **see** Test/Adjust (page 62).

Menu structure

Main menu	Submenu	Further information
Equipment	Test weight 1	see Equipment (page 48)
	Test weight 2	
	Temperature kit	
	External humidity sensor	
Test settings	Weight test - internal	see Test Settings (page 49)
	Weight test - external	
	Temperature test	
	SmartCal test	
Adjustment settings	Weight adjustment - internal	No adjustable settings available
	Weight adjustment - external	see Adjustment Settings (page 51)
	Temperature adjustment	
Print and output settings	Printout language	see Print and Output Settings (page 51)
	Print results automatically after test/adjustment	
FACT	no submenu	see FACT (Fully automatic adjustment) (page 52)

6.4.1.1 Equipment

Navigation: Home > Settings > Quality management > Test / Adjustment settings > Equipment

This menu item defines the equipment which are required for the following tests:

- Weight adjustment
- Weight test
- Temperature adjustment
- Temperature test
- SmartCal test (optional)

For performing temperature adjustment or temperature test, you need the optional temperature adjustment kit, **see** Accessories and Spare Parts (page 122).

Test Weight 1 and Test Weight 2

You can define the following parameters:

Parameter	Explanation	Values
Test weight name	Defines a name that is in an easily recognizable form as an alternative to the ID. A name should be unique and unambiguous.	any
Nominal weight	Defines the nominal weight in grams.	0.1...200.000 g (200 g)*
Identification	Defines the identification (ID) of the weight. The ID of the weight is provided on the Weight Certificate. The ID may contain your company specific identification number. This ID is stored in the adjustment or test history and shows up on the printouts.	any

* Factory setting

Temperature kit

You can define the following parameters:

Parameter	Explanation	Values
Temperature kit name	Defines a name that is in an easily recognizable form as an alternative to the ID. A name should be unique and unambiguous.	any
Identification	Defines the identification (ID) of the Temperature kit. The ID is provided on the Certificate. The ID may contain your company specific identification number. This ID is stored in the adjustment or test history and shows up on the printouts.	any

External humidity sensor

For performing SmartCal tests, you can use the METTLER TOLEDO RHT sensor for the automatic setting of temperature and humidity values.

Note

Not applicable for certified RHT sensors, see SmartCal test (page 69).

You can define the following parameters:

Parameter	Explanation	Values
Identification	Defines the identification (ID) of the RHT sensor. This ID is stored in the test history and shows up on the printouts.	any

6.4.1.2 Test Settings

Navigation: Home > Settings > Quality management > Test / Adjustment settings > Test settings

Weight test internal

You can define the following parameters:

Parameter	Explanation	Values
Tolerance	Defines the measuring tolerance of the internal weight.	0...1 % (0.001 %)*

* Factory setting

Weight test external

You can define the following parameters:

Parameter	Explanation	Values
Test weight	Selects the predefined Test weight . Test weight 1 / Test weight 2 = defined in the menu item Equipment	Test weight 1 Test weight 2
Tolerance	Defines the measuring tolerance of the weight.	0.0001...0.1000 g (0.0010 g)*

* Factory setting

Temperature test

You can define the following parameters:

Parameter	Explanation	Values
Temperature kit	Shows the name of the Temperature kit which was defined in the menu item Equipment .	none
Temperature 1	Defines the lower test temperature. The value is based on the lower adjustment temperature and is not editable in the temperature test.	OFF ON (100 °C)*
Tolerance	Defines the tolerance of the test temperature.	1...5 °C (3 °C)*
Temperature 2	Defines the upper test temperature The value is based on the upper adjustment temperature.	OFF ON (160 °C)*
Tolerance	Defines the tolerance of the test temperature.	1...5 °C (3 °C)*
Temperature 3 (additional)	Defines an additional test temperature.	OFF* ON 50...200 °C
Tolerance	Defines the tolerance of the test temperature.	1...5 °C (3 °C)*

* Factory setting

SmartCal test

You can define the following parameters:

Parameter	Explanation	Values
Lot number input	Defines the type of input for the Lot number. The Lot number can be found on the product label. Free input: Input is not required Guided input: Input is required.	ON* OFF Free input Guided input *
Clear lot number after measurement	Enabled or disabled the function for clearing the Lot number after test measurement.	☒ (enabled) ☐ (disabled)*
Temperature	Defines the test temperature. Note - The setting of the temperature influences the setting of the control limits. - If custom temperatur is set, the controll limits will be off.	70 °C 100 °C* 130 °C 160 °C Custom temperature... 40 °C...230 °C (105 °C*)

Use cSmartCal	Sets the controll limits to the correct values for using of cSmart-Cal. Note Enabling of cSmartCal affects only the protocol entry but has no effect to the values of the control limits, if custom temperature was set.	☒ (enabled) ☐ (disabled)*
Control limits	Defines the control limits. The values of the control limits are in dependence of the temperature setting. The recommended values are automatically set. The values can also be changed. The recommended values are (lower - upper limits): 70 °C: 3.2-4.4 %MCN (cSmartCal: 3.3 - 4.3 %MCN) 100 °C: 5.2-6.4 %MCN (cSmartCal: 5.3 - 6.3 %MCN) 130 °C: 7.4-8.8 %MCN (cSmartCal: 7.5 - 8.7 %MCN) 160 °C: 9.9-11.7 %MCN (cSmartCal: 10.0 - 11.6 %MCN) Note Control limits are switched off if custom temperature was set.	values see explanation OFF

* Factory setting

6.4.1.3 Adjustment Settings

Navigation: Home > Settings > Quality management > Test / Adjustment settings > Adjustment settings

Weight adjustment internal

No editable parameters.

Weight adjustment external

You can define the following parameters:

Parameter	Explanation	Values
Test weight	Selects the predefined Test weight . Test weight 1 / Test weight 2 = defined in the menu item Equipment	Test weight 1 Test weight 2

Temperature adjustment

You can define the following parameters:

Parameter	Explanation	Values
Temperature kit	Shows the name of the Temperature kit which was defined in the menu item Equipment .	none
Temperature 1	Defines the lower temperature.	80...140 °C (100 °C)*
Temperature 2	Defines the upper temperature	140...200 °C (160 °C)*

* Factory setting

Note

Most samples are measured in the range of 100 °C and 160 °C. This is also the factory setting. The range can be expanded to the range of 80°C and 200°C. The difference between the lower and upper temperature has to be equal or more than 60 °C.

6.4.1.4 Print and Output Settings

Navigation: Home > Settings > Quality management > Test / Adjustment settings > Print and output settings

This menu item allows you to set specific printer functions for test or adjustment printouts.

- The setting for printout language in this menu item has impact on test or adjustment printouts only. For setting the general printout language, **see** Print and Export Management (page 45).
- Not all languages are compatible with any printer, and also depending on the settings of the printer, **see** Recommended Printer Settings (Strip printer) (page 128).
- For installing and defining a printer **see** Peripherals (page 37).

You can define the following parameters:

Parameter	Explanation	Values
Printout language	Defines the printout language for test or adjustment printouts.	English Deutsch Italiano Français Español Português Polski Magyar Český
Print results automatically after test/adjustment	Activated or deactivated function.	☒ (enabled) ☐ (disabled)*

* Factory setting

6.4.1.5 FACT (Fully automatic adjustment)

Navigation: Home > Settings > Quality management > Test / Adjustment settings > **FACT**

This menu item allows you to deactivate or activate fully automatic adjustment function **FACT**.

Note

- Before performing an external weight adjustment, we recommend you to disable **FACT**. Otherwise **FACT** overwrites the adjustment.
- This function is not available for approved models – that is, **FACT** is always active.

You can define the following parameters:

Parameter	Explanation	Values
FACT	Enabled or disabled fully automatic adjustment FACT function.	☒ (enabled)* ☐ (disabled)

* Factory setting

6.5 User Management

Navigation: Home > Settings > User management

The security system of the instrument is based on a "Users and Groups" concept. Every real user of the instrument should be represented by a user account on the instrument which holds particular settings of the respective user. Every user belongs to a particular group and inherits the access rights assigned to that group.

You can define the following parameters:

Menu item	Explanation	Further Information
Groups	Defines groups and their access rights.	see Groups (page 53)
Users	Defines the user accounts.	see Users (page 55)
Account policies	Set default startup user.	see Account Policies (page 56)

6.5.1 Groups

Navigation: Home > Settings > User management > Groups

In this menu you can create, delete, or edit user groups and define their access rights.

Four predefined groups are available. The default group with all access rights is administrator. The administrator group can not be edited or deleted. All other groups can be edited or deleted.

Predefined groups	Default access rights						
	User Preference settings	System settings	Edit Methods	Run Tests	Run Adjustments	Quality management	User and data management
Administrator	☒	☒	☒	☒	☒	☒	☒
Supervisor	☒	☒	☒	☒	☒	☐	☐
Quality Manager	☒	☒	☐	☒	☒	☒	☐
Operator	☒	☐	☐	☐	☐	☐	☐

Creating a new group

► Menu **Groups** selected.

- 1 Tap **[New...]**.
- 2 Enter the name of the new group and confirm with **[OK]**. Name should be unique and unambiguous.
- 3 Edit the properties of the new group.

Deleting a group

► Menu **Groups** selected.

- 1 Select the group you wish to delete.
- 2 Tap **[Delete]**. Confirm the message box.

Note

The Administrator group as well as groups with assigned users cannot be deleted, **see** Users (page 55).

Editing a group

► Menu **Groups** selected.

- 1 Select the group you wish to edit.
- 2 Edit the group parameters. Store the settings with **[Save]**.

You can define the following parameters:

Parameter	Explanation	Values
Group properties	Name and (optional) description of the group. Name should be unique and unambiguous.	any
Access rights	Defines the access rights to menus or operations.	User Preference settings System settings Edit Methods Run Tests Run Adjustments Quality Management User and data management
Group members	Displays the users assigned to this group (no editing can be done here, the assignment is done when setting up or editing a user).	none
History	Displays the date and time of the last modifications made to this group (no editing here).	none

Influence of access rights

Access rights	Impact on
User Preference settings	User preferences Navigation: Home > Settings > User preferences See User Preferences (page 35)
System settings	System settings Navigation: Home > Settings > System settings See System Settings (page 37)
Edit Methods	Method definition Navigation: Home > Method definition See Method Definition (page 73)
	Application settings Navigation: Home > Settings > Application settings See Applications Settings (page 44)
Run Tests	Tests Navigation: Home > Test/Adjust > Tests See Tests (page 67)
Run Adjustments	Adjustments Navigation: Home > Test/Adjust > Adjustments See Adjustments (page 62)
Quality management	Quality management Navigation: Home > Settings > Quality management See Quality Management (page 48)
User and data management	User management Navigation: Home > Settings > User management See User Management (page 53)
	System and data management Navigation: Home > Settings > System and data management See System and Data Management (page 57)
	Deleting results See Graphical Evaluations of Measurement Results (page 99)

Note

- If password protection is mandatory requirement, the access right **User Preference settings** must be disabled.
- If a user does not have the appropriate access rights, the respective navigation elements are not shown (on Home screen) or are disabled (greyed out).

6.5.2 Users

Navigation: Home > Settings > User management > Users

In this menu you can create, edit, or delete user accounts.

Attention

Important note for the Administrator password.

We recommend that you note down your administrator password and keep it in a safe place. If you forgot the password for the default administrator access, it is not possible to regain access to the moisture analyzer. In this case contact your METTLER TOLEDO representative.

Creating a new user account

Note

The user account name must be unique.

► Menu **Users** selected.

- 1 Tap [**New...**].
- 2 Enter the name of the new user account and confirm with [**OK**].
- 3 Edit the properties of the new user account.

Editing an existing user account

► Menu **Users** selected.

- 1 Select the user account you wish to edit.
- 2 Select menu **User properties**.
- 3 Edit the user account parameters.
- 4 To store the settings, tap [**Save**].

You can define the following parameters:

Parameter	Explanation	Values
User name	Name of the user account. Name should be unique and unambiguous.	any
Full name	Full name of the user (optional).	any
Description	Descriptive text for the user account (optional).	any
Account active	Activate or deactivate the user account. Inactive user accounts are not available for login.	☐ (deactivated)* ☒ (activated)
Password	Create a new or change an existing password. OFF = no password protection.	OFF* ON any
Group	Assign the user account to a particular group.	available groups

* Factory setting

Note

- To view the date and time of last modification made to a user account, select the respective account and tap [**History**].
- If password protection is mandatory requirement, the access right **User Preference settings** must be disabled, **see** Groups (page 53).

Deleting a user account

► Menu **Users** selected.

- 1 Select the user account you wish to delete.
- 2 Tap [**Delete**]. Confirm the message box.

Note

You cannot delete the Administrator account or the account that is currently active.

6.5.3 Account Policies

Navigation: Home > Settings > User management > Account policies

In this menu you can set the default startup user.

You can define the following parameters:

Parameter	Explanation	Values
Default startup user	Select the default user account for login at startup.	OFF ON* available user accounts

* Factory setting

Note

If a non-standard user logs out, the default startup user profile is automatically activated.

6.6 System and Data Management

Navigation: Home > Settings > System and data management

This menu item allows you to export or import settings and methods to or from an external USB storage device, as well as to restore the system (depending on the user rights).

Menu structure

Main menu	Submenu	Further information
Result management	Auto-delete	see Result Management (page 57)
Export / Import	Export settings and methods to external storage devices...	see Export / Import (page 57)
	Import settings and methods from external storage device...	
Back up / Restore	Back up your system on external storage device...	see Back up / Restore (page 59)
	Restore your system from selected back up...	
Export event logs	Export all event information	see Export event logs (page 59)
Reset	no submenu	see Reset (page 60)
Update	Update your instrument software	see Update (page 60)

Note for using USB memory sticks

- Memory sticks up to 32 GB are supported.
- Recommended formatting: FAT32 (NTFS not supported).
- Max. cluster size: 32 KB.
- Max. file size: 32 MB.
- Make sure that the stick is fully inserted.
- Make sure that the write protection is deactivated if present.

6.6.1 Result Management

Home > Settings > System and data management > Result management

This menu item allows you to manage your results. The Auto-delete functionality prevents the instrument from filling up with old measurement results by removing them. We advise you to use this function to improve the performance of your Moisture Analyzer. Is the Auto-delete functionality activated, you can define the number of stored results with the result limit.

Attention

The default value of the result limit is 1000. If there are more than 1000 results on your Moisture Analyzer, the system automatically deletes the oldest results. Make sure that important data will periodically be saved on an external storage.

You can define the following parameters:

Parameter	Explanation	Values
Auto-delete	Activates the auto-delete functionality and defines the maximal number of stored result via Result limit .	OFF ON* 1...3000 (1000)*

* Factory setting

6.6.2 Export / Import

Navigation: Home > Settings > System and data management > Export / Import

This menu item allows you to export or import settings and methods to or from an external storage device. These data can be transferred to other instruments of the same type and can be useful to configure multiple instruments with the same specifications.

Note

- Software Version must be equal or higher.
- Data transfer from HX models to HS models is limited. Features which are not supported by HS models cannot be imported, e.g. Step drying or high resolution.
- Export / import of methods can also be performed from the menu **Method Definition**.

Export settings and methods to external storage devices...

You can define the following parameters:

Parameter	Explanation	Values
File name	Defines the name of the new data file.	any
Location	Finds the location on the storage device.	Browse and select
Export selection	Defines the data for export. Note • User management and user preferences contains: User management settings, user preferences • System, application and QM settings contains: System settings, application settings, quality management settings	All* ☒ User management and user preferences ☒ System, application and QM settings ☒ Methods

* Factory setting

- ▶ An external storage device is connected e.g. memory stick.
- ▶ Settings are made.
- To start, tap [**Export**].
To abort, tap [**Cancel**].

Import settings and methods from external storage device...

You can define the following parameters:

Parameter	Explanation	Values
Import from file	Defines the data to be imported.	Browse and select
Location	Shows the location of the import file.	–
Import selection	Defines the data for import.	All* ☒ User management and user preferences ☒ System, application and QM settings ☒ Methods

* Factory setting

- ▶ Import settings and methods from external storage devices – is selected.
- ▶ An external storage device is connected e.g. memory stick.
- 1 Tap **Browse and select**
⇒ **Import from file** appears.
- 2 Select Device and import file.
⇒ Full menu of Import settings and methods from external storage devices appears.
- 3 Tap Import selection and select the data for import if needed.
- 4 To start, tap Import button and follow the instructions in the message window.
To abort, tap [**Cancel**].

6.6.3 Back up / Restore

Navigation: Home > Settings > System and data management > Back up / Restore

This menu item allows you to create a recovery point of the system and to store them on an external storage device. We advise you to create a recovery point of the system regularly. So you are able to restore the system with all data in fault.

Attention

- After restoring, the system is in backup status. That means, that newer data such as settings, methods or results are lost.
- Restore is only possible on the same instrument as the back up was carried out.

Backup

You can define the following parameters:

Parameter	Explanation	Values
File name	Defines the name of the new data file.	any
Location	Finds the location on the storage device.	Browse and select

Procedure:

- ▶ An external storage device is connected e.g. memory stick.
 - ▶ **Back up your system on external storage device...** – is selected.
- 1 Define the name of the new data file.
 - 2 Define the location on the storage device.
 - 3 To confirm, tap **[OK]**.
 - 4 To start, tap **[Backup]**.
To abort, tap **[Cancel]**.

Restore

You can define the following parameters:

Parameter	Explanation	Values
Restore from file	Finds the location on the storage device and selects the file for restoring.	Browse and select

Procedure:

- ▶ **Restore your system from selected back up...** – is selected.
 - ▶ An external storage device is connected e.g. memory stick.
- 1 Tap **Restore from file... (Browse and select)**.
 - ⇒ Storage device appears.
 - 2 Select the file for restoring.
 - 3 To start, tap **[Restore]**.
To abort, tap **[Cancel]**.
 - ⇒ Confirmation window appears.
 - 4 Confirm with **[Restore]**.
 - ⇒ The system will restore the data and then perform a restart. Wait until the system has completely started up again.

6.6.4 Export event logs

Navigation: Home > Settings > System and data management > Export event logs

This menu item allows you to export a log file (Zip format) with all relevant data to an external storage device. These data can be used for example, to support METTLER TOLEDO for troubleshooting.

You can define the following parameters:

Parameter	Explanation	Values
File name	Defines the name of the new data file.	any
Export target	Defines the destination of data transfer.	Peripheral Network (FTP)

Export target: Peripheral

Parameter	Explanation	Values
Location	Finds the location on the storage device.	Browse and select

* Factory setting

Export target: Network (FTP)

For data transfer over a network (Ethernet and WLAN) the File Transfer Protocol FTP is used. The receiving server is defined by entry of the appropriate IP address. If necessary, contact the network administrator of your IT department or your IT support.

Parameter	Explanation	Values
IP address	Defines the IP address of the receiving server.	any
Port	Defines the Port for communication with the server. Note Usually the default value is used.	21* any
Password	Entry for the password needed for server login.	any
User name	Entry for the user name needed for server login.	any
File type	Defines the type of the data file.	CSV* PDF

* Factory setting

6.6.5 Reset

Navigation: Home > Settings > System and data management > Reset

This menu item allows you to reset the Instrument to factory settings. Date, time, weight and temperature adjustments/history are not affected.



CAUTION

Data loss after system reset.

After resetting, the instrument is in state as delivered. That means, that all data such as settings, methods or results are lost.

- We recommend you to save first the appropriate data on an external storage device, **see** Export / Import (page 57) and Back up / Restore (page 59).

You can define the following parameters:

Parameter	Explanation	Values
Reset	Resets the instrument to factory settings.	none

6.6.6 Update

Navigation: Home > Settings > System and data management > Update

Note

This function is not available with approved instruments. For update, contact your METTLER TOLEDO representative.

METTLER TOLEDO is continuously improving its instrument firmware (software) for the benefit of customers. So that the customer can benefit quickly and easily from further developments, METTLER TOLEDO makes the latest

firmware versions available on the Internet. The firmware made available on the Internet has been developed and tested by Mettler-Toledo AG using processes that meet the guidelines of ISO 9001. Mettler-Toledo AG does not, however, accept liability for consequences that might arise from using the firmware.

Update procedure

- 1 Connect to the Internet.
- 2 Go to the site <http://www.mettler-toledo-support.com>.
- 3 Log in to the **METTLER TOLEDO Balance Support Site** (registration with the serial number of a METTLER TOLEDO instrument required).
- 4 Click **Customer Support**.
- 5 Click appropriate product folder e.g. HX204 or HS153.
- 6 Click **Latest Firmware and Update Support**.
- 7 **Important:** Before updating read the downloadable supplementary sheet carefully. This document provides you with the relevant information for updating to the new firmware version.

7 Test/Adjust

Navigation: Home > Test/Adjust

This function offers you to adjust or test the integrated balance and the heating module. For the relevant settings **see** Test / Adjustment settings (page 48).

Menu structure

Main menu	Submenu	Further information
Adjustments	Weight adjustment - internal	see Weight Adjustment - Internal (page 63)
	Weight adjustment - external	see Weight Adjustment - External (page 63)
	Temperature adjustment	see Temperature Adjustment (page 64)
Tests	Weight test - internal	see Weight test - internal (page 67)
	Weight test - external	see Weight test - external (page 67)
	Temperature test	see Temperature test (page 68)
	SmartCal test	see SmartCal test (page 69)
History	Weight test history	see History (page 72)
	Weight adjustment history	
	Temperature adjustment history	
	Temperature test history	
	SmartCal test history	

7.1 Adjustments

Navigation: Home > Test/Adjust > Adjustments

In this menu item you can adjust the balance as well as the heating module of your instrument. For relevant settings **see** Test / Adjustment settings (page 48).

Attention

- To obtain accurate results, the balance must be adjusted at the point of use under measuring condition in order to match the gravitational acceleration at its location. The instrument must be connected to the power supply for approximately 60 minutes in order to reach operating temperature before adjusting. Adjusting is necessary:
 - before the instrument is used for the first time.
 - after a change of location.
- We advice you to wait at least 30 minutes after a drying operation (or previous heating module adjustment) before performing the adjustment.

Fully automatic adjustment FACT

This instrument is equipped with fully automatic adjustment with two internal weights called Fully Automatic Calibration Technology (**FACT**).

The balance adjusts itself automatically:

- after the warm-up phase after connection to the power supply.
- when a change in the ambient conditions (temperature) could lead to a noticeable deviation in the measurement.
- on a fixed time interval

FACT is activated by default and can be switched off as required e.g. when performing an external adjustment, **see** FACT (Fully automatic adjustment) (page 52).

Navigation: Home > Settings > Quality management > Test / Adjustment settings > FACT

Note

With approved models **FACT** cannot be switched off.

7.1.1 Weight Adjustment - Internal

Navigation: Home > Test/Adjust > Adjustments > Weight adjustment - internal

This function allows you to adjust the balance with the built-in weights. The internal adjustment is equivalent to the FACT adjustment.

Perform this function as follows:

- 1 Tap the menu item **Weight adjustment - internal**.
 - ⇒ The adjustment screen appears.
- 2 Clear pan and tap [**Start adjustment**].
 - ⇒ The balance adjusts itself automatically.
- ⇒ The instrument indicates when the adjustment process is done. The balance is adjusted and ready to use for further measurements. The display shows the adjustment results. The results can be printed out by tapping [].

7.1.2 Weight Adjustment - External

Navigation: Home > Test/Adjust > Adjustments > Weight adjustment - external

This function allows you to adjust the balance using an external weight. We recommend to use certified weights so that full traceability is guaranteed.

Note

We recommend you to disable FACT. Otherwise FACT overwrites the adjustment.

Perform this function as follows:

- ▶ FACT is off.
 - ▶ Test weight and Adjustment settings are defined.
see Test / Adjustment settings (page 48)
- 1 Have required test weight ready.
 - 2 Tap the item **Weight adjustment - external**.
 - ⇒ The adjustment screen appears.
 - 3 Clear sample pan holder and tap [**Start adjustment**].

Note
We recommend not to use a pan.

 - ⇒ The required value flashes and prompts you to load the weight, e.g. "100.00 g".
 - 4 Load the required test weight in the center of sample pan holder.
 - ⇒ The balance indicates the adjustment process, "———" appears.
 - 5 When the display flashes "**000.00**" remove the test weight.
 - ⇒ The balance is adjusted and ready to use for further measurements. The display shows the adjustment results. The results can be printed out by tapping []

Note

- ▶ After a timeout the error message appears **Weight out of range. The adjustment has been aborted** The adjustment was not executed.
- 1 Confirm with [**OK**].
 - 2 Remove the test weight for unloading.
 - 3 Repeat the adjustment.

7.1.3 Temperature Adjustment

Navigation: Home > Test/Adjust > Adjustments > Temperature adjustment

This function allows you to adjust the temperature control of the heating module. You need the optional **Temperature kit** for performing this function, **see** Accessories and Spare Parts (page 122). To learn when an adjustment of the heating module is necessary, **see** Notes on Adjustment of Balance and Heating Module (page 127). We advise you to wait at least 30 minutes after a drying operation (or previous heating module adjustment) before performing the adjustment.

Note

- The duration of the temperature adjustment is 30 minutes. (15 minutes for each measuring temperature).
- The sample chamber must be less than 50 °C before a further adjustment or test can be done.
- It is also possible to perform a test first, and then to transform the test into an adjustment if required. Only possible for tests conducted with 2 temperatures, **see** Temperature test (page 68).
- For temperature adjustment settings **see** Adjustment Settings (page 51).



CAUTION

Danger of burns

Sample chamber may still be hot.

- Cool down instrument before setting up.

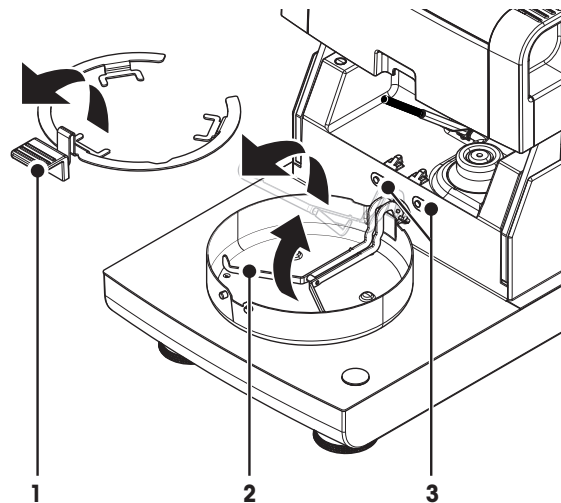
- ▶ Adjustment settings are defined, **see** Adjustment Settings (page 51).
- Tap menu item **Temperature adjustment**.
 - ⇒ Work screen **Temperature adjustment** appears.

Setting up Drying Unit

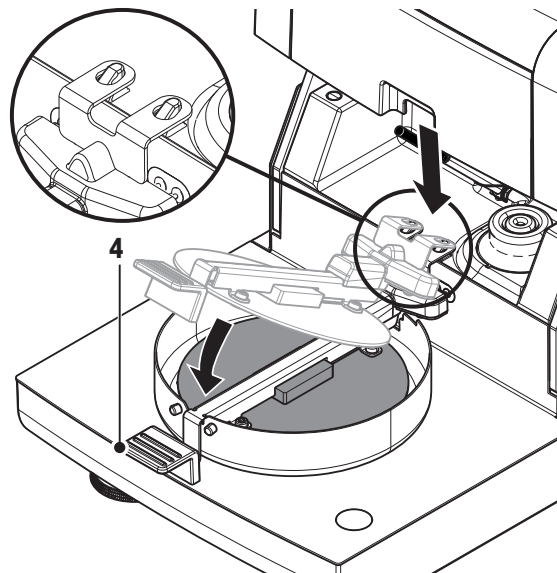
- ▶ Have required **Temperature kit** ready.
 - ▶ Instrument is on
 - ▶ Sample chamber is open.
 - ▶ Instrument is cooled down.
- 1 Remove sample pan handler (with pan) (1).
 - 2 Remove sample pan holder (2).
 - 3 Contact areas (3) should be clean. Clean them if needed.

Note

Do not remove draft shield.



- Insert **Temperature kit** (4).



Start adjustment

- ▶ Draft shield is installed.
- ▶ **Temperature kit** is inserted.

- 1 Tap [**Start adjustment**].

Note

The adjustment process takes 30 minutes.

- ⇒ The instrument starts the adjustment process. The screen shows the current temperature and the remaining time.
- ⇒ The instrument heats up to the target temperature 1 for determining the predefined lower temperature.
- ⇒ After 15 minutes, the instrument adjusts the lower temperature and continues with the process.
- ⇒ The instrument heats up to the target temperature 2 for determining the predefined upper temperature.
- ⇒ After 15 minutes, the instrument adjusts the upper temperature and concludes the adjustment.

- 2
 - To print out the results, tap [🖨️].
 - To display or print out the results of previous adjustments, tap [**History**].
 - To return to the home screen, tap [🏠].
- 3 After cooling down, remove **Temperature kit** and set up drying unit for measurement.

See History (page 72) and Setting up Drying Unit (page 26)

Note

Your calibrated temperature adjustment kit can be re-calibrated. Please contact your local METTLER TOLEDO representative for more information.

7.1.4 Adjustment Printout Examples

Weight adjustment internal

```
-WEIGHT ADJUST INTERNAL-  
  
      03.07.2013      11:51  
  
METTLER TOLEDO  
Halogen MoistureAnalyzer  
  
Type                HX204  
SNR(Drying Unit)    0123459789  
SNR(Terminal)       9876543210  
  
Nominal Weight      100.000 %  
Cell temperature     24.60 °C  
Balance is levelled  
Adjustment          Done  
  
Last Weight Test    02.05.2014 11:34  
Last Weight Adjustment 02.05.2014 16:38  
  
Signature:  
.....  
  
----- END -----
```

Weight adjustment external

```
-WEIGHT ADJUST EXTERNAL-  
  
      03.07.2013      11:51  
  
METTLER TOLEDO  
Halogen MoistureAnalyzer  
  
Type                HX204  
SNR(Drying Unit)    1234567890  
SNR(Terminal)       0987654321  
  
User name Administrator  
  
Weight ID           ECW-100/1  
Nominal Weight      100.000 g  
Cell temperature     25.20 °C  
Balance is levelled  
Adjustment          Done  
  
Last Weight Test    02.05.2014 11:34  
Last Weight Adjustment 02.05.2014 16:38  
  
Signature  
.....  
  
----- END -----
```

Temperature adjustment

```
-TEMPERATURE ADJUSTMENT-  
  
      03.07.2013      11:51  
  
METTLER TOLEDO  
Halogen MoistureAnalyzer  
  
Type                HX204  
SNR(Drying Unit)    1234567890  
SNR(Terminal)       0987654321  
  
User name Administrator  
  
Temp Kit ID         MT-414/A  
Temp1 target        100.0 °C  
Temp1 actual         97.0 °C  
Temp2 target        160.0 °C  
Temp2 actual         162.0 °C  
Adjustment          Done  
  
Last Temperature Test 02.05.2014 11:34  
Last Temperature Adjustment 02.05.2014 16:38  
  
Signature:  
.....  
  
----- END -----
```

7.2 Tests

Navigation: Home > Test/Adjust > Tests

In this menu item you can check the adjustment of the balance as well the temperature of the heating module. For relevant settings **see** Test / Adjustment settings (page 48).

7.2.1 Weight test - internal

Navigation: Home > Test/Adjust > Tests > Weight test - internal

This function allows you to test the balance with the built-in weight. Perform this function as follows:

- ▶ Test settings are defined (Factory setting of the tolerance is 0.0010 %) **see** Test / Adjustment settings (page 48)
 - 1 Open sample chamber
 - 2 Tap the item **Weight test - internal**.
 - ⇒ The test screen appears.
 - 3 Clear pan and tap [**Start test**].
 - ⇒ The balance is tested automatically.
 - ⇒ When the test process is done, the display shows the test results and indicates whether the test is passed or failed according to defined tolerances.

The test results can be printed out by tapping [].

7.2.2 Weight test - external

Navigation: Home > Test/Adjust > Tests > Weight test - external

This function allows you to test the balance using an external weight. Perform this function as follows:

- ▶ Test weight and test settings are defined. **see** Test / Adjustment settings (page 48)
 - 1 Have required test weight ready.
 - 2 Open sample chamber.
 - 3 Tap the item **Weight test - external**.
 - ⇒ The test screen appears.
 - 4 Clear sample pan holder and tap [**Start test**].

Note
We recommend not to use a pan.

 - ⇒ The required value flashes and prompts you to load the weight.
 - 5 Load the required test weight.
 - ⇒ The balance indicates the test process, "———" appears.
 - 6 When the display flashes "**000.00**" remove the test weight.
 - ⇒ When the test process is done, the display shows the test results and indicates whether the test is passed or failed according to defined tolerances. The results can be printed out by tapping [].

Timeout

- ▶ After a timeout of 75 seconds the error message appears **Weight out of range. The test has been aborted.** The test was not executed.
 - 1 Confirm with [**OK**].
 - 2 Repeat the test.

7.2.3 Temperature test

Navigation: Home > Test/Adjust > Tests > Temperature test

This function allows you to test the temperature control of the heating module. You need the optional Temperature Kit for performing this function, **see** Accessories and Spare Parts (page 122). To learn when an test of the heating module is necessary, **see** Notes on Adjustment of Balance and Heating Module (page 127). We advise you to wait at least 30 minutes after a drying operation (or previous heating module adjustment) before performing a test.

Note

- The duration of the temperature test is 15 minutes for each measuring temperature.
- The sample chamber must be cooled down below 50 °C before a further adjustment or test can be done.
- It is possible to perform a test and then subsequently to transform the test into an adjustment if required. This is only possible for tests conducted with temperatures 1 and 2.
- Temperature test settings **see** Test Settings (page 49).

Setting up the Drying Unit for temperature test

Same procedure as for temperature adjustment, **see** Temperature Adjustment (page 64).



CAUTION

Danger of burns

Sample chamber may still be hot.

- Cool down instrument before setting up.

Performing temperature test

- ▶ Test settings are defined, **see** Test Settings (page 49).
- ▶ Setting up the drying unit for temperature testing has been carried out.

1 Tap menu item **Temperature test**.

⇒ Work screen **Temperature test** appears.

2 Tap [**Start test**].

Note

The testing process takes 15 minutes for each temperature (depending on the settings).

- ⇒ The instrument starts the testing process. The screen shows the current temperature and remaining time.
- ⇒ The instrument is heating up to the target temperature 1 for determining the predefined lower temperature.
- ⇒ After 15 minutes, the instrument displays and stores the measured lower temperature and continues with the process.
- ⇒ The instrument is heating up to the target temperature 2 for determining the predefined upper temperature.
- ⇒ After 15 minutes, the instrument displays and stores the measured upper temperature.
- ⇒ If temperature 3 is defined, the procedure has an additional step.
- ⇒ The test is done and indicates whether the test has passed or failed according to defined tolerances.

3 Decide if an adjustment is needed.

4 • To return to the menu **Tests**, tap [**Tests**].

- To print out the results, tap [.
- To display or print the results at any time, tap [**History**].
- To return to the home screen, press [.

5 After cooling down, remove temperature adjustment kit and set up drying unit for measurement, **see** Setting up Drying Unit (page 26).

7.2.4 SmartCal test

What is SmartCal™?

Proper operation of the Halogen Moisture Analyzer can be quickly verified with a SmartCal test. SmartCal is a granular test material with a defined moisture value after a specified time at a defined drying temperature.

SmartCal is available for four test temperatures. There are control limits for each test temperature. If the measured and normalized moisture value is within the control limits, the instrument has passed the functional test. If value is outside the control limits, there may be a problem with the instrument or the test conditions may not have been met.

Test temperatures and control limits

cSmartCal (certified) and SmartCal can be used for testing at any temperature between 70 °C and 230 °C. METTLER TOLEDO provides established control limits for testing at 70 °C, 100 °C, 130 °C and 160 °C.

Select the temperature closest to the drying temperature normally used.

Temperature	cSmartCal™	SmartCal™
70 °C	3.3 - 4.3 %MCN	3.2 - 4.4 %MCN
100 °C	5.3 - 6.3 %MCN	5.2 - 6.4 %MCN
130 °C	7.5 - 8.7 %MCN	7.4 - 8.8 %MCN
160 °C	10.0 - 11.6 %MCN	9.9 - 11.7 %MCN

MCN = Normalized moisture content (Calculated value in consideration of temperature and humidity)

Note about using SmartCal

- Store SmartCal at room temperature.
- Do not open the blister pack before use.
- Shelf life: The expiry date is printed on the blister pack and on the stick (e.g. Exp08.2013).
- Disposal: Can be disposed of as normal waste. Observe local and national environmental regulations.

Additional information on using SmartCal, e.g. SmartCal user guide, **see**

► www.mt.com/smartcal

Performing SmartCal™ test

Navigation: Home > Test/Adjust > Tests > SmartCal test

The test is carried out in the same way as a measurement. The duration of the test is 10 minutes.

- ▶ The Instrument is acclimatized in working area, cooled and connected to AC power for at least one hour.
- ▶ The SmartCal package is acclimatized in working area.
- ▶ The expiration date of the Smart Cal sample has not been reached.
- ▶ The METTLER TOLEDO RHT sensor is properly installed and acclimatized in working area after installing for at least one hour. If no RHT sensor is available, room temperature and humidity must be enter manually, **see** Install METTLER TOLEDO RHT sensor (page 29).

Alternatively:

- you can use a certified RHT sensor from Testo, **see** note at the end of this Chapter.
 - you can enter the temperature and humidity manually.
- ▶ Identification of the METTLER TOLEDO RHT sensor if needed, **see** Equipment (page 48)
 - ▶ Test temperature and control limits are set, **see** Test Settings (page 49)
- 1 Tap **SmartCal test**.
 - ⇒ Sample chamber opens.
 - 2 If no RHT sensor is used, enter current room temperature and humidity. Tap **[OK]**
 - 3 Enter SmartCal lot number if needed. Tap **[OK]**
 - 4 Place the sample pan handler with the aluminum sample pan in the instrument.

- 5 Tap [->0/T<-] to tare the balance.
 - 6 Remove a SmartCal stick from the blister pack, tear it open, and distribute the entire contents evenly over the sample pan. If necessary, carefully rotate and tilt the sample pan until it is fully and uniformly covered with granulate.
 - 7 Start the test immediately by tapping start drying.
 - ⇒ SmartCal test is performing.
 - ⇒ The test is done and indicates whether the test has passed or failed. The display shows the normalized result.
- **Passed:** Your instrument works properly and is ready for other measurements.
 - **Failed:** Your instrument works not properly. Action are required:
 - See What to Do if... (page 114)
 - We recommend not to use this instrument further until the problem is resolved. In this case contact your METTLER TOLEDO representative.
-
- To return to the menu **Tests**, tap [**Tests**].
 - To print out the results, tap [].
 - To display or print the results at any time, tap [**History**].
 - To return to the home screen, press [].

Note

SmartCal tests do not supersede the recommended periodic weight or temperature tests.

Certified RHT sensors

The RHT sensor from METTLER TOLEDO is not available as a certified version. If a certified version is required, the following RHT models from testo are supported: testo 435-2, testo 435-4, testo 635-2, testo 735-2.

The Testo RHT sensor has to be connected on the USB port on the terminal. If both RHT sensors are connected, the testo RHT sensor has priority.

For more information about Testo, **see**

► <http://www.testo.com>

7.2.5 Test Printout Examples

Weight test internal

```
--WEIGHT TEST INTERNAL--  
  
      03.07.2013      11:51  
  
METTLER TOLEDO  
Halogen MoistureAnalyzer  
  
Type                HX204  
SNR(Drying Unit)    1234567890  
SNR(Terminal) 0987654321  
  
Nominal Weight      100.0000 %  
Actual Weight       99.9994 %  
Difference           -0.0006 %  
Tolerance            +/-0.0010 %  
Cell temperature    24.30 °C  
Balance is levelled  
Test                passed  
  
Last Weight Test  
      02.05.2014 11:34  
Last Weight Adjustment  
      02.05.2014 16:38  
  
Signature:  
  
.....  
----- END -----
```

Weight test external

```
--WEIGHT TEST EXTERNAL--  
  
      03.07.2013      11:51  
  
METTLER TOLEDO  
Halogen MoistureAnalyzer  
  
Type                HX204  
SNR(Drying Unit)    1234567890  
SNR(Terminal) 0987654321  
  
User name Administrator  
  
Weight ID           ECW-100/1  
Nominal Weight      100.0000 g  
Actual Weight       101.3198 g  
Difference           1.3198 g  
Tolerance            +/- 0.001 g  
Cell temperature    23.83 °C  
Balance is levelled  
Test                Failed  
  
Last Weight Test  
      02.05.2014 11:34  
Last Weight Adjustment  
      02.05.2014 16:38  
  
Signature:  
  
.....  
----- END -----
```

Temperature test

```
----TEMPERATURE TEST----  
  
      03.07.2013      11:51  
  
METTLER TOLEDO  
Halogen MoistureAnalyzer  
  
Type                HX204  
SNR(Drying Unit)    1234567890  
SNR(Terminal) 0987654321  
  
User name Administrator  
  
Temp Kit ID         MT-414/A  
Temp1 target        100.0 °C  
Temp1 actual         99.0 °C  
Temp1 tolerance      +/- 3.0 °C  
Temp2 target        160.0 °C  
Temp2 actual         159.0 °C  
Temp2 tolerance      +/- 3.0 °C  
Test                passed  
  
Last Temperature Test  
      02.05.2014 11:34  
Last Temperature Adjustment  
      02.05.2014 16:38  
  
Signature:  
  
.....  
----- END -----
```

7.3 History

Navigation: Home > Test/Adjust > History

The instrument always records all test or adjustment operations that have been made and saves them in a memory which is specially protected against power failure (limited to the last 50 records per history selection).

Note

Aborted adjustments or tests will not be saved.

When the History is selected, a list appears with the history selection as follows:

- **Weight test history**
- **Weight adjustment history**
- **Temperature test history**
- **Temperature adjustment history**
- **SmartCal test history**

The following information is displayed:

Weight test history

Weight test external		Weight test internal	
Calibration weight identification		Nominal weight	100 %
Nominal weight	g	Actual weight	%
Actual weight	g	Difference	%
Difference	g	Tolerance	%
Tolerance	g	Temperature	°C
Temperature	°C	Leveling information	
Leveling information			

Weight adjustments history

Weight adjustment external		Weight adjustment internal	
Calibration weight name		Temperature	°C
Calibration weight identification		Leveling information	
Nominal weight	g		
Temperature	°C		
Leveling information			

Temperature adjustment/test history

Temperature adjustment		Temperature test	
Adjustment kit identification		Adjustment kit identification	
Temperature 1	target - actual	Temperature 1	target - actual
Temperature 2	target - actual	Temperature 2	target - actual
		Free temperature	target - actual
		Tolerances for each temperature	°C

SmartCal test history

SmartCal test	
User name	
SmartCal Identification	
Start weight	g
Total test time	minutes
Dry weight	%MCN
Test result	passed / failed
Date and time	
Leveling information	

The selected individual results can be printed out by tapping [🖨️].

8 Method Definition

Navigation: Home > Method definition

What is a method?

Methods simplify and accelerate your daily work. A method contains all settings for measuring the moisture content of a particular sample (substance). It can be recalled by tapping a measurement button or a shortcut and your instrument immediately operates with the corresponding settings.

The **optimal setting of parameters and the drying time** depends on the type and size of the sample and the desired accuracy of the measurement result. The exact parameters can only be determined experimentally, **see** How to Obtain Best Results (page 127). Your Instrument helps you to determine the settings.

Method definition

This function offers you to define new methods and to change, delete, export or import existing methods depending on the user rights. It allows you to store up to 300 individual methods.

All parameters using in a drying method, can be defined under this menu item.

Note

- In most cases it is sufficient to set **Main measurement parameters**.
- You will find more information about defining methods in the delivered application brochure «Guide to Moisture Analysis» or **see** How to Obtain Best Results (page 127).

Prerequisite

The parameter menu appears only if a method already exists and is selected in the method definition dialog or create a new method.

Menu structure

Main menu	Submenu	Further information
Main measurement parameters	Drying program (including temperature and switch-off criteria)	see Settings of Drying Program (page 75)
	Display mode	see Settings of Display Mode (page 82)
	Start weight	see Settings of Start Weight (page 85)
Result and value handling	Control limits	see Control limits (page 86)
	Resolution	see Resolution (page 87)
	Free factor	see Free Factor (page 88)
Workflow handling	Starting mode	see Starting Mode (page 89)
	Preheating	see Preheating (page 89).
General method properties	Method name	see Method name (page 91)

Defining a new Method

Note

A method name can only exist once.

- ▶ **Method definition** is selected.

- 1 Tap [**New...**].
 - ⇒ The keypad appears.
- 2 Enter a name for the new method. Name should be unique and unambiguous. 1 to 30 characters are possible (including spaces).
- 3 Confirm with [**OK**].
 - ⇒ The parameter menu for the new method appears.
- 4 Set the desired parameters e.g. **Main measurement parameters**.
- 5 To store the method, tap [**Save**].

Editing an existing method

- ▶ **Method definition** is selected.

- 1 Tap the method in the list you want to edit.
 - ⇒ The parameter menu for the selected method appears.
- 2 Edit the desired parameters e.g. **Main measurement parameters**.
- 3 To store the method, tap [**Save**].

Copying an existing method

- ▶ **Method definition** is selected.

- 1 Tap the method in the list you want to copy.
 - ⇒ The parameter menu of the desired method appears.
- 2 Tap [**Save as...**].
 - ⇒ The keypad appears.
- 3 Enter a new name for the copied method. Name should be unique and unambiguous. 1 to 30 characters are possible.
- 4 To store the method, tap [**Save**].

Removing an existing method

Note

All results of this method are also removed.

- ▶ **Method definition** is selected.

- 1 Tap the method in the list you want to remove.
 - ⇒ The parameter menu of the desired method appears.
- 2 Tap [**Delete**].
 - ⇒ A message box appears.
- 3 Confirm with [**Delete**].
 - ⇒ The desired method is removed.

Note

A possible shortcut to this method is not removed.

For removing shortcuts **see** Using Shortcuts (page 97)

8.1 Main Measuring Parameters

Navigation: Home > Method Definition > Method name > Main measurement parameters

These parameters are relevant for measuring and must be determined for all samples. Most samples can be determined with these parameters.

You can define the following parameters:

Menu item	Explanation	Further Information
Drying program	Defines the drying program which is best suited for the specific sample. (including temperature and switch-off criteria)	see Setting the Drying Program (page 75)
Display mode	Defines the type of value for displaying and printing.	see Settings of Display Mode (page 82)
Start weight	Defines a start weight so that the samples always have approximately the same weight .	see Settings of Start Weight (page 85)

8.1.1 Settings of Drying Program

Navigation: Home > Method Definition > Method name > Main measurement parameters > Drying program

This function offers you different predefined drying programs for optimum matching of the drying characteristics to the sample used. Furthermore, the drying temperature, switch-off criterion as well as the start weight can be set. Carry out a test measurement for determining the appropriate switch-off criterion, if you do not know the behavior of the sample. For carry out a test measurement **see** Testing a Method (page 92).

Note

You will find more information about defining methods in the delivered application brochure «Guide to Moisture Analysis».

You can define the following parameters:

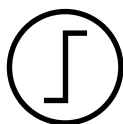
Parameter	Explanation	Values
Drying program	Defines the drying program which is best suited for the specific sample.	Standard* Rapid Gentle Steps

* Factory setting

Further information:

- **Standard** drying **see** Settings of Standard drying (page 75)
- **Rapid** drying **see** Settings of Rapid drying (page 76)
- **Gentle** drying **see** Settings of Gentle drying (page 77)
- **Steps** drying **see** Settings of Step drying (page 79)

8.1.1.1 Settings of Standard drying



Standard drying

This drying program is set in the factory and is suitable for most samples. The sample is heated to the drying temperature.

Factory setting

Drying temperature = 105 °C, **Switch-off criterion** 3 = 1 mg / 50 s

You can define the following parameters:

Parameter	Explanation	Values
Drying temperature	Defines the drying temperature.	40...230 °C (105 °C *)
Switch-off criterion	Defines the criterion when the instrument should end the drying.	1 (1 mg / 10 s) 2 (1 mg / 20 s) 3 (1 mg / 50 s)* 4 (1 mg / 90 s) 5 (1 mg / 140 s) Timed... Free (mg / s)... Free (% / s)...
SOC delay	Activates and defines the delay for the starting time of the switch-off criterion. OFF = no delay applied.	OFF* ON 0...99 min

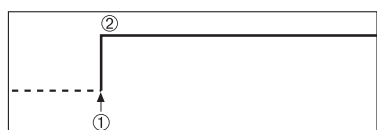
Setting the Drying Temperature

Navigation: Method Definition > Name of method > **Main measurement parameters** > **Drying temperature**

In this menu item you can set the final drying temperature.

Note:

The admissible input range for the drying temperature is indicated.

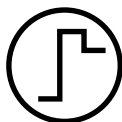


- 1 Start of drying
- 2 Final temperature

Setting Switch-off criterion

see Settings of Switch-off Criterion (SOC) (page 80)

8.1.1.2 Settings of Rapid drying



Rapid drying

This drying program is primarily suitable **for samples with a moisture content over 30%**. Following the start, the selected temperature is exceeded by 40% for 3 minutes (however, is possible up to maximum 230°C) to compensate for the cooling due to vaporization and accelerate the drying process. The drying temperature is then lowered to the set value and maintained.

Factory setting

Drying temperature = 105 °C, **Switch-off criterion** 3 = 1 mg / 50 s

You can define the following parameters:

Parameter	Explanation	Values
Drying temperature	Defines the drying temperature.	40...230 °C (105 °C *)
Switch-off criterion	Defines the criterion when the instrument should end the drying.	1 (1 mg / 10 s) 2 (1 mg / 20 s) 3 (1 mg / 50 s)* 4 (1 mg / 90 s) 5 (1 mg / 140 s) Timed... Free (mg / s)... Free (% / s)...

* Factory setting

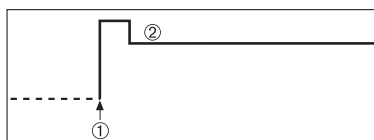
Setting the Drying Temperature

Navigation: Method Definition > Name of method > **Main measurement parameters** > **Drying temperature**

In this menu item you can set the final drying temperature.

Note:

The admissible input range for the drying temperature is indicated.



- 1 Start of drying
- 2 Final temperature

Setting Switch-off criterion

see Settings of Switch-off Criterion (SOC) (page 80)

8.1.1.3 Settings of Gentle drying



Gentle drying

This drying program is suitable for the gentle drying of **substances which tend to form a skin** (e.g. substances containing sugar or volatile substances). With this program the temperature is continuously increased and attains the selected drying temperature **only after the elapse** of the so-called ramp time. The «ramp», i.e. the time that should elapse between the start of drying and attainment of the final temperature can be pre-set.

Factory setting

Drying temperature = 105 °C, **Ramp time** = 3:00 min,

Switch-off criterion 3 = 1 mg / 50 s

You can define the following parameters:

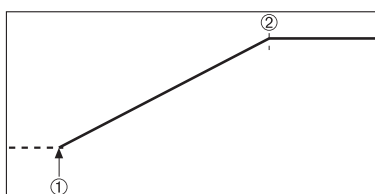
Parameter	Explanation	Values
Drying temperature	Defines the drying temperature.	40...230 °C (105 °C *)
Ramp time	Defines the time in minutes that should elapse between the start of drying and attainment of the final temperature for Gentle drying.	0...480 min
Switch-off criterion	Defines the criterion when the instrument should end the drying.	1 (1 mg / 10 s) 2 (1 mg / 20 s) 3 (1 mg / 50 s)* 4 (1 mg / 90 s) 5 (1 mg / 140 s) Timed... Free (mg / s)... Free (% / s)...

* Factory setting

Setting the Drying Temperature

Navigation: Method Definition > Method name > Main measurement parameters > Drying programGentleDrying temperature

In this menu item you can set the desire final drying temperature.

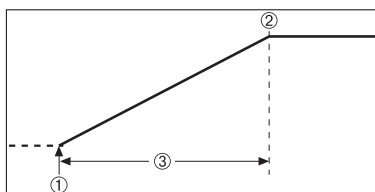


- 1 Start of drying
- 2 Final temperature

Setting the ramp time

Navigation: Method Definition > Method name > Main measurement parameters > Drying programGentle > Ramp time

In this menu item you can define the **Ramp time**, i.e. the time that should elapse between the start of drying and attainment of the final temperature. The ramp is started once the temperature reaches 50 °C.



- 1 Start of drying
- 2 Final temperature
- 3 Ramp time

Setting Switch-off criterion

see Settings of Switch-off Criterion (SOC) (page 80)

8.1.1.4 Settings of Step drying



Step drying

This drying program is suitable for the drying of substances composed of several components which vaporize at different temperatures (e.g. ethereal oils). With this program drying is performed stepwise, i.e. the sample is preheated to a particular temperature (1st step) and kept at this temperature for the selected switch-off criterion e.g. time or weight loss per unit of time. The temperature can be increased for the next step, lowered or remain the same. At the last step, the switch-off criterion terminates the measurement. Maximum 5 steps are selectable.

Step drying can also be used for faster determining of samples with high moisture content. In this case the first step should be set with a higher temperature than for the last step. This reduces evaporation.

Tip for the measurement of plastic granules:

Setting of switch-off criterion "timed" for the first step (possibly same temperature setting as for step 2). The larger sample (e.g. 30 g) has to be heated first before it gives off the small amount of moisture. This prevents that the measurement is terminated prematurely. Setting of switch-off criterion "5" for the second step which terminates the measurement automatically.

Factory setting

Step 1, **Temperature** = 50 °C, **Switch-off criterion** = 5:00 min

Step 2, **Temperature** = 105 °C, **Switch-off criterion** 3 = 1 mg / 50 s

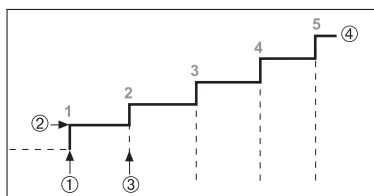
You can define the following parameters:

Parameter	Explanation	Values
Drying steps	Defines the number of steps for Step drying. This dialog includes the setting of temperature and switch-off criterion for each step.	1...5 Steps
Temperature	Defines the drying temperature for each Step.	50...230 °C
Switch-off criterion	Defines the criterion when the instrument should end the drying or should start the next step.	1 (1 mg / 10 s) 2 (1 mg / 20 s) 3 (1 mg / 50 s)* 4 (1 mg / 90 s) 5 (1 mg / 140 s) Timed... Free (mg / s)... Free (% / s)...

* Factory setting

Drying steps

Navigation: Method Definition > Method name > Main measurement parameters > Drying programSteps > Drying steps



- 1 Start of drying
- 2 Step temperature
- 3 Next step, **Switch-off criterion**
- 4 Last step

Setting Switch-off criterion

see Settings of Switch-off Criterion (SOC) (page 80)

8.1.1.5 Settings of Switch-off Criterion (SOC)

Navigation: Home > Method Definition > Method name > Main measurement parameters > Drying program

This function offers you different switch-off criteria. A switch-off criterion defines when the instrument should end the drying or should start the next step ("Step drying"). Switch-off criteria ensure that measurements are finished in the same condition (weight loss per unit of time) every time, ensuring repeatable measurements.

Note

Accurate results require same start weights, **see** Settings of Start Weight (page 85).

You can define the following parameters:

Parameter	Explanation	Values
Switch-off criterion	Defines the criterion when the instrument should end the drying.	1 (1 mg / 10 s) 2 (1 mg / 20 s) 3 (1 mg / 50 s)* 4 (1 mg / 90 s) 5 (1 mg / 140 s) Timed... Free (mg / s)... Free (% / s)...

* Factory setting

- Weight loss per unit of time (5 preprogrammed settings)
- Free switch-off criterion (2 different settings)
- Timed switch-off

Weight loss per unit of time

The switch-off defines in which drying condition, the measurement is completed. This switch-off is based on a weight loss per unit of time. As soon as the mean weight loss is less than a preset value during a specified time, the instrument considers drying as complete and automatically discontinues the measurement process.

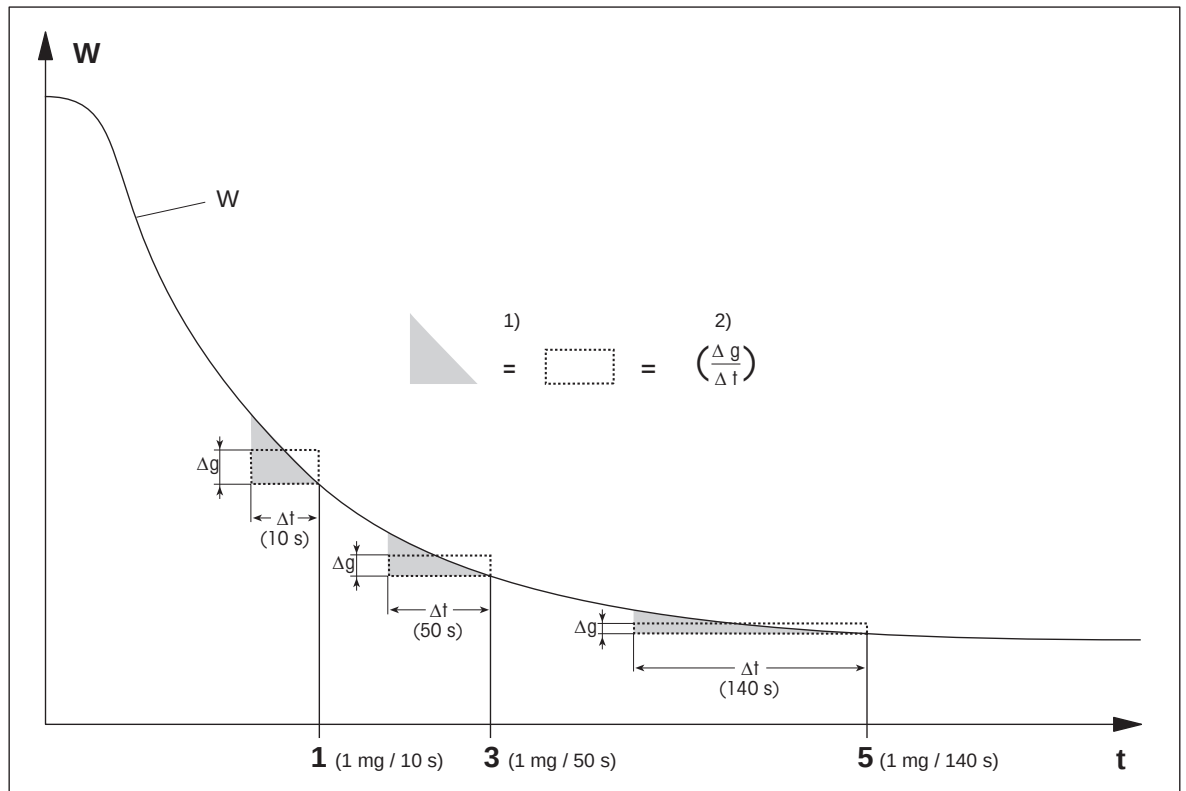
Note

The switch-off criterion is inactive during the first 30 sec.

The following 5 preprogrammed settings are available

Switch-off criterion	Δg	Δt	Description
1	1 mg	10 s	This setting is suitable for fast measurements to determine a trend.
2	1 mg	20 s	This setting is suitable for quick drying samples.
3	1 mg	50 s	This is the factory setting . It is suitable for most kind of samples.
4	1 mg	90 s	This setting is suitable for samples which dry moderately quickly or for samples with higher accuracy requirements.
5	1 mg	140 s	This setting is suitable for samples which dry very slowly (trapped moisture, skin formation) or for samples with very low moisture content e.g. plastics. Not suitable for very temperature sensitive samples.

The graph below exemplifies the mode of operation of the switch-off operation (not to scale).



t = time
 W = weight of sample
 1, 3, 5 = switch-off criteria shown as an example

1) = equal area
 2) = mean weight loss per unit of time

Free switch-off criterion

The free switch-off criterion is based on a user-defined mean weight loss per unit of time.

The following 2 settings are available:

- **Switch-off criterion > Free (mg / s)...** (weight loss per unit of time)
- **Switch-off criterion > Free (% / s)...** (weight loss in percent per unit of time)

Timed switch-off

With this switch-off criterion the measurement lasts until the pre-set drying time has elapsed. The display provides you with continuous information in the drying time.

Performing Test measurements

For performing test measurements to determine a correct switch-off criterion, **see** Testing a Method (page 92).

8.1.1.6 Settings of SOC Delay

Navigation: Home > Method Definition > Method name > Main measurement parameters > SOC delay

With this function you can delay the starting time of the switch-off criterion. This can be useful for samples with low moisture content which need extended time until they evaporate moisture e.g. plastics.

Note

SOC delay is only available for the following drying programs:

- **Standard**
- QuickPredict

You can define the following parameters:

Parameter	Explanation	Values
SOC delay	Activates and defines the delay for the starting time of the switch-off criterion. OFF = no delay applied.	OFF* ON 0...99 min

* Factory setting

8.1.2 Settings of Display Mode

Navigation: Home > Method Definition > Method name > Main measurement parameters > Display Mode

With this function you can select the desired type of result display. You also define what type of values are printed on the records.

You can define the following parameters:

Parameter	Explanation	Values
Display Mode	Defines the type of value for displaying and printing.	%MC* %DC %AM %AD g g/kg MC g/kg DC -%MC

* Factory setting

- **%MC** – Moisture Content (calculated value)
- **%DC** – Dry Content (calculated value)
- **%AM** – ATRO Moisture Content (calculated value)
- **%AD** – ATRO Dry Content (Wet weight, calculated value)
- **g** – Weight in Grams
- **g/kg MC** – Moisture Content (calculated value)
- **g/kg DC** – Dry Content (calculated value)
- **-%MC** – Moisture Content (negative value, calculated value)

Note

Calculated values are indicated with an asterisk in the display.

Detailed information:

%MC – Moisture Content

The moisture content of the sample is displayed (and printed out) as a percentage of the wet weight (WW = initial weight = 100 %). This is the **factory setting**.

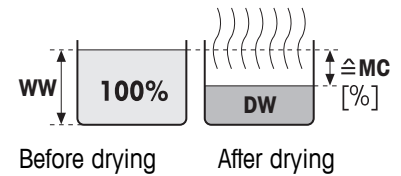
During the measurement the value is constantly displayed in percent and as a graphical drying curve. The measured value is marked by "%MC" (Moisture Content, e.g. 11.35 %MC) also for the printed results.

$$MC = \frac{WW - DW}{WW} \cdot 100 \%$$

MC = Moisture Content [0...100 %]

WW = wet weight

DW = dry weight



%DC – Dry Content

The dry content of the sample is displayed (and printed out) as a percentage of the wet weight (WW = initial weight = 100 %).

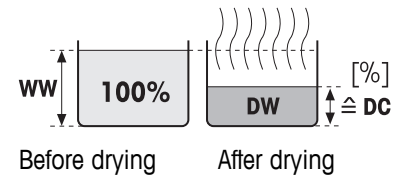
During the measurement the value is constantly displayed in percent and as a graphical drying curve. The measured value is marked by "%DC" (Dry Content, e.g. 88.65 %DC) also for the printed results.

$$DC = \frac{DW}{WW} \cdot 100 \%$$

DC = dry content [100...0 %]

WW = wet weight

DW = dry weight



%AM – ATRO Moisture Content ¹⁾

The moisture content of the sample is displayed (and printed out) as a percentage of the dry weight (DW = final weight = 100 %)

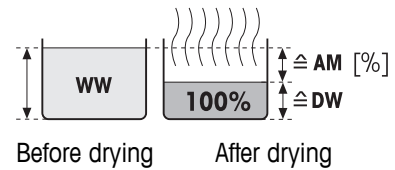
During the measurement the value is constantly displayed in percent and as a graphical drying curve. The measured value is marked by "%AM" (ATRO Moisture Content, e.g. 255.33 %AM) also for the printed results.

$$AM = \frac{WW - DW}{DW} \cdot 100 \%$$

AM = ATRO moisture content [0...1000 %]

WW = wet weight

DW = dry weight



%AD – ATRO Dry Content (Wet weight) ¹⁾

The wet weight of the sample is displayed (and printed out) as a percentage of the dry weight (DW = final weight = 100 %)

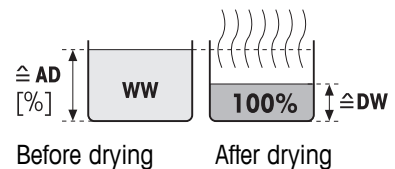
During the measurement the value is constantly displayed in percent and as a graphical drying curve. The measured value is marked by "%AD" (ATRO Dry Content, e.g. 312.56 %AD) also for the printed results..

$$AD = \frac{WW}{DW} \cdot 100 \%$$

AD = ATRO dry content [100...1000 %]

WW = wet weight

DW = dry weight



¹⁾ Comment on the ATRO display mode

If the current measured value in the ATRO display mode is greater or less than the predefined limit value (i.e. greater than 999.99 %AD or less than -999.99 %AM), the ATRO result values are limited to 999.99%.

g – Weight in Grams

The weight of the sample is displayed (and printed out) in grams. With this setting, the Moisture Analyzer is used as a precision balance.

During the measurement current weight is constantly displayed in grams.

g/kg MC – Moisture Content

The moisture content of the sample is displayed (and printed out) in g/kg of the wet weight (WW = initial weight = 1000 g/kg).

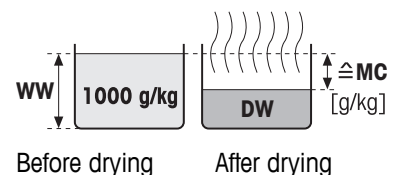
During the measurement the current measured value is constantly displayed in percent and as a graphical drying curve. The measured value is marked by "g/kg MC" (Moisture Content, e.g. 11.35 g/kg MC) also for the printed results.

$$MC = \frac{WW - DW}{WW} \cdot 1000 \text{ g/kg}$$

MC = Moisture Content [0...1000 g/kg]

WW = wet weight

DW = dry weight



g/kg DC – Dry Content

The dry content of the sample is displayed (and printed out) in g/kg of the wet weight (WW = initial weight = 1000 g/kg).

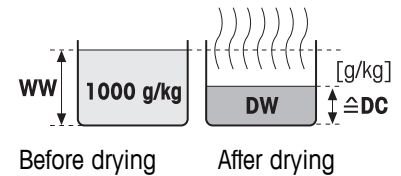
During the measurement the current measured value is constantly displayed in percent and as a graphical drying curve. The measured value is marked by "g/kg DC" (Dry Content, e.g. 88.65 g/kg DC) also for the printed results.

$$DC = \frac{DW}{WW} \cdot 1000 \text{ g/kg}$$

DC = dry content [1000...0 g/kg]

WW = wet weight

DW = dry weight



-%MC – Moisture Content

The moisture content of the sample is displayed (and printed out) as a percentage of the wet weight (WW = initial weight = 100 %).

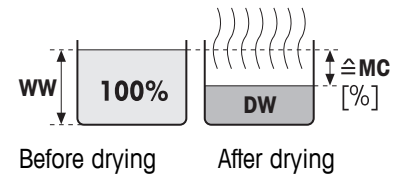
During the measurement the value is constantly displayed in percent and as a graphical drying curve. The measured value is marked by "-%MC" (Moisture Content, e.g. -11.35 -%MC) also for the printed results and shown as a negative value.

$$MC = - \frac{WW - DW}{WW} \cdot 100 \%$$

MC = Moisture Content [0...100 %]

WW = wet weight

DW = dry weight



8.1.3 Settings of Start Weight

Navigation: Home > Method Definition > Method name > Main measurement parameters > Start weight

The start weight affects both the duration of the measurement as well as the accuracy of the results. A small weight results in a shorter duration of the measurement but will decrease the accuracy of the result. With this function you can define a start weight so that the samples are approximately of equal weight thus improving the repeatability of the measurements. The weighing-in aid assists you in weighing-in the sample. At most of the samples the target weight is in the range of 2–5 g (plastics 30 g). We recommend to cover the whole surface of the sample pan with a thin uniform layer of sample.

You can define the following parameters:

Parameter	Explanation	Values
Start weight	Defines a start weight so that the samples always have approximately the same weight .	OFF* ON (0.100...200.000 g)
Start weight tolerance	Defines the tolerance of the Start weight . 10 %, recommended for sample weights up to 5 g.	1...25 % (10 %)*
Weigh-in aid	Defines monitoring of target weight and tolerance. Passive = tolerance is displayed. Active = tolerance is monitored. If the startweight is out of tolerance, the measurement cannot be started.	Passive* Active

* Factory setting

8.2 Result and Value Handling

Navigation: Home > Method Definition > Method name > Result and value handling

You can define the following parameters:

Menu item	Explanation	Further Information
Control limits	Defines the acceptable range for measuring results in the unit of the selected display mode.	see Control limits (page 86)
Resolution	Defines the readability of the built-in analytical balance.	see Resolution (page 87)
Free factor	Defines a method-specific multiplication factor to calculate a corrected final result.	see Free Factor (page 88)

8.2.1 Control limits

Navigation: Home > Method Definition > Method name > Result and value handling > Control limits

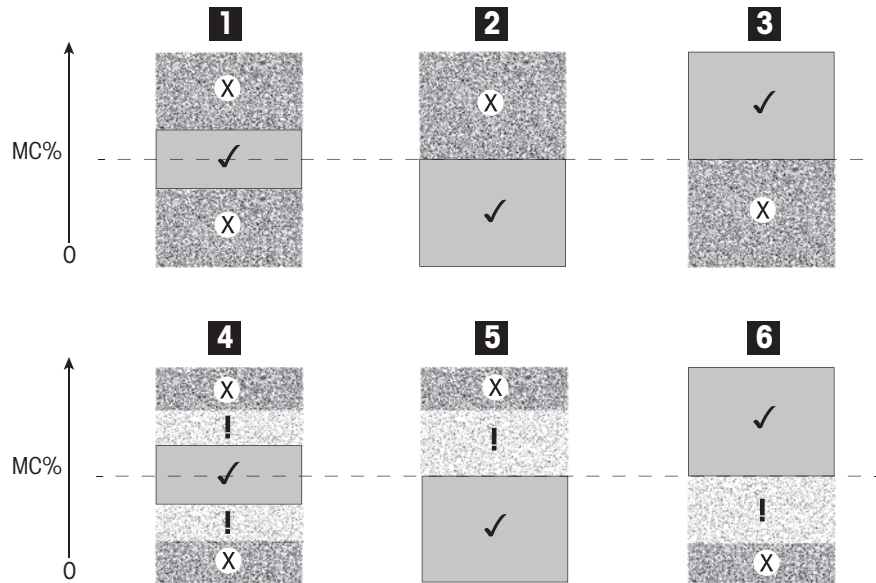
With this function you can set warning and intervention limits for the quality and process monitoring. You can also set limits for distinguish passed or failed.

You can define the following parameters:

Parameter	Explanation	Values
Control limits	Defines the acceptable range for the measuring results in the unit of the selected display mode. OFF = no limits applied.	OFF* ON (0.00...100.00)
T1+ (Upper warning limit)	Defines the upper warning limit in the selected unit of the display mode. Measurement passed with warning (!). Value range in ATRO mode (%AD, %AM) up to 1000.	OFF* 0.01...100.0 (1000)
T1- (Lower warning limit)	Defines the lower warning limit in the unit of the selected display mode. Measurement passed with warning (!). Value range in ATRO mode (%AD, %AM) up to 1000.	OFF* 0.01...100.0 (1000)
T2+ (Upper intervention limit)	Defines the upper intervention limit in the unit of the selected display mode. Measurement failed (X). Value range in ATRO mode (%AD, %AM) up to 1000.	OFF* 0.01...100.0 (1000)
T2- (Lower intervention limit)	Defines the lower intervention limit in the unit of the selected display mode. Measurement failed (X). Value range in ATRO mode (%AD, %AM) up to 1000.	OFF* 0.01...100.0 (1000)

* Factory setting

Setting the limits



Limits		1	2	3	4	5	6
T2+ (Upper intervention limit)	(X)			Off			Off
T1+ (Upper warning limit)	(!)	Off	Off	Off			Off
T1- (Lower warning limit)	(!)	Off	Off	Off		Off	
T2- (Lower intervention limit)	(X)		Off			Off	

(✓) Measurement is within the warning limits: passed (displayed in green)

(!) Measurement is between the warning limits and the intervention limits: passed with warning (displayed in yellow)

(X) Measurement is outside the intervention limit: failed (displayed in red)

Control limits is off = no limits applied (displayed in blue)

8.2.2 Resolution

Navigation: Home > Method Definition > Method name > Result and value handling > Resolution

Traditionally, moisture analyzers are operated at a maximum resolution of 1 mg / 0.01 %. The following setting allows you to determine moisture content using the resolution of the built-in analytical balance, which is ten times higher (0.1 mg weight / 0.001 % moisture). The high resolution is especially suitable for samples with low moisture content (lower than 1 %). Careful sample preparation is extremely important for these kind of samples!

Note

High resolution is not available for the display mode **ATRO moisture content** or **ATRO dry content**, see Settings of Display Mode (page 82).

You can define the following parameters:

Parameter	Explanation	Values
Resolution	Defines the readability of the built-in analytical balance.	Standard* High

* Factory setting

8.2.3 Free Factor

Navigation: Home > Method Definition > Method name > Result and value handling > Free factor

This function allows you to multiply the result by a method-specific factor in the selected display mode. Furthermore, it allows you changing the result by an offset entry in the unit of the selected display mode. It is used to calculate a corrected final result (e.g. in order to compensate for systematic deviations from the reference result).

Note

- Free factor is not available in display mode "g" (weight).
- The calculation is carried out continuously during the measurement and displayed with the graphical drying curve.
- With free factor, the results are calculated and marked with an asterisk in the display.

You can define the following parameters:

Parameter	Explanation	Values
Free factor	Defines the settings for free factor. OFF = no free factor applied.	OFF* ON
Factor	Defines the multiplication factor.	-10.000...+10.000 (1.000)*
Offset	Defines the offset value in the selected display mode.	-1000.000... +1000.000 (0.000)*
Format	Defines the number of decimal places of the calculated final results, for displaying and printing.	x x.x x .xx x.xxx*

* Factory setting

8.3 Workflow Handling

Navigation: Home > Method Definition > Method name > Workflow handling

With this functions you can define the workflow for the measurement

You can define the following parameters:

Menu item	Explanation	Further Information
Starting mode	Defines how the sample chamber is operated.	see Starting Mode (page 89)
Preheating	Defines the settings for preheating the sample chamber before starting the measurement.	see Preheating (page 89)

8.3.1 Starting Mode

Navigation: Home > Method Definition > Method name > Workflow handling > Starting mode

In this menu item you can choose whether the sample chamber should operate automatically or manually (e.g. tare, stop drying).

You can define the following parameters:

Parameter	Explanation	Values
Starting mode	Defines how the sample chamber is operated.	Automatic* Manual

* Factory setting

Automatic

Your instrument is set in the factory to the automatic operating mode. This mode can be used for most kind of samples. When closing the sample chamber, the weight of the sample is recorded and the measurement is started.

Manual

We advise you to use the manual operating mode for samples which contain readily volatile substances. In contrast to the automatic operating mode, in the manual operating mode the sample chamber does not automatically close when the **[Start drying]** button is tapped. However, the initial weight (wet weight) important for the determination of the moisture content is recorded. In the manual operating mode you have time for further preparation of the sample (e.g. mixing with quartz sand or even distribution of the sample) while weight losses due to evaporation during the preparation time are measured from the outset. As soon as the sample is ready for drying, press **[↑]** key. The automatic sample chamber closes and drying starts. In the manual operating mode you can open the sample chamber during a drying operation. In contrast to the automatic mode, drying will not be stopped but simply interrupted until the automatic sample chamber is closed again.

8.3.2 Preheating

Navigation: Home > Method Definition > Method name > Workflow handling > Preheating

If necessary, you can activate preheating function to preheat the sample chamber. Because of the short heat-up times with halogen moisture analyzers, no preheating is usually required for standard applications. A preheated instrument can improve the reproducibility and accuracy of the results, since the instrument is in the same climatic state with every measurement. Preheating will only executed until time of measurement (**Manual**), in standby mode even after the measurement.

You can define the following parameters:

Parameter	Explanation	Values
Preheating	Defines the settings for preheating the sample chamber (standby temperature) before starting the measurement. OFF = no preheating applied.	OFF* ON Manual Standby

* Factory setting

Manual

Before a measurement can be started, the instrument prompts the user to activate the function preheating. The instrument indicates, when it is enough preheated to start the measurement. The measurement can also be performed during the preheating phase by tapping **Stop preheating**. In this case, the duration of preheating will be recorded. After a maximal preheating time of 1 hour, preheat will be off.

You can define the following parameters:

Parameter	Explanation	Values
Temperature	Defines the standby temperature.	40...100 °C (40 °C)*

* Factory setting

Standby

The instrument indicates, when it is enough preheated to start the measurement. The measurement can also be performed during the preheating phase by tapping **Stop drying**. In this case, the duration of preheating will be recorded.

The function **Duration** allows you to deactivate preheating at a preselected time period after measurement. The feature **Turn-off time** allows to disable preheating at a defined clock time.

You can define the following parameters:

Parameter	Explanation	Values
Temperature	Defines the standby temperature.	40...100 °C (40 °C)*
Duration	Defines the time which the instrument is maintained at standby temperature for a limited amount of time after measurement (time out).	Hours Minutes (3 h)*
Turn-off time	Defines the clock time at which Preheating will be off.	OFF* Hours : Minutes

* Factory setting

8.4 General Method Properties

Navigation: Home > Method Definition > Method name > General method properties

8.4.1 Method name

Navigation: Home > Method Definition > Method name > General method properties > Method name

You can define the following parameters:

Parameter	Explanation	Values
Method name	Renaming a method. Name should be unique and unambiguous.	any

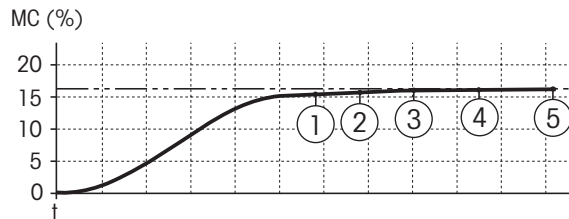
8.5 Testing Methods

8.5.1 Testing a Method

Test a method

This function allows you to test the settings during the definition phase of a method. Testing is possible at any time. Test measurements are recorded in the journal and marked as test results. If you wish to work with the switch-off criterion «weight loss per unit of time» and are not familiar with the behavior of the sample, the test measurement will help you select the suitable setting. For more information about switch-off criteria, **see** Settings of Switch-off Criterion (SOC) (page 80).

The diagram illustrates the progress of a drying. The points at which the individual switch-off criteria were reached (1-5) are marked.



During this measurement, the measured values are recorded the switch-off criteria 1 to 5 and maybe one of a free-defined switch-off criterion. The measurement terminates after the defined time (default 30 minutes). Note that the start weight affects the switch-off criterion. The test time can also be switched off. In this case the measurement terminates after reaching the predefined switch-off criterion. Nevertheless, all switch-off criteria are recorded.

By comparing the measured values with values of a reference method e.g. drying oven and the standard deviation, you can define the appropriate parameter settings. For more information see the application brochure «Guide to Moisture Analysis».

Carry out all measurements under **Home > Method Definition > Method name > Tests** before releasing a method. The results of test measurements are specifically marked.

Attention

Before releasing the method, check if the determined switch-off criterion is set.

Performing a Test measurement

You perform a test measurement just like any other measurement. How to perform the measurement **see** Measurement (page 94).

Changing the test time

- ▶ Work screen for testing is displayed.

1 Tap **Test time** in the work screen.

⇒ **Test time in min** appears.

2 Confirm with **OK**.

Displaying the parameter

- ▶ Work screen for testing is displayed.

– Tap the parameter in the work screen.

⇒ The parameter list appears.

Printing results of the determined switch-off criteria

For a printout example, **See** Information on Printouts (page 101) section «Special events».

Displaying results of the determined switch-off criteria

See Graphical Evaluations of Measurement Results (page 99)

8.5.2 Testing with AutoMet

AutoMet test measurement helps you to save time when developing methods. It also enables you to obtain a result with your Moisture Analyzer that corresponds most closely to the reference value for the chosen temperature and sample quantity.

With this function you can use the **AutoMet** button to enter the reference value and the instrument automatically establishes an appropriate switch-off criterion for the defined method. This will be the point at which the measurement most accurately matches the reference value. This is the free switch-off criterion **Free (mg / s)...**. The AutoMet test measurement only proposes the switch-off criterion **Free (mg / s)...** if the drying trend means you can expect a precise moisture determination. The range for the switch-off criterion **Free (mg / s)...** is therefore limited to between 1 mg/20 seconds and 1 mg/140 seconds.

Note

If the target value is reached but the switch-off criterion is still outside 1 mg/20-140 seconds, you have not selected the optimum temperature. If **Free (mg / s)...** is below 1 mg/20 seconds, reduce the temperature. If **Free (mg / s)...** is above 1 mg/140 seconds, either switch-off criterion 5 will provide a good approximation or it might be a good idea to increase the drying temperature. If the target value is not reached, increasing the drying temperature may obtain the desired result.

Performing a test with AutoMet

- 1 Perform a test measurement with the same sample as for the reference measurement.
- 2 Tap [**AutoMet**].
- 3 Enter Measurement reference in display unit (target value) and tap [**OK**].
 - ⇒ **AutoMet** suggests you an appropriate switch-off criterion for the temperature selected or gives further instructions on the display.
- 4 Tap [**Save**] to assume this switch-off criterion into the method.

9 Measurement

This function offers you to perform a measuring using previously defined methods. After choosing the measurement method, the measurement process can be started. The work screen guides you step by step through the measuring process.

The following functions are available:

- On the work screen, you can create a shortcut. This allows you to start a method directly from the home screen.
See Using Shortcuts (page 97).
- After completion of a measuring you can perform a further measuring with same method by tapping [**Next Sample**] or with an other method by tapping [**Measurement**].
- For graphical evaluations of measurement results tap [**Results**]. If you have defined the control limits in the method, passed, warned or failed status is shown.
See Results (page 99)
- A detailed overview of the method parameter can you call up by tapping the parameter panel.
See Work Screen (page 22).

See also

- Using Shortcuts (page 97)
- Results (page 99)
- Work Screen (page 22)

9.1 Performing a Measurement

You are now familiar with all the parameters of your instrument and have defined all values for your sample. The instrument is now ready for the determination of your samples. In this Section you will learn how to perform measurements and how to stop the measurement process.

Switching on

- ▶ The instrument must be connected to the power supply for approximately 60 minutes in order to reach operating conditions.
- 1 To switch on the Instrument, press [ϕ].
 - 2 Log-in with your password, if needed.



Selecting method for measurement

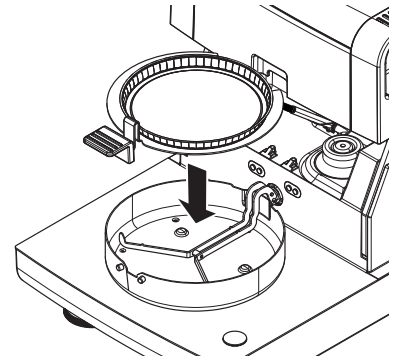
- 1 Tap **Measurement**.
 - ⇒ The methods list is displayed.
- 2 Select the method to determine your sample.
 - ⇒ The work screen of the method appears.
 - ⇒ The sample chamber opens automatically.

Placing sample pan

- ▶ The display now prompts you to load the empty sample pan and tare the balance.
- 1 Place the empty sample pan in the sample pan handler.
- 2 Place the sample pan handler in the sample chamber. Ensure that the tongue of the pan handler fits exactly in the slot of the draft shield element. The sample pan must lie flat in the pan holder.

Note

We advise you to work with the sample pan handler at all times. The pan handler is ergonomic, automatic positioning, safe and provides protection against possible burns due to the hot sample pan.



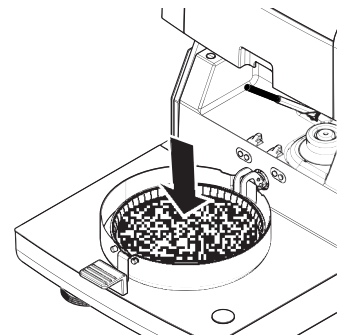
Taring balance

- Tap [->0/T<-].
 - ⇒ The sample chamber closes automatically for taring.
 - ⇒ When taring is completed, the sample chamber opens automatically.



Starting measurement

- ▶ After taring, the display prompts you to add the sample to the sample pan.
- 1 Add the sample to the sample pan. If you have defined a start weight, weigh the sample using the weighing-in aid.
- 2 Tap [Start drying].
 - ⇒ The sample chamber closes automatically.
 - ⇒ The drying process starts automatically.



Drying process

You can follow the measurement process on the display, **see** Work Screen (page 22).

- The drying process is continuously displayed graphically.
- The current temperature in the heating module is displayed as well as the elapsed drying time and the current drying value.
- The display shows the selected settings.
- The drying process can be aborted by tapping [Stop drying].

At the end of the drying process you can read the moisture content of your sample on the display. If you have defined the control limits in the method, passed, warned or failed status is shown.

Removing Sample

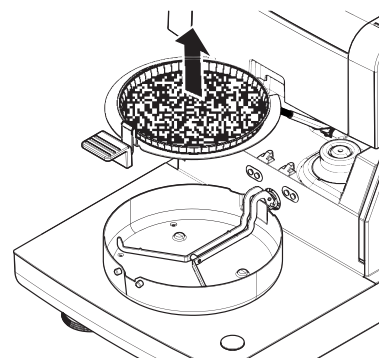


CAUTION

Danger of burns

Sample, sample pan and sample pan holder may still be hot.

- ▶ The drying process is completed.
 - ▶ The sample chamber is open (opens automatically).
- 1 Carefully remove the sample pan handler from the sample chamber.
Note
 To remove the sample pan from the handler, lift the pan slightly from below and remove it from the handler.
 - 2
 - To perform a further measurement with the current method, tap **[Next Sample]**.
 - To perform a measurement with a new method, tap **[Measurement]**.
 - To return to the home screen, press **[↶]**.



Stop drying

When you stop the measuring process by tapping **[Stop drying]**, you can choose between two possibilities:

- **Abort without saving**
 Abort without saving of any data collected until now.
- **Abort measurement and save data**
 Already sampled data are stored and an entry will be made in the results. The result is marked as aborted.

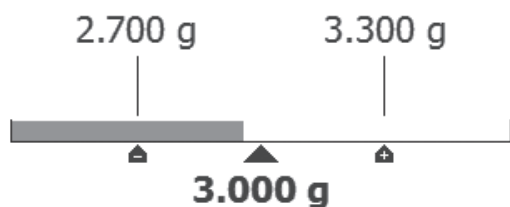
Adding a comment

At the end of a measurement you can add a comment to the result of the measurement. This comment is stored in the measurement result and can be printed. A comment can only be entered before leaving the current measurement.

- 1 To create a comment, tap **[Note]**.
 ⇒ The keyboard dialog appears.
- 2 Enter a comment.
- 3 Confirm with **[OK]**.

9.2 Work with Weighing-in Aid

The weighing-in aid can be defined for each method and facilitates weighing in of the sample to a target value. This is particularly necessary if you require all the samples of a method you wish to process to have same weight in order to improve the repeatability of the measurement results. In addition, the weighing-in aid can be set as active so that the drying process cannot be started if the weight of sample is outside the set tolerance. You are therefore compelled to weigh-in the correct quantity of sample. If all sample weighed are within tolerances then this will improve the repeatability. The weighing-in aid is available only if start weight has been activated. For further information **see** Settings of Start Weight (page 85).



Icon	Function
	Lower weight limit (tolerance range)
	Target weight

Icon	Function
	Upper weight limit (tolerance range)

9.3 Using Shortcuts

Navigation for managing shortcuts: **Home > Measurement > Method name > [⌂]**

Shortcuts allow you to start methods directly from the home screen. Shortcuts are user-specific, i.e. each individual user can have his own shortcuts for the most common tasks.

You can define the following parameters:

Parameter	Explanation	Values
Shortcut name	Defines a name for the shortcut. It is recommended to choose a short name because only approx. 8 characters are displayed on the shortcut. Name should be unique and unambiguous.	any
Method name	Shows the name of the method. The name of the method is the target of the shortcut and can not be changed.	none

Defining a shortcut

- ▶ Menu **Measurement** is activated.
 - ▶ The method is selected.
 - 1 Tap [⌂].
 - ⇒ Window **My shortcuts** appears.
 - 2 Tap **Add a shortcut to my home for this method....** (To abort, tap [X].)
 - ⇒ **New shortcut** appears.
 - 3 To enter a name for the shortcut if needed, tap **Shortcut name**.
- Note**
It is recommended to choose a short name because only approx. 8 characters are displayed on the shortcut. Name should be unique and unambiguous.
- 4 Confirm with [OK].
 - 5 To store the setting, tap [Save].
To abort, tap [Cancel].
 - ⇒ The shortcut is added to the home screen.

Editing a shortcut

- ▶ The method is selected.
- 1 Tap [⌂].
 - ⇒ Window **My shortcuts** appears.
- 2 Tap **Edit this shortcut....** (To abort tap [X].)
 - ⇒ **Shortcut properties** appears.
- 3 To edit the name of the shortcut, tap **Shortcut name**.
- 4 Confirm with [OK].
- 5 To store the setting, tap [Save].
To abort, tap [Cancel].

Removing a shortcut

- ▶ The method is selected.
- 1 Tap [⌂].
 - ⇒ Window **My shortcuts** appears.

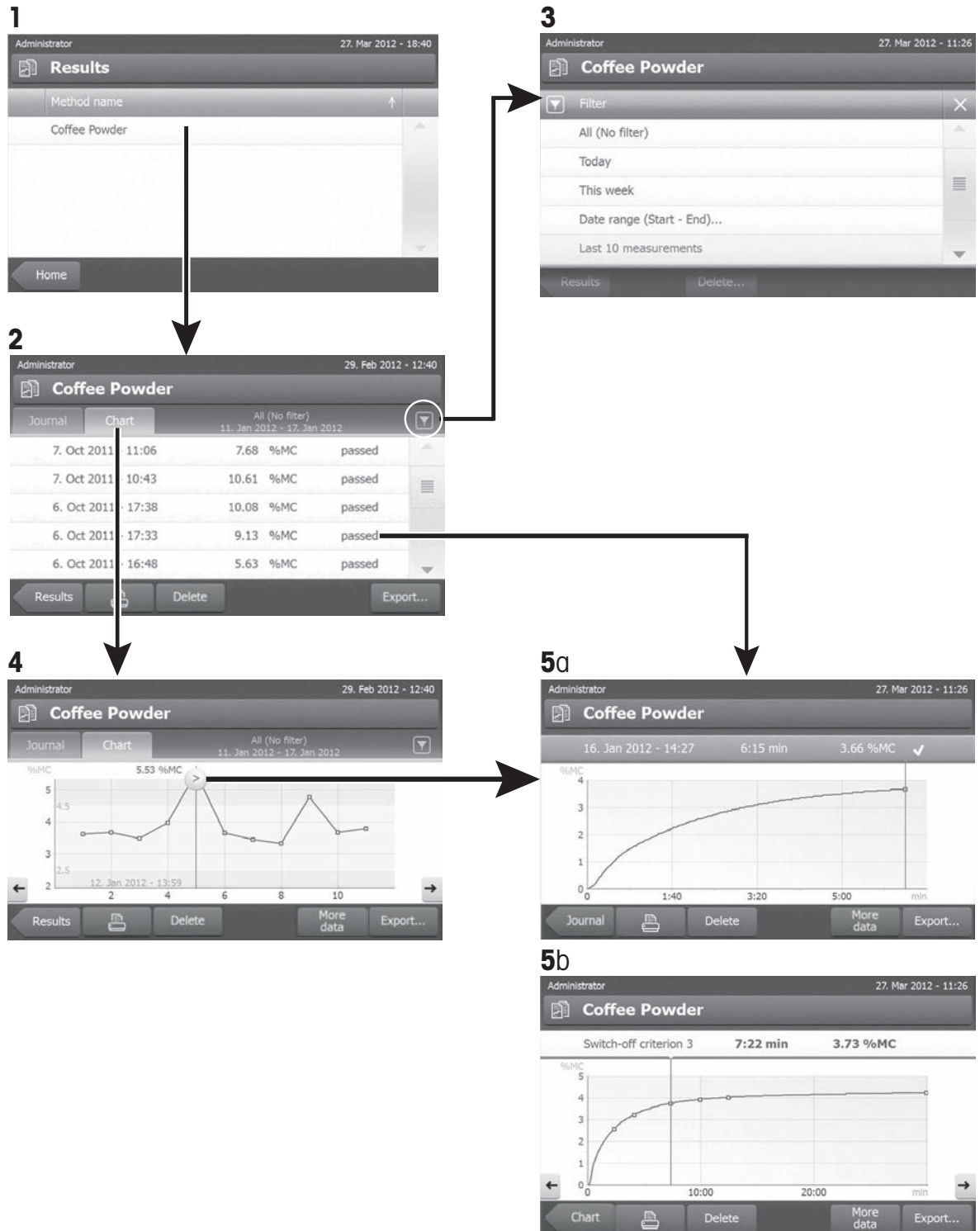
- 2 Tap **Edit this shortcut...** (To abort, tap [**X**].)
⇒ **Shortcut properties** appears.
- 3 To remove the shortcut, tap [**Remove**].
⇒ A message window appears.
- 4 Confirm with [**Remove**]. (To abort, tap [**Cancel**].)
⇒ The shortcut is removed from the home screen.

10 Results

10.1 Graphical Evaluations of Measurement Results

Navigation: Home > Results

This function allows you to manage and evaluate measurement results.



1 Method list

- Tap the method you want to evaluate.
 - ⇒ The journal view appears.

2 Journal view

The journal view allows you to start different graphical evaluations from a measurement series. You can perform the following functions:

- To start **filter function**, tap [▼].
 - ⇒ The filter menu appears.
- To start **chart view**, tap [Chart].
 - ⇒ The chart view of the series of measurement appears (4).
- To start **graphic view**, tap a measurement result.
 - ⇒ The graphic view appears (5).

3 Filter menu

The filter function allows you to evaluate the measurement series according to various criteria. You can select the following criteria:

- **All (No filter)**
- **Today**
- **This week**
- **Date range (Begin - End)**
- **Last 10 measurements / tests***
- **Last 20 measurements / tests**

- Tap [X] to close filter menu.

* Factory setting

4 Chart view

This view allows you to display the results of a series of measurements depending on the filter settings. If you have defined the control limits in the method, they are diagrammed in the results.

The  marked measuring point is displayed with date, time, and the measurement result. You can perform following functions:

- To jump forward to the next measurement result, tap [->].
- To jump backwards to the previous measurement result, tap [<-]. Alternatively, tap directly on the desired measurement point.
- To recall the full measurement curve of the corresponding measurement, tap [].

5a Graphics view

With this function you can display a graphical view of the detailed results of a single measurement. If you have defined the control limits in the method, passed, warned or failed status is shown, **see** Control limits (page 86).

You can perform the following functions:

Printing result

- To print the result, tap [].

Exporting result

- To export the result, tap [Export].

See Exporting Results (page 103).

Deleting result

- To delete this measurement result, tap [**Delete**] (depending on user rights).

With the auto-delete functionality the oldest results will automatically be removed, **see** Result Management (page 57).

More data

- To display more measurement data, tap [**More data**]

5b Displaying results of test measurements

For test measurements **see** Testing a Method (page 92).

This view allows you to display the results of the determined switch-off criteria. Each criterion is marked as a measuring point and displayed with its result. You can perform following functions:

- To jump forward to the next measurement result, tap [->].
- To jump backwards to the previous measurement result, tap [<-]. Alternatively, tap directly on the desired measurement point.

10.2 Information on Printouts

The illustrations in this chapter show examples of a standard measurement printout (factory setting) and a short measurement printout. The level of detail on the printouts depends on the selected settings in the menu.

Structure of the standard printout type

```

-MOISTURE DETERMINATION-

METTLER TOLEDO
Halogen MoistureAnalyzer

Type                HX204
SNR(Drying Unit)    1234567890
SNR(Terminal)       0987654321
SW(Drying Unit)     2.00
SW(Terminal)        2.00

User name  Administrator

Weight Adjustment
           03.08.2013 15:18
Temperature Adjustment
           18.04.2013 08:09

Method Name      NEG
Drying Prog      Standard
Drying Temp      105 °C
Switch Off       2:00 min
Display mode     %MC
Start Weight     OFF
Control Limits   OFF
Resolution       Standard
Start Mode       Automatic
Preheating       OFF

Start Weight     6.336 g
Total Time       2:00 min
Dry Weight       6.199 g

Moisture Content
                0.137 g

End Result       2.16 %MC
Note

Signature:

.....

      23.09.2013    12:01

----- END -----

```

Structure of the short printout type

```

-MOISTURE DETERMINATION-

METTLER TOLEDO
Halogen MoistureAnalyzer

Type                HX204
SNR(Drying Unit)    1234567890
SNR(Terminal)       0987654321
SW(Drying Unit)     2.00
SW(Terminal)        2.00

Method Name:        NEG
Drying Prog         Standard
Drying Temp         105 °C
Switch Off          2:00 min

Start Weight        6.336 g
Total Time          2:00 min
End Result          2.16 %MC
ID
Note

Signature:

.....

      23.09.2013    12:01

----- END -----

```

Special events

The **sample chamber was opened and closed** during drying process. When the sample chamber is open, drying is interrupted and then continued when the chamber is closed.

01:00 min	2.26 %MC
01:20 min	opened
01:28 min	closed
02:00 min	3.49 %MC

The **drying process has been aborted** by tapping **[Stop drying]** and the measurement result at the time of the abort is not recorded as it could be wrong.

01:00 min	2.26 %MC
02:00 min	3.49 %MC
ABORTED	
14.10.2011	12:01

Printing the test measurement

During the test measurement a record is printed which explains exactly when and with what measurement result each switch-off criterion was reached.

Switch-off crit. 3	
Time	01:21 min
Dry weight	3.385 g
Moisture content	0.53 %MC

10.3 Exporting Results

Results can be exported to an external storage device e.g. memory stick as well via network to a server. Data in CSV format can be imported into e.g. MS Excel for further evaluations.

Note

Results cannot be imported in a Moisture Analyzer.

Sample procedure

- ▶ Result is activated.
- ▶ An external storage device is connected e.g. memory stick.
- 1 To start, tap **[Export]**
 - ⇒ **Export results** appears.
- 2 Tap Export target > Peripheral
- 3 Tap **Location > Browse and select.**
 - ⇒ **Location** appears.
- 4 Select location for file and confirm with **[OK]**.
- 5 Enter new file name, if needed.
- 6 To start export, tap **[Export]**.

10.3.1 Single Result Export

You can define the following parameters:

Parameter	Explanation	Values
File name	Defines the name of the new data file.	any
File type	Defines the type of the data file.	CSV* PDF

Language	Defines the language for export files.	English Deutsch Italiano Français Español Português Polski Magyar Český Русский 日本語 中文 한국어/조선말
Export target	Defines the destination of data transfer.	Peripheral Network (FTP)

Peripheral

Parameter	Explanation	Values
Location	Finds the location on the storage device.	Browse and select

* Factory setting

Network (FTP)

For data transfer over a network (e.g. Ethernet, WLAN) the File Transfer Protocol FTP is used. The receiving server is defined by entry of the appropriate IP address. If necessary, contact the network administrator of your IT department or your IT support.

Parameter	Explanation	Values
IP address	Defines the IP address of the receiving server.	any
Port	Defines the Port for communication with the server. Note Usually the default value is used.	21* any
Password	Entry for the password needed for server login.	any
User name	Entry for the user name needed for server login.	any

* Factory setting

10.3.2 Multiple Result Export

- **Summary export**
 - The summary export contains method parameters and the end results per measurement. The intermediate values are not exported.
 - If the method has step drying, the summary export also includes the results per step.
 - If the test measurement was executed, then the summary export also includes the results for each switch-off criteria reached during the test measurement.
- **Multiple export**
 - The multiple export contains method parameters, method data, intermediate values and end results per measurement.
 - Each measurement is exported as a single file.

You can define the following parameters:

Parameter	Explanation	Values
File name prefix	Defines the name of the new data file. Date and time are automatically added by the system.	any
File type	Defines the type of the data file.	CSV* PDF

Language	Defines the language for export files.	English Deutsch Italiano Français Español Português Polski Magyar Český Русский 日本語 中文 한국어/조선말
Export selection	Custom selection of multiple results for the export from selected measurements. The selection can be made via the filter function.	Selection: ☐ ☒
Export content	Defines the content of the multiple results export. Summary export (one file, no intermediate values): Results as summary report in one file. Multiple export (one file per measurement): Results with intermediate measurement values as separate files with the same layout as for the single result export.	Summary export Multiple export
Export target	Defines the destination of data transfer.	Peripheral Network (FTP)

Peripheral

Parameter	Explanation	Values
Location	Finds the location on the storage device.	Browse and select

* Factory setting

Network (FTP)

For data transfer over a network (e.g. Ethernet, WLAN) the File Transfer Protocol FTP is used. The receiving server is defined by entry of the appropriate IP address. If necessary, contact the network administrator of your IT department or your IT support.

Parameter	Explanation	Values
IP address	Defines the IP address of the receiving server.	any
Port	Defines the Port for communication with the server. Note Usually the default value is used.	21* any
Password	Entry for the password needed for server login.	any
User name	Entry for the user name needed for server login.	any

* Factory setting

10.4 Information on Exports

The illustrations in this chapter show examples for the level of details of a PDF single result export and a PDF summary result export.

METTLER TOLEDO
Method
 Measured Values & Drying Curve

Moisture Determination
Standard

End Results
 %MC

Summary report
 17 Feb 2012 / 12-46 : 8 Apr 2013 / 09:52

Sample	%MC
1	0
2	10
3	10
4	0
5	0
6	100

Measured data	1	2	3	4	5
Date	8 Apr 2013	20 Jan 2013	9 Jan 2013	3 Jul 2012	17 Feb 2012
Time	09:52	10:15	11:12	16:00	24:00
Batch (ID01)					TEST
Samples (ID02)					
ID3					
ID4					
End result measured %MC		0.11	12.86	12.32	0.15
Status					
Duration min		00:46 min	11:10 min	10:44 min	00:42 min
Start weight g		11.110	12.494	12.513	5.886
Inclonometer		levelled	levelled	levelled	levelled
Username		Administrator	Administrator	Administrator	Administrator

Measured data	1	2	3	4	5
Date	17 Feb 2012				
Time	12:46				
Batch (ID01)					
Samples (ID02)					
ID3					
ID4					
End result measured %MC		100.00			
Status					
Duration min		03:59 min			
Start weight g		0.361			
Inclonometer					
Username					

Comment

Date / Signature

Review Date / Signature

Page 1/2

print 12 Aug 2013 / 05:56

Version 1.00 (2012_03)

11 Maintenance



WARNING

Risk of electric shock

- The instrument must be disconnected from the power supply, before cleaning or other maintenance work to be performed.
 - Use only the power cord from METTLER TOLEDO, if it needs replacing.
-

Note

- The thermal overload protection can not be reset by the user.
- The halogen lamp can not be replaced by the user.

In such cases, contact your METTLER TOLEDO representative.

11.1 Cleaning



CAUTION

Danger of burns

The interior parts of the heating module as well as the parts in the sample chamber could be very hot.

- Wait until the heating module has cooled down completely.
-

To obtain precise measurement results, we recommend you to clean the temperature sensor and the protective glass of the halogen lamp regularly. Please note the following directions for cleaning your instrument.

General

Your Moisture Analyzer is made from high quality, resistant materials and can therefore be cleaned with a commercially available, mild cleaning agent e.g. isopropanol.

Attention

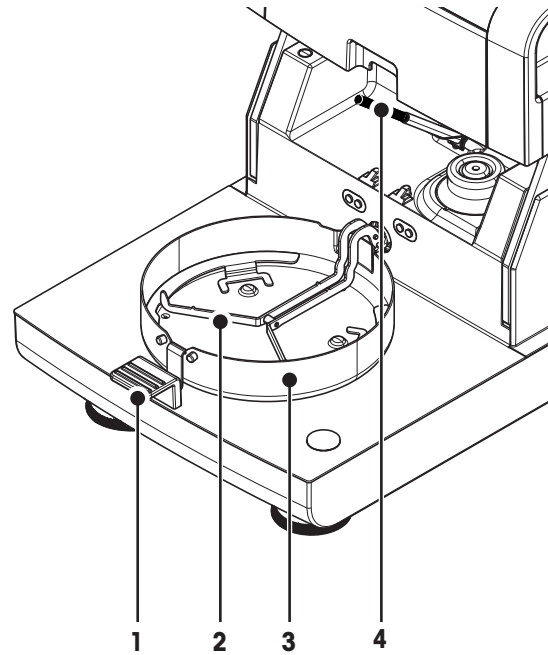
- Use a lint-free cloth for cleaning.
- Ensure that no liquids enters the interior of the instrument.
- **Heating Module**
Clean the exterior of the heating module with a mild cleaning agent although the housing is extremely rugged and resistant to solvents.
- **Terminal**
On no account use cleaning agents, which contain solvents or abrasive ingredients, as this can result in damage to the terminal overlay.
- Never open the housing of the instrument – it contains no components, which can be cleaned, repaired or replaced by the user.

Note

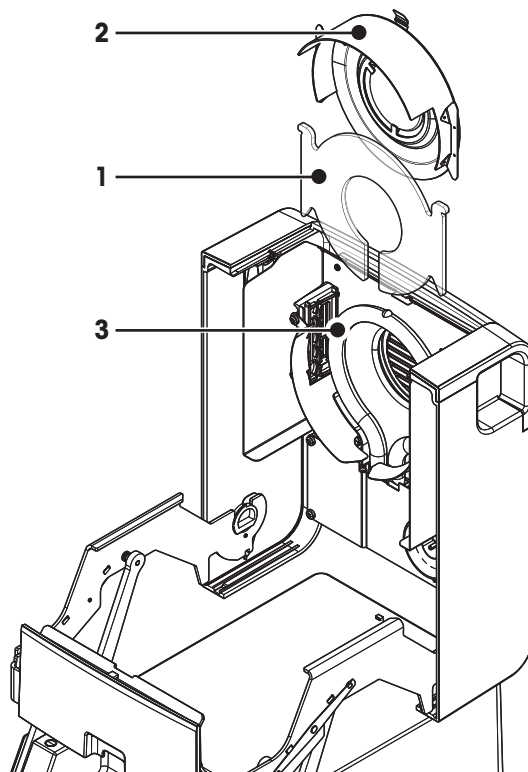
After the temperature sensor or the protective glass have been cleaned, we recommend to adjust the heating module using the temperature adjustment kit, **see** Temperature Adjustment (page 64).

11.1.1 Sample Chamber

- ▶ Sample chamber is open.
- 1 Remove the sample pan handler (1), the sample pan holder (2) and the draft shield (3) for cleaning.
- 2 Carefully remove any deposits from the black temperature sensor (4).



11.1.2 Heating Module



- 1 Protective glass
- 2 Reflector with inspection window glass
- 3 Halogen lamp

To clean the protective glass, the reflector as well as the reflector ring, you should first open the heating module.

Attention

Avoid touching the round halogen lamp. If you have to remove any splashes, deposits, or spots of grease from

the halogen lamp, it is advisable to use a weak organic solvent such as, for example ethanol. Ensure that the lamp is cool. **Do not remove the halogen lamp!**

Opening heating module for cleaning

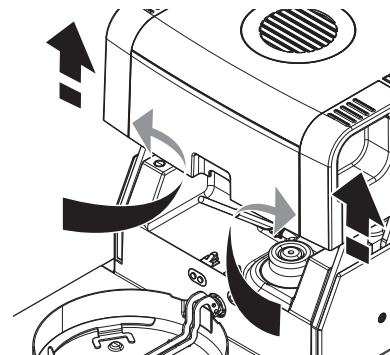
- ▶ Sample chamber is open.
- 1 Inside, there is a locking device on each side. Push both outwards (together) for unlocking.

Note

Do not hold the module while unlocking it.

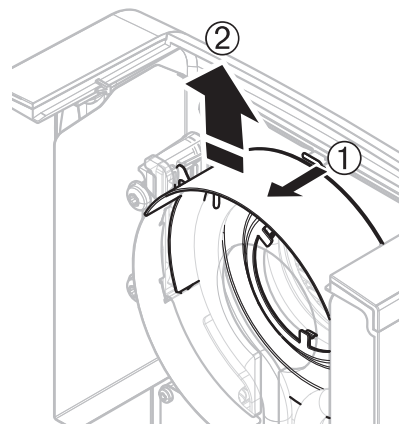
⇒ The upper part has been unlocked on both sides.

- 2 Open the heating module.



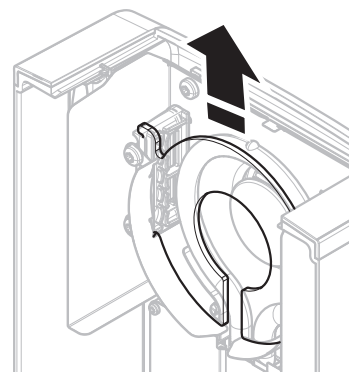
Removing reflector with inspection window glass for cleaning

- 1 To unlock, pull the spring clip forwards.
- 2 Pull out the reflector upwards off the bracket.



Removing protective glass for cleaning

- ▶ Reflector is removed.
- Pull out the protection glass upwards off the bracket.



Reassembling after cleaning

Reassemble all parts in the opposite order.

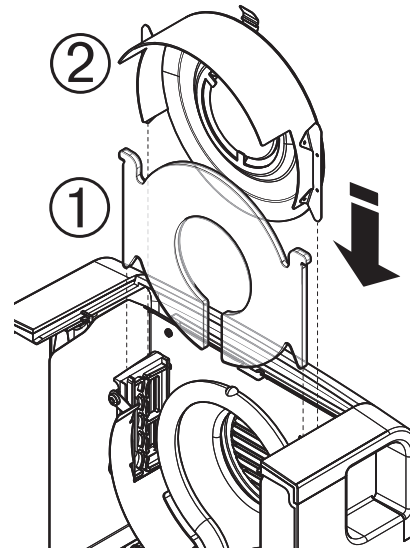
- ▶ All parts are cleaned.

- 1 Insert the protective glass.
- 2 Insert the reflector with inspection window glass (until it clicks).

Note

The lamp must be located in front of the reflector. Do not touch the lamp with your fingers.

- 3 Close the heating module (until it clicks).



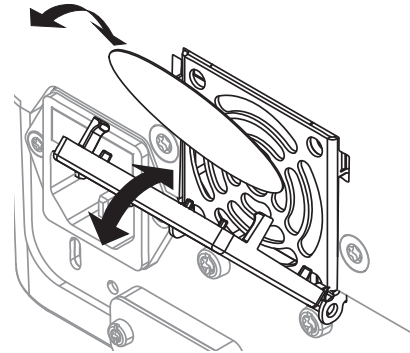
11.1.3 Fan Grill

The air inlet of the fan is located at the rear of the instrument and its exterior should be cleaned from time to time to free it from any dust deposits.

11.2 Dust Filter

If you use the Moisture Analyzer in a dusty environment with an dust filter, check the filter at regular intervals. Replacement Filters **see** Accessories and Spare Parts (page 122).

- Replace the filter, if needed.



11.3 Replacing Power Line Fuse



CAUTION

Safety risk or damage of the instrument

Do not use a fuse of a different type or rated value, or short out (bridge) the fuse, because this can put your safety at risk and damage the instrument!

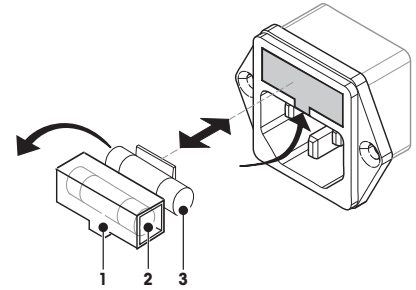
If the display of your terminal remains dark after switching on, in all probability the power line fuse of the drying unit is blown.

The power line fuse is located on the back of the drying unit. To change the fuse, proceed as follows:

- 1 Pull out the power cord.
- 2 Slide out the fuse holder (1) with a suitable tool such as a screw-driver.
- 3 Remove the fuse (3) and check its condition.
- 4 If the fuse is blown, replace the fuse with one of the same type and the same rated value (5 x 20 mm, T6.3H 250 V).

Note:

A spare fuse is located in the holder (2).



Fuse **see** General Technical Data (page 116)

11.4 Disposal

In conformance with the European Directive 2002/96/EC on Waste Electrical and Electronic Equipment (WEEE) this device may not be disposed of in domestic waste. This also applies to countries outside the EU, per their specific requirements.

Please dispose of this product in accordance with local regulations at the collecting point specified for electrical and electronic equipment. If you have any questions, please contact the responsible authority or the distributor from which you purchased this device. Should this device be passed on to other parties (for private or professional use), the content of this regulation must also be related.

Thank you for your contribution to environmental protection.



12 Troubleshooting

Errors can occur during operation of your instrument. This section describes how you can rectify these errors.

12.1 Error Messages

Most error messages appear in plain text directly in the respective application, and usually accompanied by a text describing how to correct the error. Error messages of this type are self-explanatory and therefore are not mentioned below. The following error messages can appear instead of the weighing result.

Error Message	Cause	Rectification
Weight display		
	Overload – The weight on the pan exceeds the weighing capacity of the balance.	– Reduce the weight of the sample.
	Underload – Sample pan holder is missing.	– Insert sample pan holder. If needed, reboot the system by disconnecting and connecting to the mains.
	Weight display flashes / Out of zero range – When the instrument was switched on or upon zeroing, one or more limits were exceeded. The usual reason for this message to appear is when there is a weight on the weighing pan when the balance is switched on.	– Remove the weight.
Adjustment		
Unstable weight.	No stability during adjustment.	• Ensure ambient conditions and an optimum location. • Take care that no part of the sample or the sample pan touches the draft shield or the sample pan handler. • Ensure that the sample pan holder is correctly installed as well as not defect. • Highly volatile substances in the sample also prevent a stable weighing result being detected since the sample is continuously losing weight.
Weight out of range.	Either no weight , or the wrong weight , has been placed on the sample pan during adjustment. (This message is also displayed if you do not remove the weight when prompted to do so by the instrument.)	– Repeat the adjustment process and load the required adjustment weight.

12.2 Status Messages

Status messages are displayed by means of small icons in the status bar of the display. For further information see Status Message Icons (page 17). The status icons indicate the following:

	Cause	Rectification
	Hot surface Indicates that the temperature inside the sample chamber is over approximately 50°C. The parts in the sample chamber and the sample itself can be very hot and there is a risk of burns.	The status icon will disappear as soon the temperature inside the sample chamber is less than approximately 50°C.
	The built-in level sensor has determined that the instrument is not correctly leveled.	– Level the instrument immediately. See Leveling Drying Unit (page 27) ⇒ If the instrument is leveled correctly, the icon disappears.
	The instrument would like to carry out a fully automatic FACT adjustment but is unable because another operating sequence is running.	The adjustment is carried out as soon as the balance is unloaded, the display becomes stable and no key has been pressed for 2 minutes. The status icon disappears after the adjustment is successfully completed.
	The battery in your instrument must be replaced. This battery ensures that the date and time are not lost when the instrument is disconnected from the power supply.	– Contact your METTLER TOLEDO representative as soon as possible. ⇒ A service technician will change the battery.
	Your instrument is due for servicing.	– Contact your METTLER TOLEDO representative as soon as possible to have a technician service your instrument.
	Quick Access icon is displayed continuously.	Quick setting of brightness and sound.
	WLAN connection status.	The instrument is connected to the network. If the device is connected to a network, the WiFi icon indicates the signal strength. The more bars displayed, the stronger the connection.
	WLAN connection status.	WLAN connection failed. The instrument is not connected to the network.

12.3 What to Do if...

Symptom	Countermeasure
Display remains dark after switching on	• Ensure that the terminal is properly connected to the drying unit. • Ensure that the instrument is connected to the power supply and the power is on. • Check the power line fuse of the drying unit and replace if necessary, see Replacing Power Line Fuse (page 110). • If the problem persists, contact a METTLER TOLEDO representative.
Keys and buttons do not respond	• Reboot the system by disconnecting and connecting to the mains. • If the problem persists, contact a METTLER TOLEDO representative.
Printer that is connected does not print	• Ensure that the printer is properly connected to the terminal (do not connect the printer to the drying unit). • Ensure that the printer is on and enabled in the menu. See Peripherals (page 37) • Check the printer settings. See Recommended Printer Settings (Strip printer) (page 128).
Incorrect characters are printed	• Change the bit/parity setting of the printer and the instrument to 8/NO. • Check if both instruments have the same baud rate setting, see Peripherals (page 37). • Use the correct character sets, see Recommended Printer Settings (Strip printer) (page 128).
Measurement takes too long	• You have selected an unsuitable switch-off criterion, see Settings of Switch-off Criterion (SOC) (page 80). • An excessive amount of sample can also be the cause of slow drying, likewise samples which tend to form a skin which hinders vaporization. Perform a measurement at higher temperature. • Enlarge the surface of the sample, e.g. by crushing or grinding. • Use absorbent glass fiber filters for liquids. • If the sample is very temperature sensitive and decomposes, reduce the temperature. • If measuring is unstable, check correct positioning of pan, sample handler, sample, draft shield or sample holder.
No further measures are possible	• Activate Auto-delete and reduce the result limit via Settings > System and data management > Result management.
Instrument does not heat after the start	• Halogen lamp is defective or the heating module is overheated and the thermal overload protection has switched off the heating. In this case, contact your METTLER TOLEDO representative.

Symptom	Countermeasure
Measurement results are not repeatable	• Support on which the instrument is standing is not sufficiently stable. Use a stable support. • Surroundings are very unstable (e.g. vibrations, air draft, humidity). Provide better environmental conditions. • Sample has more or less moisture between sampling and starting the drying process. • Sample is not evenly dispersed on the pan. • Start weight has not always the same value. • Samples are not homogeneous, i.e. they have different compositions. The more inhomogeneous a sample, the larger the amount of sample needed to obtain a repeatable result. • Selected drying time is too short for the "Timed switch-off" criterion. Extend the drying time or select a suitable switch-off criterion "Weight loss per unit of time". • Sample does not become completely dry (e.g. owing to skin formation). Dry the sample with the aid of glass fiber discs. • Selected temperature is too high and the sample has oxidized or decomposed. Lower the drying temperature. • Sample boils and the splashed drops continuously change the weight. Lower the drying temperature. • Granulation is not homogeneous or too large. • Insufficient heating power because the protective glass of the halogen lamp is dirty. Clean the protective glass, see Cleaning (page 107). • Temperature sensor is contaminated or faulty. Clean the temperature sensor, see Cleaning (page 107). • If the problem persists, contact a METTLER TOLEDO representative.
SmartCal test fails	• Ensure that the test prerequisites are met. • Ensure that entries of room temperature and humidity are correct (check RTHSensor) • Ensure that the sample pan is fully and uniformly covered with the test granulate. • Perform the temperature test, see Temperature test (page 68) • Perform the weight test, see Tests (page 67) • Repeat the test with new sample. • If the problem persists, contact a METTLER TOLEDO representative.
WLAN usage is not possible.	• Ensure you have set the correct country setting, see Regional Settings (page 37). • If the problem persists, possibly there is no authorization in your country for WLAN usage with this instrument. For more information, contact a METTLER TOLEDO representative.

13 Technical Data

13.1 General Technical Data

Drying Unit

Heating Module	Halogen ring-shaped radiator
Temperature range	40–230 °C
Temperature step	1 °C
Temperature programs	standard, rapid, gentle, step

Balance

Maximum load	200 g
Readability	1 mg / 0.1 mg
Minimum sample weight	0.1 g
Weighing technology	Monobloc
Adjustment	FACT, internal weight, external weight

Moisture Content

Readability	0.01%
Readability, fine range	0.001%
Repeatability (sd) with 2 g sample	max. 0.05%
Repeatability (sd) with 10 g sample	max. 0.01%

Materials

Drying Unit

Housing	Plastic, PBT, Crastin SO653-GB20
Inspection window grill	Plastic, PEEK-HT G22 (UL94-V0)
Protective glass	Glass ceramics
Halogen lamp	Quartz glass
Reflector	Stainless steel, X2CrNiMo17-2 (1.4404)
Reflector bracket	Plastic, PEEK-HT G22 (UL94-V0)
Draft shield, interior bottom plate	Stainless steel, X2CrNiMo17-2 (1.4404)

Terminal

Top housing	EN ZL-ZnAl4Cu1 (EN ZI-0410)
Bottom housing	PA12 GB30

Protection and Standards

Overvoltage category	II
Degree of pollution	2
Standards for safety and EMC	see Declaration of Conformity (part of standard equipment)
Range of application	for use in dry interior rooms

Environmental Conditions

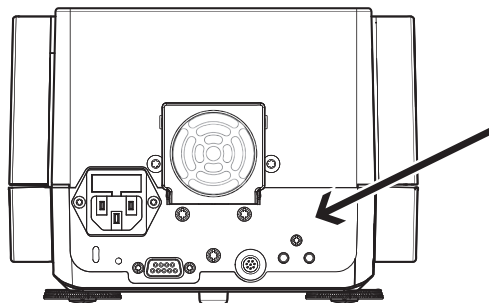
Height above sea level	up to 4000 m
Ambient temperature range	5 °C to 40 °C
Relative air humidity	10% to 80% at 31 °C, linearly decreasing to 50 % at 40 °C, noncondensing

Warm-up time	At least 60 minutes after connecting the instrument to the power supply; when switched on from standby, the instrument is ready for operation immediately.
Power Supply	
110 V AC Version	100 V–120 V, 50/60 Hz, 4 A
230 V AC Version	220 V–240 V, 50/60 Hz, 2 A
Voltage fluctuations	-15%+10%
Power load	max. 450 W during drying process
Power line fuse	5 x 20 mm, T6.3H 250 V
Interfaces	
Drying Unit	• 1x System (Terminal – drying unit) • 1x Connection for METTLER TOLEDO RHT sensor
Terminal	• 1 x RS232C (9-pin socket) • 2x USB Host (Type A socket), USB 1.1 Supported memory sticks up to 32 GB • 1x USB Device (Type B socket), USB 1.1 • 1x Slot for memory cards SD/SDHC up to 32 GB (SDXC not supported) • Ethernet: TCP/IP, DHCP, static IP, 100 Mbit/s, RJ45 • WLAN: TCP/IP, DHCP, static IP, IEEE 802.11b/g/n, open, WEP, WPA-PSK, WPA2-PSK (tkip only)
Data	
Export data format of results	• CSV (Comma-separated values) • PDF
Hardware	
Drying unit	
Sample chamber opening/closing	Motorized
Leveling	2 leveling screws, Level indicator and Inclinator
Sample pan	Ø 90 mm
Maximum height of sample	15 mm
Thermal overload protection	Bimetallic-element switch in heating module
Dimensions (w x h x d)	199 x 139 x 428 mm (Stand for terminal mounted, 530 mm See Dimensions (page 119))
Weight, ready to measure	6.8 kg
Terminal	
Display	WVGA color display, resistive touch-sensitive
Reading angle	Adjustable, 2 folding feet
Dimensions (w x h x d)	200 x 63.5/79.5 x 134.5 mm See Dimensions (page 119))
Weight	1.2 kg

13.1.1 Explanatory note to periodic inspections in accordance with EU Directive 2001/95/EC

This instrument is provided with a 3-pin plug. The earth conductor is led into the instrument and connected to the bottom and the rear panel. The other accessible metal parts are not connected to the earth conductor. These parts are reinforced insulated and must not be connected to the earth conductor according to current European standards.

The earth connection must be checked on the metallic rear panel.



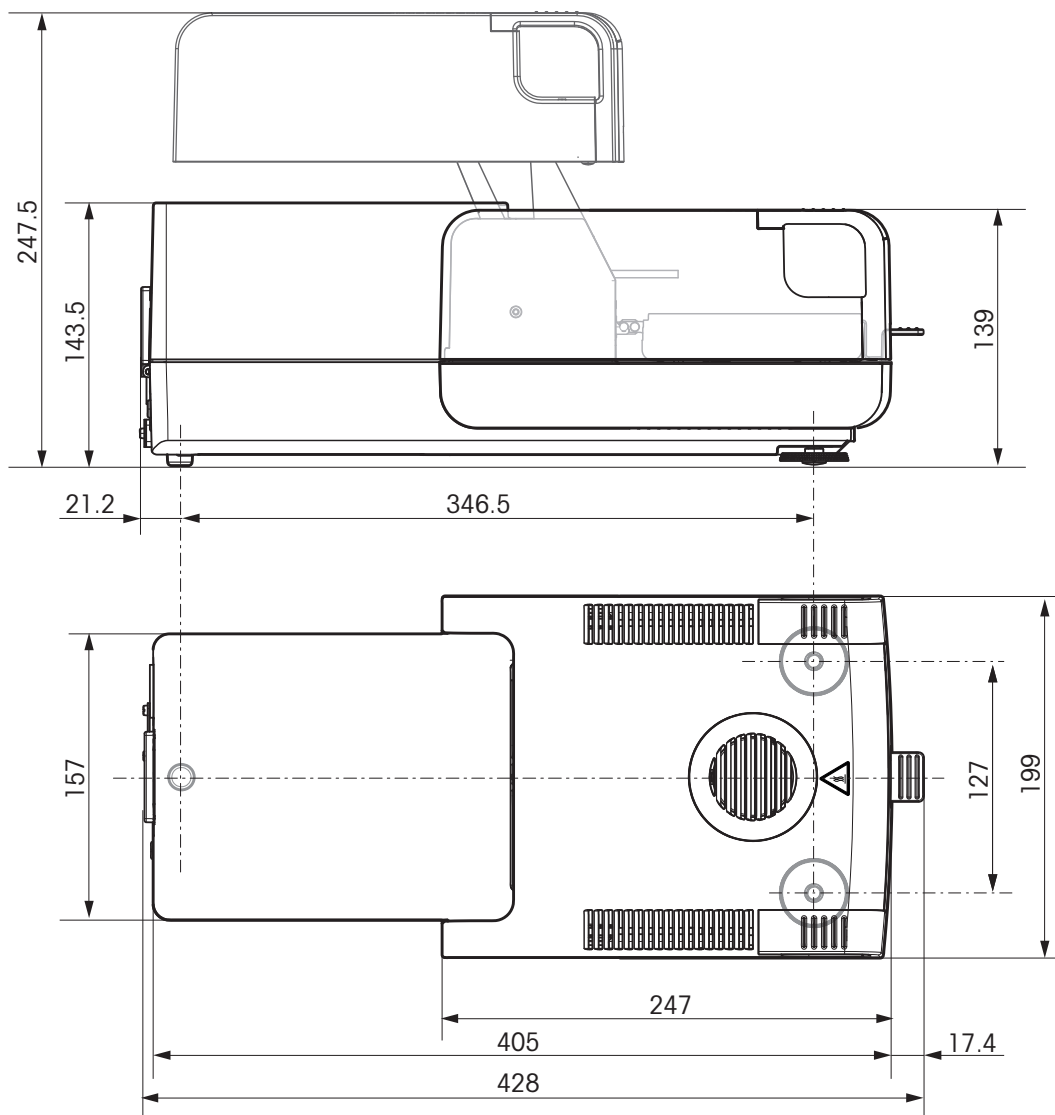
Further information about conformance of this instrument can be found in the Declaration of Conformity which is coming with the product or can be downloaded from the internet.

► www.mt.com/hxhs

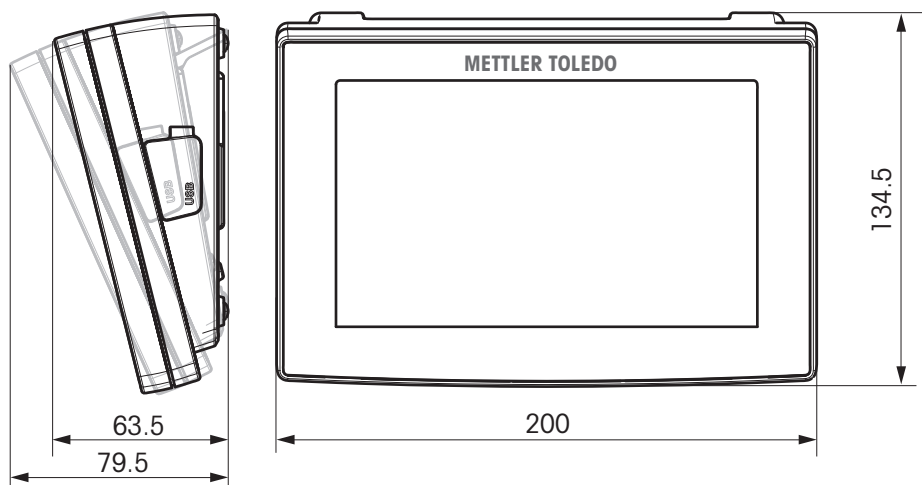
13.2 Dimensions

(all dimensions in mm)

13.2.1 Drying Unit



13.2.2 Terminal

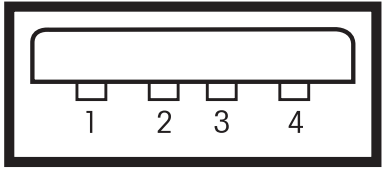


13.3 Interfaces Specification

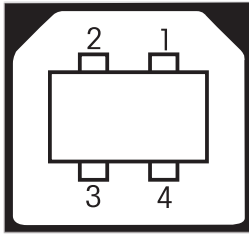
13.3.1 RS232C

Schematic	Item	Specification
	Interface type	Voltage interface according to EIA RS-232C/DIN66020 CCITT V24/V.28)
	Max. cable length	15 m
	Signal level	Outputs: +5 V ... +15 V (RL = 3–7 kΩ) –5 V ... –15 V (RL = 3–7 kΩ) Inputs: +3 V ... +25 V –3 V ... –25 V
	Connector	Sub-D, 9-pole, female
	Operating mode	Full duplex
	Transmission mode	Bit-serial, asynchronous
	Transmission code	ASCII
	Baud rates	see setting options
	Bits/parity	see setting options
	Stop bits	see setting options
	Handshake	None, XON/XOFF, RTS/CTS (software selectable)

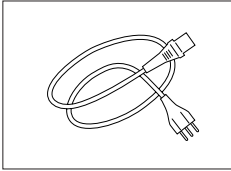
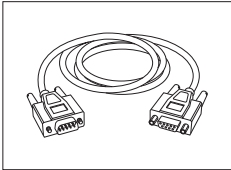
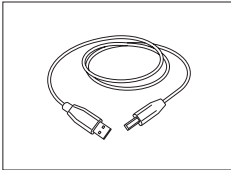
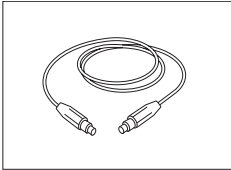
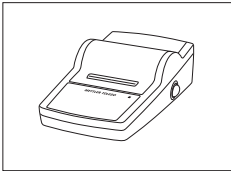
13.3.2 USB Host

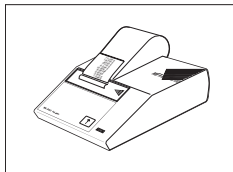
Schematic	Item	Specification
	Standard	In conformity with USB Specification Revision 1.0/1.1
	Speed	Full speed 12 Mbps (requires shielded cable)
	Power usage	Max. 500 mA
	Connector	Type A
	Pin assignment	1 VBUS (+5 V DC)
		2 D- (Data -)
		3 D+ (Data +)
		4 GND (Ground)
		Shell Shield

13.3.3 USB Device

Schematic	Item	Specification										
 <table data-bbox="394 993 681 1157"><tr><td>1</td><td>VBUS (+5 VDC)</td></tr><tr><td>2</td><td>D- (Data -)</td></tr><tr><td>3</td><td>D+ (Data +)</td></tr><tr><td>4</td><td>GND (Ground)</td></tr><tr><td>Shield</td><td>Shield</td></tr></table>	1	VBUS (+5 VDC)	2	D- (Data -)	3	D+ (Data +)	4	GND (Ground)	Shield	Shield	Standard	In conformity with USB Specification Revision 1.1
	1	VBUS (+5 VDC)										
	2	D- (Data -)										
	3	D+ (Data +)										
	4	GND (Ground)										
	Shield	Shield										
Speed	Full speed 12 Mbps (requires shielded cable)											
Function	CDC (Communication Device Class) serial port emulation											
Power usage	Suspended device: Max 10 mA											
Connector	Type B											

14 Accessories and Spare Parts

	Description	Part No.
Power supplies		
	Country-specific 3-Pin power cable with grounding conductor.	
	Power cable AU	00088751
	Power cable BR	30015268
	Power cable CH	00087920
	Power cable CN	30047293
	Power cable DK	00087452
	Power cable EU	00087925
	Power cable GB	00089405
	Power cable IL	00225297
	Power cable IN	11600569
	Power cable IT	00087457
	Power cable JP	11107881
	Power cable TH, PE	11107880
	Power cable US	00088668
	Power cable ZA	00089728
Cables for RS232C interface		
	RS9 – RS9 (m/f): connection cable for PC, length = 1 m	11101051
Cables for USB interface		
	USB (A –B) connection cable for connection to PC, length = 1 m	12130716
Cables for terminal		
	Terminal cable, length = 0.68 m	30003971
Printers		
	RS-P25 printer with RS232C connection to instrument	11124300
	Paper roll, set of 5 pcs	00072456
	Paper roll, self-adhesive, set of 3 pcs	11600388
	Ribbon cartridge, black, set of 2 pcs	00065975



RS-P42 printer with RS232C connection to instrument

00229265

Paper roll, set of 5 pcs

00072456

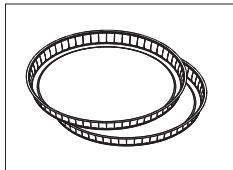
Paper roll, self-adhesive, set of 3 pcs

11600388

Ribbon cartridge, black, set of 2 pcs

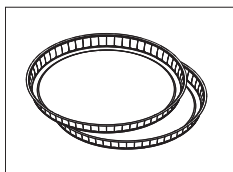
00065975

Sample Pans



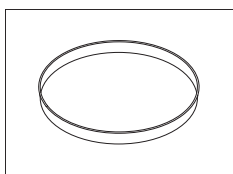
Aluminum sample pan, HA-D90, set of 80 pcs

00013865



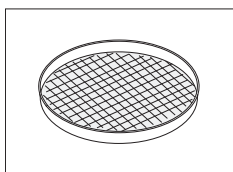
Professional aluminum sample pan, extra strong, set of 80 pcs

11113863



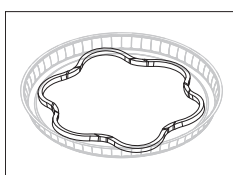
Stainless steel reusable sample pan 6 mm, DA-DR1, set of 3 pcs

00214462



Textile weighing pan HA-CAGE, 1 pcs

00214695



Down holder, prevents slipping of the sample during measurement, 1 pcs

00214758

Adjustment / Test Parts

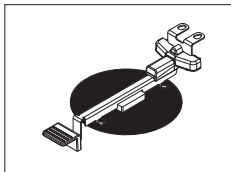


Certified adjustment weight, 100 g (F1)

11119531

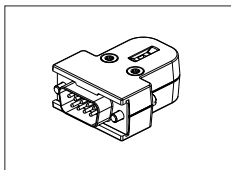


OIML / ASTM Weights (with calibration certificate) see www.mt.com/weights



Temperature adjustment set HA-ETCC, certified

30020851



RHT sensor for detecting of humidity and temperature

30020850



SmartCal™, Moisture Analyzer Reference Substance

cSmartCal™, certified, 12 tests

30005793

cSmartCal™, certified, 24 tests

30005791

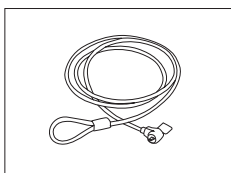
SmartCal™, 12 tests

30005792

SmartCal™, 24 tests

30005790

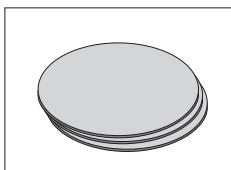
Anti-theft devices



Steel cable

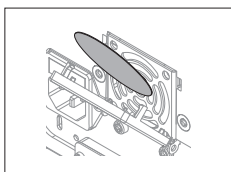
11600361

Miscellaneous



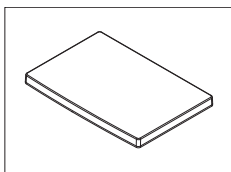
Glass fiber filter (for Liquids), set of 100 pcs

00214464



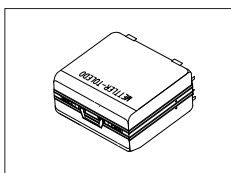
Dust filter, set of 50 pcs

30020838



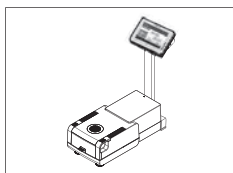
Protective cover for Terminal

30003957



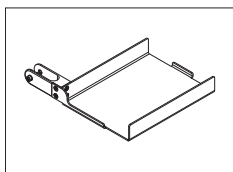
Transport case, 452 mm x 360 mm x 250 mm

30020836



Stand for terminal

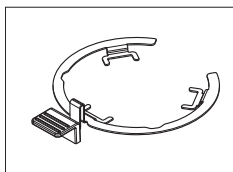
30018474



Printer holder for mounting on the stand for terminal.

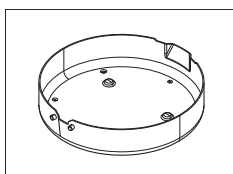
30066692

Spare Parts



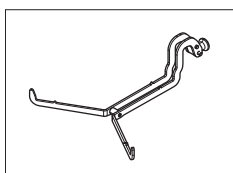
Sample pan handler

30020852



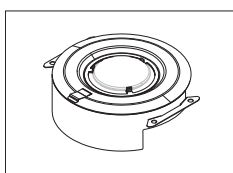
Draft shield

30128349



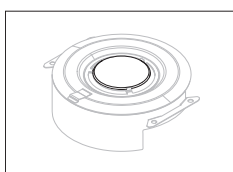
Sample pan holder

11148108



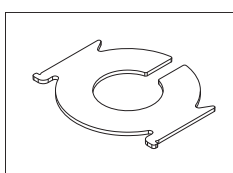
Reflector without inspection glass

11148330



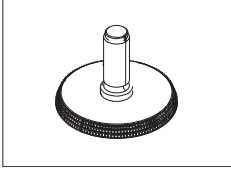
Inspection glass for reflector

11148421



Protective glass

11148416



Leveling foot

11106323

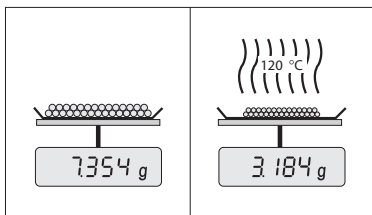
15 Appendix

15.1 How to Obtain Best Results

In this section you will find important information on how to obtain optimum results. You will discover which parameters influence the measurement process and how you can optimize the settings of your instrument to achieve the best measurement results.

15.1.1 Measurement Principle of Halogen Moisture Analyzer

Your instrument performs measurements based on the **thermogravimetric principle**, i.e. the moisture is determined from the weight loss of a sample dried by heating.



In principle, your instrument thus comprises two instruments: An analytical balance and a heating module. In contrast to other thermogravimetric methods (drying oven, infrared, microwave), the Halogen Moisture Analyzer operates with halogen heating technology. This ensures fast heating of the sample and thus guarantees rapid availability of the measurement results.

Irrespective of the measurement method, a correct preparation of the sample and the correct choice of the following measurement parameters improve the quality of the measurement result:

- Sample size
- Drying temperature
- Switch-off mode
- Drying time

Attention

Inappropriate setting of these parameters can cause the results to be incorrect or misleading. For this reason, check that the results for each type of sample are what might reasonably be expected.

You will find detailed information on the relationships between these parameters in the delivered application brochure «Guide to Moisture Analysis» or **see** Accessories and Spare Parts (page 122).

In practice not only the quality of the measurement results, but also the speed of the measurement process is important. Thanks to its drying principle (heat generated by a halogen radiator), the Halogen Moisture Analyzer is very fast. You can increase the speed even further through optimum setting of the instrument, e.g. using the **Drying program Rapid**.

The optimum drying temperature and the drying time are dependent on the nature and size of the sample and on the desired accuracy of the measurement results. These can be determined only by experiment. The Halogen Moisture Analyzer supports you in this task: It offers result recording of test measurements in the menu **Method Definition**.

15.1.2 Notes on Adjustment of Balance and Heating Module

The balance and the heating module in your drying unit can be adjusted using the appropriate accessories, **see** Accessories and Spare Parts (page 122). The heating module and the balance can also be tested (to check the adjustment). The user can define a test weight or the test temperature, together with permissible tolerances for this test. A test report can be printed out showing the test result has passed or failed.

A Moisture Analyzer is typically used in place of or in addition to the drying oven method. In an oven heat energy is transferred by the flow of air, which establishes an equilibrium between the sample temperature and the ambient temperature. This is not the case in a Moisture Analyzer. The actual sample temperature primarily depends on the specific absorption properties of the sample (dark samples absorb more heat), which can change during the measurement process. There can also be differences between the temperature at the surface of the sample and the temperature inside the sample. The heat output is therefore not dependent on the true sample temperature but instead is regulated by a temperature sensor underneath the halogen heating module.

For the reasons explained above, the sample temperature will be slightly different than the temperature shown on the instrument display. By regularly testing or adjusting the drying unit, you will ensure a consistent and reproducible heat output for the entire lifetime of your instrument.

Note:

- METTLER TOLEDO offers an adjustment service – please contact your METTLER TOLEDO representative.
- We recommend to adjust the instrument exclusively under operating conditions.
- After the temperature sensor or the protective glass have been cleaned, we recommend adjusting the heating module using the temperature adjustment kit, **see** Accessories and Spare Parts (page 122). For the procedure for testing or adjusting the drying unit (balance/heating module), **see** Test/Adjust (page 62).

15.1.3 Optimum Sample Preparation

Preparation of the sample is decisive for the speed of the measurement process and the quality of the measurement results.

Note:

Basic rules for the preparation of your sample:

The amount of sample you select should be as small as possible and only as large as necessary.

Excessive amounts of sample require more time for drying and thus prolong the measurement process. If the amount of sample is too small, the measurement result may possibly not be representative of the true moisture content. The following always holds: The greater the inhomogeneity of the sample, the larger the amount of sample needed to obtain a repeatable result.

Distribute the sample evenly over the sample pan

You thus increase the surface area of the sample and facilitate heat absorption. The base of the pan should be evenly covered.

With liquid, fat-containing, melting and highly reflecting samples, you should use the sample with the glass fiber filter available as optional equipment, **see** Accessories and Spare Parts (page 122). This also applies to samples which form a skin on their surface when heated. The glass fiber filter ensures even and rapid heat distribution and prevents the formation of a skin on the sample surface.

15.1.4 Further Information on Moisture Determination

You will find further information on the moisture determination, the importance of the parameters and the preparation of the samples in the delivered application brochure «Guide to Moisture Analysis» dealing with moisture determination, **see** Accessories and Spare Parts (page 122).

Useful tips and a variety of example methods (comparison between Halogen Moisture Analyzer results and the oven drying method) can be downloaded from:

- www.mt.com/moisture
- www.mt.com/moisture-methods
- www.moisture-guide.com

If you require information on specific applications, your METTLER TOLEDO Customer Service will be pleased to help you.

Attention

Moisture determination applications must be optimized and validated by the user according to local regulations. Application-specific data provided by METTLER TOLEDO is intended for guidance only.

15.2 Recommended Printer Settings (Strip printer)

English, German, French, Spanish, Italian, Polish, Czech, Hungarian

Printer			
Model	Char Set	Auto baudrate	Balance feature
RS-P25	IBM/DOS	Off	Disable

Printer				
RS-P42	IBM/DOS ¹⁾	—	—	—
Instrument / Printer				
Model	Baudrate	Bit / Parity	Stop Bits	Handshake
RS-P25	9600	8/NO	1	Xon/Xoff
RS-P42	1200	8/NO	1	Xon/Xoff

Brazil Portuguese

Printer			
Model	Char Set	Auto baudrate	Balance feature
RS-P25	IBM/DOS	Off	Disable
RS-P42	— 2)	—	—

Instrument / Printer				
Model	Baudrate	Bit / Parity	Stop Bits	Handshake
RS-P25	9600	8/NO	1	Xon/Xoff
RS-P42	— 2)	— 2)	— 2)	— 2)

¹⁾ Printer settings not available.

²⁾ Required font for this language not available.

15.3 Installing USB Device Driver

For transmitting MT-SICS commands via USB Device interface, a driver needs installing on the connected PC. You can find the USB driver installer for download on the METTLER-TOLEDO website:

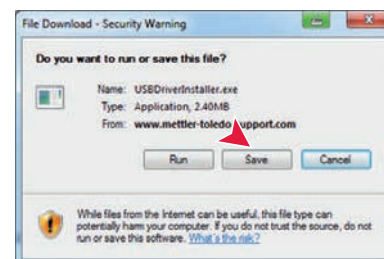
► www.mettler-toledo-support.com

Requirements

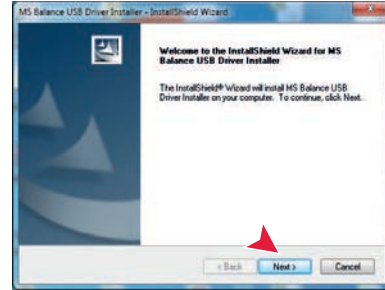
- PC with one of the Microsoft Windows® operating system 32-Bit & 64-Bit: Windows XP (SP3), Vista (SP2), Win 7 (SP1) or Win 8.
- Administrator rights for installing software.
- PC to instrument USB connection cable.

Installing USB Driver on the PC:

- 1 Connect to the Internet
- 2 Go to the site <http://www.mettler-toledo-support.com>.
- 3 Log in to the METTLER TOLEDO Balance Support Site (registration with the serial number of a METTLER TOLEDO instrument required).
- 4 Click **Customer Support**.
- 5 Click appropriate product folder e.g. HX204 or HS153.
- 6 Click **USB Driver**.
- 7 Click **USBDriverInstaller.exe**.
- 8 Click **Save** to download to your specified location.
- 9 Right-click on the downloaded install program: **USBDriverInstaller.exe** and select **Run as Administrator** from the menu.
- 10 If a safety warning appears, allow Windows to install.



11 Click **Next** and follow the installer's instructions.



Installing Instrument

- 1 Switch off moisture analyzer
- 2 Connect the moisture analyzer to the preferred USB Port on the PC.
- 3 Switch the moisture analyzer on.
- 4 Follow the instructions of the wizard and install the software automatically (recommended).



Note

The wizard appears again for each USB port, either on your PC or if another moisture analyzer is connected.

Attention

Do not click [Cancel] as for the connected USB port, it might not be possible anymore to perform the installation process.

15.4 Third Party License/Notice

This section contains Third Party Software Notices and/or Additional Terms and Conditions for licensed third party software components included within SOFTWARE PRODUCT.

This SOFTWARE PRODUCT is based in part on the work of:

- **Qwt project**
For user's guide **see** <http://qwt.sf.net>
For LGPL license V2.1 **see** <http://www.gnu.org/licenses/old-licenses/lgpl-2.1.html>
- **KompexSQLite**
For LGPL license V3 **see** <http://www.gnu.org/licenses/lgpl.txt>
- **Qt library 4.8**
For LGPL license V2.1 **see** <http://qt-project.org/doc/qt-4.7/lgpl.html>
For GPL license V3 **see** <http://qt-project.org/doc/qt-4.7/gpl.html>
- **decNumber**
For ICU license V3.68 **see** <http://source.icu-project.org/repos/icu/icu/trunk/license.html>
- **RapidXML**
For MIT license **see** <http://www.opensource.org/licenses/mit-license.php>

Index

A

Access rights	53, 55
Accessories	122
Account	
Policies	56
User	55
Adding comments	96
Adjustment	16, 29
Balance	62
FACT	62
Heating module	62
Notes	127
Printouts	66
Settings	48
Touch screen	40
Weight	63
Weight external	63
Weight internal	63
Anti-theft	11, 28
Appendix	127
Asterisk	82, 88
ATRO	87
Auto-delete functionality	57
Automatic sample chamber	89
AutoMet	93

B

Back up	59
Basic elements	17
Battery	113
Brightness	35, 40

C

Calculated values	82
Certified RHT sensor	69, 70
Chart	99
Chart view	23
Cleaning	107
Color	35, 40
Comments to measurement	96
Commissioning	24
Company information	40
Connecting the Instrument	25
Connectivity	11, 13
Control limits	86
Copy	
Method	74
Country	28, 37, 37
CSV	117
CSV format	103

D

Data	117
Data management	57
Date	20, 28, 37
Default settings	39

Define

Method	74
Shortcut	97

Delete

see Remove	74
------------	----

Dimensions

119

Display language	35, 39
------------------	--------

Display mode	82
--------------	----

Disposal	111
----------	-----

Draft shield element	11
----------------------	----

Drying

Steps	79
-------	----

Temperature	76, 77, 78
-------------	------------

Unit	11, 26, 116, 117
------	------------------

Unit dimensions	119
-----------------	-----

Drying program

Gentle	77
--------	----

Rapid	76
-------	----

Standard	75
----------	----

Dust filter	11, 110
-------------	---------

E

Eco	90
-----	----

Edit

Method	74
--------	----

Shortcut	97
----------	----

Entering

Numerical values	19
------------------	----

Text and numbers	18
------------------	----

Error messages	112
----------------	-----

Ethernet	13, 41
----------	--------

Event logs	60
------------	----

Export	60
--------	----

Export

Result	103, 117
--------	----------

Settings and methods	58
----------------------	----

Export management	45
-------------------	----

F

FACT	29, 62, 113
------	-------------

Factor	88
--------	----

Fan	11, 110, 110
-----	--------------

Filter	99
--------	----

Firmware update	60-61
-----------------	-------

First measurement	31
-------------------	----

Free factor	88
-------------	----

Fuse	110, 117
------	----------

G

General method properties	91
---------------------------	----

General safety information	8
----------------------------	---

General settings	16
------------------	----

Gentle drying	77
---------------	----

Graphics panel	22
----------------	----

Graphics view	23, 99
---------------	--------

Groups	53
--------	----

H		
Halogen lamp	11	
Heating module	11, 11, 107	
History	72	
Home screen	14, 16	
Host name	41	
Host settings	37	
I		
Identifications	44	
Identifications panel	22	
Import		
Settings and methods	58	
Inspection window	11	
Installation	24	
Intended Use	8	
Interface	117	
Interim results	45	
Introduction	7	
J		
Journal	99	
K		
Kensington slot	11, 28	
Keyboard		
text and numbers	18	
Keyboard language	35	
Keypad		
numerical values	19	
Keys on the terminal	14	
L		
LabX Direct	38	
Languages	35, 39	
Printout	45	
Level		
Indicator	11	
Sensor	11, 28, 113	
Leveling	27	
Leveling screw	11	
Limits	86	
Lists	21	
Location	24	
Log file	60	
Login	56	
M		
Maintenance		
Cleaning	107	
Dust filter	110	
Heating module	107	
Power line fuse	110	
Measurement	16	
First	31	
Measurement principle	127	
Memory Card	13	
Memory stick	13, 57	
Method		
Copy	74	
Definition	73	
Edit	16, 74	
Export and import	58	
Name	22, 91, 91	
New	16, 74	
Properties	91	
Remove	74	
Renaming	91	
Test	92	
MT-SICS	37	
N		
Networks	43	
New method	74	
Numbers	18	
Numerical values	19	
O		
Out of zero range	112	
Overload	112	
P		
Parameter panel	22	
Password	36, 55	
PCL printer	38	
PDF	117	
Peripheral settings	37	
Power line fuse	11, 110	
Power supply	11, 117	
Power supply socket	11	
Preheating	89	
Print management	45, 45	
Printer		
Settings	37, 128	
Printout		
Intermediate results	45	
Language	45	
Short	102	
Standard	102	
Protective clothing	9	
Q		
QuickAccess	113	
R		
Ramp time	78	
Rapid drying	76	
Reading angle	28	
Remove		
Method	74	
Shortcut	97	
Replace		
Dust filter	110	
Power line fuse	110	
Reset	60	
Resolution	87	
Restore	59	

Result			Standard drying	75
Export	103, 117		Standard equipment	24
Result and value handling	86		Standby	14, 90
Result limit	57		Start measurement	16
Result management	57		Start weight	85, 96
Results	16, 99, 99		Starting mode	89
RHT sensor	12, 49, 70		Startup user	56
Installation	29		Status icon	113
RS232C	12, 13		Status messages	17, 113
S			Stop drying	96, 103
Safety information	8		Switch button	19
Intended use	8		Switch ON/OFF	14
Protective clothing	9		Switch-off criterion	80, 92, 103
Signal words	8		Switch-off time	90
Staff safety	9		System	11, 13
Warning Symbols	8		System management	57
Sample chamber	11, 14		System settings	37
Sample pan handler	11		T	
Sample pan holder	11		Tables	21
Sample preparation	128		TCP/IP	13, 41
Scope of delivery	24		Technical data	116
Screen settings	35, 39, 40		Temperatur	
SD card	13		Adjustment settings	51
Selecting the location	24		Temperature	
Setting up drying unit	26		Adjustment kit	11
Settings	16		Sensor	11
Adjustment	48		Standby	90
Date	20, 28, 37		Switch-off criterion	80
Export and import	58		Test	68
Host	37		Test settings	49
Peripheral	37		Terminal	28, 107
Regional	37		Terminal dimensions	120
Screen	35		Test	16
System	37		Balance	67
Temperature Test	49		Heating device	67
Time	20, 28, 37		Measurement	92, 103
Weight adjustment	51		Method	92, 103
Weight test	49		Temperature	68
Shortcuts	16, 17, 22, 97		Time	92
Define	97		Weight external	67
Edit	97		Weight internal	67
Remove	97		Text	18
Show results	16		Thermogravimetric	127
Signal words	8		Time	20, 28, 37
SmartCal			Touch screen	14
General	69		Touch screen adjustment	40
History	72		Troubleshooting	60, 112
RHT sensor	29		U	
Test	69		Underload	112
Test settings	50		Unpacking	24
SOC	80		USB	12
SOC delay	82		USB Device	13
Software update	60-61		USB Driver	
Sound	36		Installation	129
Spare parts	122		USB Host	13
Staff Safety	9		USB Memory stick	
Stand for terminal	27			

	see memory stick	57
User		
	Account	55
	Groups	53
	Home screen	16
	Interface	15
	Management	53
	Shortcuts	16, 97
V		
	Value panel	22
W		
	Warning Symbols	8
	Weighing-in aid	85, 96
	Weight	
	Adjustment external	63
	Adjustment internal	63
	Test external	67
	Test internal	67
	Weight adjustment	
	Settings	51
	Weight test	
	Settings	49
	Wireless LAN interface	13, 42
	WLAN	13, 37, 42,
		113
	Settings	37
	Work screen	22
	Workflow handling	89

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- reduce costs by optimizing testing procedures
- comply with the most common regulatory requirements

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For more information

Mettler-Toledo AG, Laboratory Weighing

CH-8606 Greifensee, Switzerland

Tel. +41 (0)44 944 22 11

Fax +41 (0)44 944 30 60

www.mt.com

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